Commutative Algebra and Algebraic Geometry

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This summary is based on the course "Fundamentals of Commutative Algebra and Algebraic Geometry" taught in the Hebrew University of Jerusalem in the Spring semester of 2026 by Prof. Zev Rosengarten.

While the summary covers the entirety of the material taught, it does not do so in the same order. In particular, while the course taught elements of algebraic geometry and commutative algebra simultaneously, these two topics were split into distinct parts in this summary. Within each part, the sections are kept in-order. This was done for ease of reference, but not ease of reading. Thus this summary should be used as a reference, and not read in-order.

Issues may arise when reading this summary in-order, for example topics on algebraic geometry may be referenced before they are defined. Furthermore, a more full picture of the topic is apparent when you read about geometric aspects of algebra concurrently with reading about the algebra itself. Thus if you would like to use this summary to learn about the material yourself, it is recommended you follow the syllabus of the course.

My personal recommendation for the order to read this summary is as follows:

(1) Sections 1 and 2 of Commutative Algebra (Rings and Modules; Localization).

(2) Sections 1.1 and 1.2 of Algebraic Geometry (Affine Schemes, Basic definitions; Topological properties of affine schemes).

(3) Section 3 of Commutative Algebra (Properties of Modules).

(4) Section 2.1 of Algebraic Geometry (Varities; Affine varieties); this is essentially just more commutative algebra, there is little geometry here.

(5) Section 4 of Commutative Algebra (Tensoring and Flatness).

(6) Second 1.3 of Algebraic Geometry (Affine Schemes, Categorical constructions); this is a short section which requires tensor products.

(7) Continue with the rest of the summary in order, from section 2.2 of Algebraic Geometry (Varieties; Algebraic varieties) onward.

I. Commutative Algebra

1 Rings and Modules

1.1 Rings and ideals

We recall some basic notions regarding rings.

1.1.1 Definition

A **ring** is a set R equipped with two operations $+$ and $\cdot$, making it into a commutative group and a monoid respectively. Furthermore, they satisfy the distributivity properties:

$$a \cdot (b + c) = ab + ac, \qquad (a + b) \cdot c = ac + bc$$

for all $a, b, c \in R$. The additive and multiplicative identities are denoted 0 and 1 respectively.

Note:
In this course all rings are assumed to be **commutative**: $\cdot$ forms a commutative monoid.

1.1.2 Definition

A **ring morphism** is a function between two rings, $f\colon R \to S$, which preserves structure: $f(x + y) = f(x) + f(y)$, $f(xy) = f(x)f(y)$, and $f(1) = 1$. A **ring isomorphism** is a ring morphism with an inverse (equivalently a bijective ring morphism).

A **subring** of a ring R is a subset $S \subseteq R$ equipped with a ring structure making the inclusion $S \to R$ a ring morphism.

An **ideal** of a ring R is a nonempty subset $I \subseteq R$ closed under addition and multiplication by R: for $r \in R$ and $x \in I$, $rx \in I$. For an ideal $I \subseteq R$, we define its **quotient ring** R/I to be the set of all cosets $[r] = r + I$ with ring structure given by

$$0 = [0] = I, \qquad 1 = [1], \qquad [x] + [y] = [x + y], \qquad [x] \cdot [y] = [xy],$$

equivalently the unique structure making the projection map $\pi\colon R \to R/I, r \mapsto [r]$ a ring morphism.

1.1.3 Proposition

Let $f\colon R \to S$ be a ring morphism, then

(1) For every ideal $J \leq S$, $f^{-1}(J) \leq R$ is an ideal.

(2) In particular f's **kernel**, $\ker f = f^{-1}0$, is an ideal of R. And f's **image**, $\mathrm{im} f = f(R)$ is a subring of S. f then quotients to give a ring isomorphism $R/\ker f \cong \mathrm{im} f$.

(3) For an ideal $I \leq R$, consider the projection $\pi\colon R \to R/I$. Then $J \mapsto \overline{J} = \pi(J)$ and $\overline{J} \mapsto \pi^{-1}\overline{J}$ gives a one-to-one correspondence between ideals of R containing I and ideals of R/I.

1.1.4 Definition

For a ring R and a subset $S \subseteq R$, we denote the smallest ideal containing S by (S). This exists as the arbitrary intersection of ideals is an ideal. When $S = \{x\}$ is a singleton, we denote $(\{x\}) = (x)$; such an ideal is called a **principal ideal**.

1.1.5 Proposition

The ideal generated by S, (S), is the set of all finite R-combinations of elements from S:

$$(S) = \left\{ \sum_{s \in S} r_s s \;\middle|\; r_s \in R, \text{ all but finitely many being zero} \right\}.$$

In particular, $(x) = Rx$ is the set of all multiples of x.

Proof

It is not hard to see that this is an ideal, and clearly every ideal containing S must contain it. ■

1.1.6 Definition

Let R be a ring, an element $x \in R$ is

(1) a **zero divisor** if there is a $y \in R$ such that $xy = 0$;

(2) a **unit** if there is a $y \in R$ such that $xy = 1$;

(3) **nilpotent** if $x^n = 1$ for some $n > 0$.

The set of units of R is denoted $R^\times$ and forms a group.

1.1.7 Definition

A ring $R \neq 0$ is

(1) **reduced** if every $x \neq 0$ is not nilpotent;

(2) an **integral domain** if every $x \neq 0$ is not a zero divisor;

(3) a **field** if every $x \neq 0$ is a unit.

It is easy to see that a field is an integral domain is reduced.

1.1.8 Proposition

For a ring R, the following are equivalent:

(1) R is a field,

(2) R has no non-trivial ideals,

(3) Every ring morphism out of R to a non-zero ring is injective.

Proof

If R is a field, then every nonzero element is a unit and thus the ideal generating it must be all of R. Conversely if R has no non-trivial ideals, then for $x \neq 0$ we must have that $1 \in (x)$ and thus there is a $y \in R$ such that $xy = 1$ so it is a unit.

If R has no non-trivial ideals then for every $f\colon R \to S \neq 0$ we must have $\ker f = 0$, as it is an ideal not containing 1. Thus f is injective. Conversely if $I \neq R$ is an ideal of R, then $\pi\colon R \to R/I$ is injective, but its kernel is I, so $I = 0$. ■

1.1.9 Proposition

If $\{I_\alpha\}_\alpha$ is a collection of ideals from R, then

(1) $\bigcap_\alpha I_\alpha$ is an ideal, the largest ideal contained in each I_α.

(2) $\sum_\alpha I_\alpha = \{\sum_\alpha x_\alpha \mid x_\alpha \in I_\alpha \text{ all but finitely many are zero}\}$ is an ideal, the smallest ideal containing each I_α.

Proof

The first point is easily verified, and the second one follows because

$$\sum_\alpha I_\alpha = \left(\bigcup_\alpha I_\alpha\right).$$

■

1.1.10 Definition

If I, J are ideals then we define their **product** to be the ideal generated from all products of elements from I and J:

$$IJ = (\{xy \mid x \in I, y \in J\}).$$

Since I, J are ideals and are thus closed under multiplication from R, for $x \in I$ and $y \in J$ we have that $xy \in I \cap J$. Thus $IJ \subseteq I \cap J$.

1.1.11 Definition

For an ideal $I \subseteq R$, we define its **radical** to be

$$\sqrt{I} = \{r \in R \mid r^n \in I \text{ for some } n \geq 0\}.$$

1.1.12 Lemma

The radical of an ideal is an ideal.

Proof

If $a^n \in I$ and $r \in R$ then $(ra)^n = r^n a^n \in I$ so that $ra \in \sqrt{I}$. And if $a^n, b^m \in I$ then

$$(a+b)^{n+m} = \sum_{i=0}^{n+m} \binom{n+m}{i} a^i b^{n+m-i} \in I,$$

since for each i, if $i < n$ then $n+m-i > m$, so $a+b \in \sqrt{I}$. ■

1.2 Prime and maximal ideals

1.2.1 Definition

A proper ideal $I \subset R$ is

(1) **prime** if $ab \in I$ implies $a \in I$ or $b \in I$. Equivalently if $a, b \notin I$ then $ab \notin I$.

(2) **maximal** if it is not contained in any other proper ideal.

1.2.2 Proposition

An ideal I is

(1) prime iff R/I is an integral domain;

(2) maximal iff R/I is a field.

Consequently, a maximal ideal is prime.

Proof

R/I is an integral domain iff $[a][b] \neq [0]$ when $[a], [b] \neq [0]$. This means that $ab \notin I$ when $a, b \notin I$, as desired. R/I is a field iff it has no nontrivial ideals. Since ideals of R/I are in bijection with ideals containing I, this is iff I is maximal. ■

In particular the zero ideal $0 = (0) = \{0\}$ is prime iff $R \cong R/(0)$ is an integral domain.

1.2.3 Corollary

Let I be an ideal, then the correspondence $J \leftrightarrow \overline{J}$ between ideals containing I and ideals of R/I preserves primality and maximality: J is prime/maximal iff $\overline{J}$ is.

Proof

We have that

$$(R/I)/\overline{J} = (R/I)/(J/I) \cong R/J,$$

so that $(R/I)/\overline{J}$ is an integral domain/field iff R/J is. The result follows. ■

1.2.4 Definition

(1) An element $0 \neq x \in R$ is **irreducible** if it is not a unit, and if $x = yz$ then either y or z is a unit.

(2) An integral domain R is a **principal ideal domain** (or a **PID**) if all of its ideals are principal.

1.2.5 Proposition

If R is a PID and $0 \neq x$, then (x) is prime iff x is irreducible iff (x) is maximal.

Proof

If (x) is prime, x is not a unit since it is a proper ideal. And if $x = yz$ we have that y or z is in (x), say $y = xt$. Then $x = yz = xtz$, and so $x(1 - tz) = 0$, and since R is an integral domain this means $tz = 1$ so z is a unit.

Now if x is irreducible, then suppose $(x) \subseteq I$ is a proper ideal containing (x). Since R is a PID, $I = (y)$ so that $x \in (y)$ meaning $x = yz$. (y) is a proper ideal so y is not a unit, thus z is, and then $y = xz^{-1}$ so that $y \in (x)$ and thus $(x) = (y)$. So (x) is maximal, as desired. ■

1.2.6 Example

The ring of integers $\mathbb{Z}$ is a PID. Its irreducible elements are precisely its prime elements, so that its prime ideals are (0) and (p) for $p \in \mathbb{Z}$ prime.

1.2.7 Example

If k is a field, then its ring of polynomials $k[x]$ is a PID. Indeed, if $0 \neq I \leq k[x]$ is an ideal, then let $f \in I$ have minimal degree. By polynomial division we see that $I = (f)$: for $g \in I$ we have that $g = fq + r$, but since $r = g - fq \in I$ has smaller degree than f, it must be zero.

Its prime ideals are (0) and (f) for an irreducible monic polynomial f.

1.2.8 Lemma

If $f\colon R \to S$ is a ring morphism, and $\mathfrak{q} \leq S$ is prime, then $f^{-1}\mathfrak{q} \leq R$ is prime.

Proof

A direct proof is easy, instead consider the composition:

$$\overline{f}\colon R \xrightarrow{f} S \xrightarrow{\pi} S/\mathfrak{q}.$$

By definition $f^{-1}\mathfrak{q} = \ker \overline{f}$, and so $\overline{f}$ induces an embedding $R/f^{-1}\mathfrak{q} \hookrightarrow S/\mathfrak{q}$. Subrings of integral domains are integral domains, so $R/f^{-1}\mathfrak{q}$ is an integral domain, thus $f^{-1}\mathfrak{q}$ is prime. ■

Subrings of fields are not fields, so the above proof does not show that the preimage of a maximal ideal is maximal. Indeed, this is not true: consider the unique map $f\colon \mathbb{Z} \to \mathbb{Q}$. Then $0 \leq \mathbb{Q}$ is maximal, but $f^{-1}0 = \ker f = 0$ is not maximal in $\mathbb{Z}$.

1.2.9 Theorem

For every ring R and every proper ideal $I < R$, there is a maximal ideal $\mathfrak{m}$ containing I.

Proof

Consider the set of all proper ideals containing I. This is nonm-empty, as it contains I. Furthermore, the union of every chain of proper ideals is a proper ideal; it being an ideal is easy to see, and it is proper since 1 is not in any element of the chain. Thus by Zorn, this set has a maximal element, which must be a maximal ideal. ∎

1.2.10 Corollary

Let $x \in R$, then x is a unit iff x is not in any maximal ideal. That is,

$$R^\times = \bigcap_{\mathfrak{m}} \mathfrak{m}^c,$$

where the intersection is over all maximal ideals of R.

Proof

An element is a unit iff it is not contained in any proper ideal. This is clear enough: if x is not a unit then (x) is proper, and proper ideals don't contain units. Every proper ideal can be extended to a maximal ideal, so the result follows. ∎

1.2.11 Definition

Let R be a ring. Then define its **nilradical** to be $\mathrm{nil}(R) = \sqrt{0}$, i.e. the ideal of all nilpotent elements.

Clearly R is reduced iff $\mathrm{nil}(R) = 0$.

1.2.12 Proposition

Let $I \leq R$ be an ideal, and let $\pi \colon R \to R/I$ be the projection. Then $\sqrt{I} = \pi^{-1}\mathrm{nil}(R/I)$.

Proof

We note that if $x^n \in I$ then $\pi(x)^n \in I$ so that $\pi(x) \in \mathrm{nil}(R/I)$ as desired. Conversely, if $\pi(x) \in \mathrm{nil}(R/I)$ then $\pi(x^n) = I$ so that $x^n \in I$ as desired. ∎

1.2.13 Proposition

For a ring R,

(1) $R/\mathrm{nil}(R)$ is reduced;

(2) The nilradical is the intersection of all prime ideals:

$$\mathrm{nil}(R) = \bigcap_{\mathfrak{p}} \mathfrak{p}.$$

Proof

We noted above that $\mathrm{nil}R = \sqrt{\mathrm{nil}R} = \pi^{-1}\mathrm{nil}(R/\mathrm{nil}(R))$. But we have also that $\mathrm{nil}R = \pi^{-1}0$, so $\pi^{-1}0 = \pi^{-1}\mathrm{nil}(R/\mathrm{nil}(R))$ and since π is surjective this means $\mathrm{nil}(R/\mathrm{nil}(R)) = 0$ so that $R/\mathrm{nil}(R)$ is reduced.

Clearly if $x \in \mathrm{nil}(R)$ then $x \in \mathfrak{p}$ for every prime $\mathfrak{p}$ since $x^n = 0 \in \mathfrak{p}$. Now if $x \notin \mathrm{nil}(R)$ then we claim there exists a prime ideal $\mathfrak{p}$ such that $x \notin \mathfrak{p}$. We prove this using Zorn. Let $T = \{1, x, x^2, \ldots\}$ and let S be the poset of all ideals I such that $I \cap T = \emptyset$; note that all ideals in S are proper. As explained before the union of a chain of ideals is an ideal, and clearly if all of the ideals in the union don't intersect T, neither does the union. Thus by Zorn, S has a maximal element $\mathfrak{p}$.

We claim that $\mathfrak{p}$ is prime. Indeed, suppose that $ab \in \mathfrak{p}$ but $a, b \notin \mathfrak{p}$. Then by maximality we have that $x^n \in (a, \mathfrak{p})$ and $x^m \in (b, \mathfrak{p})$ for some $n, m \geq 0$. But then $x^{n+m} \in (a, \mathfrak{p})(b, \mathfrak{p}) \subseteq \mathfrak{p}$, a contradiction.

Thus there is a prime ideal $\mathfrak{p}$ such that $\mathfrak{p} \cap T = \emptyset$, in particular $x \notin \mathfrak{p}$. So if $x \notin \mathrm{nil}(R)$ then $x \notin \bigcap_{\mathfrak{p}} \mathfrak{p}$, as desired. ∎

1.2.14 Corollary

Let I be an ideal of R, then

$$\sqrt{I} = \bigcap_{I \leq \mathfrak{p}} \mathfrak{p},$$

where the intersection is over all prime ideals containing I.

Proof

We showed above that $\sqrt{I} = \pi^{-1}\mathrm{nil}(R/I)$. By above $\mathrm{nil}(R/I)$ is the intersection over all prime ideals of R/I, which are all prime ideals of R containing I:

$$\sqrt{I} = \pi^{-1} \bigcap_{I \leq \mathfrak{p}} \overline{\mathfrak{p}} = \bigcap_{I \leq \mathfrak{p}} \pi^{-1}\overline{\mathfrak{p}} = \bigcap_{I \leq \mathfrak{p}} \mathfrak{p}.$$ ∎

1.2.15 Definition

Let R be a ring, then define its **Jacobson radical** to be

$$J(R) = \bigcap_{\mathfrak{m}} \mathfrak{m},$$

the intersection of all maximal ideals.

1.2.16 Lemma

Let $\mathfrak{m} \leq R$ be maximal. Then $x \notin \mathfrak{m}$ iff there is a $y \in R$ such that $1 - xy \in \mathfrak{m}$.

Proof

$x \notin \mathfrak{m}$ iff $\overline{x} \in R/\mathfrak{m}$ is nonzero, or equivalently is a unit. This is iff there is a $y \in R$ such that $\overline{xy} = \overline{1}$. This just means $1 - xy \in \mathfrak{m}$, as desired. ∎

1.2.17 Corollary

For a ring R, its Jacobson radical $J(R)$ is the set of all elements $x \in R$ such that $1 - yx \in R^\times$ for all $y \in R$.

Proof

Now if $x \notin J(R)$ then $x \notin \mathfrak{m}$ for some $\mathfrak{m}$, and then $1 - yx \in \mathfrak{m}$ for some y and thus is not a unit. And if $1 - yx$ is not a unit, then $1 - yx \in \mathfrak{m}$ for some maximal $\mathfrak{m}$ and thus $x \notin \mathfrak{m}$ so that $x \notin J(R)$. ■

1.2.18 Definition

A ring R is **local** if it has a unique maximal ideal $\mathfrak{m}$. The field $k = R/\mathfrak{m}$ is called its **residue field**. By abuse of notation we write $(R, \mathfrak{m})$ for a local field.

1.2.19 Proposition

(1) If $(R, \mathfrak{m})$ is a local ring then $R - \mathfrak{m} \subseteq R^\times$.

(2) Conversely if $I < R$ is a proper ideal with $R - I \subseteq R^\times$ then (R, I) is a local ring.

(3) If $\mathfrak{m} \leq R$ is maximal such that $1 + \mathfrak{m} \subseteq R^\times$ then $(R, \mathfrak{m})$ is local.

Proof

If $x \notin \mathfrak{m}$ then (x) is not contained in $\mathfrak{m}$, and since $\mathfrak{m}$ is the unique maximal ideal this means that $(x) = R$ so that x is a unit. Also we noted that $R^\times$ is the intersection of all complements of maximal ideals, so this follows from that as well.

Conversely, we have that
$$R - I \subseteq R^\times \bigcap_{\mathfrak{m}} \mathfrak{m}^c \Longrightarrow I \supseteq \bigcup_{\mathfrak{m}} \mathfrak{m},$$
which means that I contains all maximal ideals, which can only happen if it is the unique maximal ideal.

We want to show that $R - \mathfrak{m} \subseteq R^\times$. Indeed, if $x \notin \mathfrak{m}$ then there is a y such that $1 - yx \in \mathfrak{m}$ so that $yx \in 1 + \mathfrak{m}$ and is thus a unit. So x is a unit, as desired. ■

1.2.20 Example

For $p \in \mathbb{Z}$ prime, define
$$\mathbb{Z}_{(p)} = \left\{ \frac{a}{b} \;\middle|\; a, b \in \mathbb{Z}, b \notin (p) \right\} \subseteq \mathbb{Q}, \qquad I = \left\{ \frac{a}{b} \in \mathbb{Z}_{(p)} \;\middle|\; a \in (p) \right\}.$$

Then $(\mathbb{Z}_{(p)}, I)$ is local. Indeed, if $\frac{a}{b} \notin I$ then it is invertible (its inverse is $\frac{b}{a}$). We can generalize this, as we will in the next section.

1.3 Modules

1.3.1 Definition

Let A be a ring, then a **module** over A is an Abelian group M along with a ring morphism $\varphi\colon A \to \operatorname{end}(M)$. We define **multiplication** to be a function $\mu\colon A \times M \to M$, written by juxtaposition, by

$$\mu(a, m) = \varphi(a)m.$$

An A-module can also be defined via multiplication: require that multiplication distributes both ways, and $1 \in A$ is an identity:

$$(a+b)m = am + bm, \qquad a(m+n) = am + an, \qquad a(bm) = (ab)m, \qquad 1m = m.$$

1.3.2 Definition

For two A-modules M and N, an **A-module morphism** is a map $f\colon M \to N$ which is an Abelian group morphism and also multiplicative:

$$f(am) = af(m)$$

for all $a \in A, m \in M$.

Note:

I'm going to skip the rest of the standard definitions, for Chrissake.

1.3.3 Definition

If M is an A-module, we define $\operatorname{ann}_A(M) \le A$ the **annihilator** of M by

$$\operatorname{ann}_A(M) = \{a \in A \mid aM = 0\} = \ker \varphi,$$

where φ is the module map $\varphi\colon A \to \operatorname{end}(M)$. This is then an ideal. For $S \subseteq M$ a subset we define $\operatorname{ann}_A(S)$ similarly.

A module M is **faithful** if $\operatorname{ann}_A(M) = 0$.

$\operatorname{ann}_A(S)$ is equal to $\operatorname{ann}_A(\langle S \rangle)$, the annihilator of the submodule generated by S (all R-combinations of elements from S), and is thus also an ideal.

1.3.4 Definition

For a ring A, an **A-algebra** is a ring B equipped with a ring morphism $\varphi\colon A \to B$.

Note that B can be embedded into $\operatorname{end}(B)$ mapping $b \in B$ to multiplication by b. Thus an A-algebra is an A-module with multiplication given by $a \cdot b = \varphi(a)b$.

1.3.5 Definition

For two A-algebras B, C, an **A-algebra morphism** is a map $f\colon B \to C$ which is both a ring morphism and an A-module morphism.

If $\varphi\colon A \to B, \psi\colon A \to C$ are the algebra maps, then $f\colon B \to C$ is an algebra morphism iff it is a ring morphism and $\psi = f \circ \varphi$:

$$\psi(a) = a1_C = af(1_B) = f(a1_B) = f \circ \varphi(a).$$

1.3.6 Definition

For an A-module M and a submodule N, its quotient module M/N is the set of cosets $m+N$ with the usual M-module structure. The usual isomorphism results hold.

1.3.7 Proposition

Let $I \leq A$ be a module, then every A/I-module is an A-module with $\mathrm{ann}_A(M) \supseteq I$ and vice-versa. In particular an A-module M can be viewed as a faithful $A/\mathrm{ann}_A(M)$-module.

Proof

Consider the projection $\pi\colon A \to A/I$. Composing with $\varphi\colon A/I \to \mathrm{end}(M)$ gives an A-module structure to an A/I-module and $\ker(\varphi \circ \pi) \supseteq \ker \varphi = I$. Conversely if $\varphi\colon A \to \mathrm{end}(M)$ has kernel containing I we can quotient it by I. ■

1.3.8 Definition

For a collection of A-modules $\{M_\alpha\}_\alpha$ we define their **direct product** and **direct sum** by

$$\prod_\alpha M_\alpha = \{(m_\alpha) \mid m_\alpha \in M_\alpha\}, \qquad \bigoplus_\alpha M_\alpha = \left\{(m_\alpha) \in \prod_\alpha M_\alpha \;\middle|\; \text{All but finitely many of } m_\alpha \text{ are zero}\right\}.$$

$\prod_\alpha M_\alpha$'s module structure is given by pointwise operations, and $\bigoplus_\alpha M_\alpha$ is viewed as a submodule. These form the products and coproducts in the category of A-modules.

1.3.9 Definition

An A-module M is **free** if

$$M \cong A^{\oplus S} = \bigoplus_{\alpha \in S} A$$

for some set S. Note that $\{e_\alpha\}_{\alpha\in S}$ (the usual standard basis vectors) forms a basis of $A^{\oplus S}$ in the sense that every element of $A^{\oplus S}$ can be written as a unique combination of standard basis vectors. A collection of elements $\{m_\alpha\}_{\alpha\in S}$ is a **basis** of M if mapping $m_\alpha \mapsto e_\alpha$ extends to an isomorphism $M \cong A^{\oplus S}$.

A basis of M is equivalently a set of elements which generate M and are also linearly independent in the usual way.

A free module is called such because they are free in the usual sense with respect to the obvious forgetful functor $\mathsf{Mod}_A \to \mathsf{Set}$. That is, $\hom_A(A^{\oplus S}, B) \cong \hom_{\mathsf{Set}}(S, B)$.

1.3.10 Proposition

If $A^{\oplus S} \cong A^{\oplus S'}$ then S and S' have the same cardinality.

Proof

Let $\mathfrak{m} \leq A$ be maximal, then the isomorphism $A^{\oplus S} \cong A^{\oplus S'}$ induces an isomorphism $(A/\mathfrak{m})^{\oplus S} \cong (A/\mathfrak{m})^{\oplus S'}$.

$A/\mathfrak{m}$ is a field, and so this is an isomorphism of $A/\mathfrak{m}$-modules (vector spaces) of dimensions $|S|, |S'|$. Since the dimension is invariant, their cardinalities must be equal. ∎

1.3.11 Definition

If M is a free A-module, then its **rank** is the cardinality of S where $M \cong A^{\oplus S}$.

1.3.12 Definition

An A-module M is **finitely generated** (**fg**) if there is a finite set $S \subseteq M$ such that $M = \langle S \rangle$.

1.3.13 Proposition

An A-module is finitely generated iff it is a quotient of A^n for some n (up to isomorphism).

Proof

If $\varphi\colon A^n \to M$ is surjective, then $\varphi(e_i)$ generate M clearly and so it is finitely-generated. Thus if $M \cong A^n/N$ is a quotient of A^n then the projection $\pi\colon A^n \to A^n/N \cong M$ is surjective and thus M is finitely-generated. Conversely let m_i generate M, and define $\varphi\colon A^n \to M$ by $\varphi(e_i) = m_i$ which extends by linearity to all of A^n since it is free. This is surjective and then $A^n/\ker\varphi \cong M$ as desired. ∎

1.3.14 Proposition

Let M be a fg A-module, and $I \leq A$ an ideal such that $M = IM$. Then there is an $x \in I$ such that $(1-x)M = 0$ ($xm = m$ for all $m \in M$).

Proof

Let $m_1, \ldots, m_n$ generate M, we induct on n. For $n = 0$ we have $M = 0$ so this is clear. Consider $N = M/(m_n)$, then $IN = (IM + (m_n))/(m_n) = M/(m_n) = N$ and N is generated by $n-1$ elements, so by induction there is an $x \in I$ such that $(1-x)N = 0$. Meaning $(1-x)M \subseteq (m_n)$, and since $m_n \in M = IM$ we have

$$(1-x)m_n \in (1-x)IM = I(1-x)M \subseteq I(m_n),$$

so $(1-x)m_n = ym_n$ for some $y \in I$ and thus $(1-x-y)m_n = 0$. Now $(1-x-y)(1-x) \in 1+I$ and satisfies

$$(1-x-y)(1-x)M \subseteq (1-x-y)(m_n) = 0$$

as desired. ∎

1.3.15 Corollary (Nakayama's lemma)

Let M be a fg A-module, such that $J(A)M = M$, then $M = 0$.

Proof

By above there is an $x \in J(A)$ such that $(1-x)M = 0$. Since $x \in J(A)$, $1-x \in A^\times$ so that $(1-x)M = M = 0$ as desired. ∎

1.3.16 Corollary

Let M be a fg A-module and $N \subseteq M$ a submodule such that $M = IM + N$ for some $I \subseteq J(A)$, then $M = N$.

Proof

Consider M/N, then $I(M/N) = (IM+N)/N = M/N$, so by above $M/N = 0$ meaning $M = N$. ∎

1.3.17 Corollary

Let $(A, \mathfrak{m})$ be local, M a fg A-module. Then $x_1, \ldots, x_n \in M$ generate M iff $\overline{x}_1, \ldots, \overline{x}_n \in M/\mathfrak{m}M$ generate $M/\mathfrak{m}M$ over $A/\mathfrak{m}$.

Proof

If x_i generate M then $\overline{x}_i$ clearly generate M. Conversely, let $N = (x_1, \ldots, x_n)$ be the submodule generated by M, and consider the composition $f\colon N \to M \to M/\mathfrak{m}M$. f's image is the span of $(\overline{x}_i)_i$, so all of $M/\mathfrak{m}$. Thus f is surjective, so $M = N + \mathfrak{m}M$, since for every $m \in M$ there is an $n \in N$ such that $f(n) - m \in \mathfrak{m}M$ by surjectivity, and by above since $J(A) = \mathfrak{m}$ this means $M = N$ as desired. ∎

1.3.18 Definition

Recall the definition of an exact sequence:

$$M \xrightarrow{f} N \xrightarrow{g} P$$

is **exact** at N if $\operatorname{im} f = \ker g$.

2 Localization

2.1 Localization of rings

Consider the following problem: for a ring A and a subset $S \subseteq A$, we want a ring $S^{-1}A$ and a morphism $\iota\colon A \to S^{-1}A$ such that any morphism $f\colon A \to B$ such that $f(S) \subseteq B^\times$, f factors uniquely through ι. That is, there is a unique $\hat{f}\colon S^{-1}A \to B$ such that $f = \hat{f} \circ \iota$:

$$\begin{tikzcd} A \arrow[r, "\iota"] \arrow[dr, "f"'] & S^{-1}A \arrow[d, dashed, "\exists!"] \\ & B \end{tikzcd}$$

This is a universal property, and as such any solution is unique up to unique isomorphism. In this section we will investigate the construction of such a solution.

2.1.1 Definition

A subset $S \subseteq A$ is **multiplicatively closed** if $1 \in S$ and for $a, b \in S$ we have $ab \in S$.

It is sufficient to consider only the case that S is multiplicatively closed. Otherwise, let $\hat{S}$ be S's multiplicative closure: the set of all finite products of elements from S. Then $\hat{S}^{-1}R$ is a solution to the problem for S. Indeed, if $f(S) \subseteq B^\times$, then $f(\hat{S}) \subseteq B^\times$ as well, as the product of units. Then there is a unique factoring through $\hat{S}$'s ι.

2.1.2 Definition

For $S \subseteq A$ multiplicatively closed, define an equivalence on $A \times S$ by $(a, s) \sim (b, t)$ iff there is a $u \in S$ such that $u(at - bs) = 0$, in which case we write $(a, s) \sim_u (b, t)$.

2.1.3 Lemma

This is indeed an equivalence relation.

Proof

The subrelations $\sim_u$ are all clearly reflexive and symmetric, thus their union $\sim$ is as well. Now if $(a, s) \sim_\alpha (b, t) \sim_\beta (c, v)$ then $\alpha at = \alpha sb$ and $\beta bv = \beta ct$. So then $\alpha t\beta av = \alpha sb\beta v = \alpha\beta tcs$ so that $(a, s) \sim_{\alpha t\beta} (c, v)$. ■

2.1.4 Example

A more naive equivalence would be $\sim_1$, i.e. $(a, s) \sim (b, t)$ iff $at - bs = 0$. This corresponds to $\sim$ when A is an integral domain, but is not generally an equivalence.

2.1.5 Definition

For $S \subseteq A$ multiplicatively closed, define $S^{-1}A$ to be the quotient of $A \times S$ by $\sim$. Write the equivalence class of (a, s) by a/s. We give this a ring structure by

$$0 = 0/1, \qquad 1 = 1/1, \qquad a/s + b/t = (at + bs)/st, \qquad a/s \cdot b/t = (ab)/(st).$$

Define $\iota\colon A \to S^{-1}A$ by $\iota(a) = a/1$.

2.1.6 Proposition

$S^{-1}A$ is indeed a ring, and ι is a ring morphism.

Proof

We need to show that addition and multiplication are well-defined. Indeed, if $(a, s) \sim_u (a', s')$ and $(b, t) \sim_v (b', t')$ then $(at + bs, st) \sim_{uv} (a't' + b's', s't')$ which can be verified. And similarly $(ab, st) \sim_{uv} (a'b', s't')$. From well-definedness it is easy to see that addition and multiplication define a ring structure and ι is a ring morphism. ■

A few properties are immediate:

(1) For $s \in S$, $\iota(s) = 1/s$ has an inverse $s/1$, so that $\iota(S) \subseteq (S^{-1}R)^\times$;

(2) We have that $a/s = 0/1$ iff there is a $t \in S$ such that $at = 0$. In particular, $\iota(a) = 0$ iff there is a $t \in S$ such that $ta = 0$.

(3) Every element of $S^{-1}A$ is of the form

$$a/s = a/1 \cdot 1/s = \iota(a) \cdot \iota(s)^{-1}.$$

2.1.7 Theorem

This construction solves the above universal property.

Proof

Let $f\colon A \to B$ be such that $f(S) \subseteq B^\times$: we need to show that there is a unique map $g\colon S^{-1}A \to B$ such that $f = g \circ \iota$. Indeed, we must define

$$g(a/s) = g(\iota(a)\iota(s)^{-1}) = (g \circ \iota(a)) \cdot (g \circ \iota(s))^{-1} = f(a)f(s)^{-1}.$$

This shows that g is unique, if it exists. We need to show that this map is well-defined and a ring morphism. If $a/s = b/t$ then we must have that $f(a)f(s)^{-1} = f(b)f(t)^{-1}$ so that g is well-defined. Indeed, if $u(at-bv) = 0$ for $u \in S$, we have

$$f(u) \cdot (f(at) - f(bv)) = 0$$

and since $f(S) \subseteq B^\times$ we have that $f(u)$ is a unit so that $f(at) - f(bv)$, meaning this is well-defined. And this is a ring morphism, as can be checked easily. ■

2.1.8 Corollary

Let $f\colon A \to B$ be a ring morphism such that

(1) $f(S) \subseteq B^\times$,

(2) if $f(a) = 0$ then $ta = 0$ for some $t \in S$,

(3) every element of B is of the form $f(a)f(s)^{-1}$ for $a \in A$ and $s \in S$.

Then there is a unique isomorphism $g\colon S^{-1}A \to B$ such that $g \circ \iota = f$.

Proof

f solves the universal problem above for S as well. If $h\colon A \to C$ is a map where $h(S) \subseteq C^\times$, then we need to define $\hat{h}\colon B \to C$ by

$$\hat{h}(f(a)f(s)^{-1}) = h(a)h(s)^{-1},$$

which is well-defined and a ring morphism for similar reasons as above. Since universal properties are unique up to unique isomorphism, the result follows.

Alternatively we can prove this directly: let $g\colon S^{-1}A \to B$ be the unique map where $g \circ \iota = f$ by the universal property. By definition $g(a/s) = f(a)f(s^{-1})$. By the third point g is surjective, and it is injective since if $g(a/s) = 0$ we have $f(a) = 0$ so that $ta = 0$, which means that $a/s = 0$. ■

2.1.9 Example

If A is an integral domain, then $S = A - 0$ is multiplicatively closed, and $S^{-1}A$ is the field of fractions of A.

2.1.10 Proposition

For $A \neq 0$, then $S^{-1}A = 0$ iff $0 \in S$.

Proof

We have that $S^{-1}A = 0$ iff $1/1 = 0/1$ iff there is a $s \in S$ such that $s \cdot 1 = 0$. ■

2.1.11 Definition

Let $\mathfrak{p} \subseteq A$ be prime. Then $S = A - \mathfrak{p}$ is multiplicatively closed, and we denote its localization by $A_{\mathfrak{p}} = S^{-1}A$.
For $a \in A$, $S = \{1, a, a^2, \dots\}$ is multiplicatively closed, and we denote its localization by $A_a = S^{-1}A$.

For $a \in A$ not nilpotent (so $A_a \neq 0$) define a map

$$\varphi\colon A[x] \to A_a, \qquad x \mapsto 1/a.$$

This map is clearly surjective, and $p(x) = \sum_k^n b_k x^k \in A[x]$ is mapped to zero iff

$$\sum_{k=0}^{n} b_k \frac{1}{a^k} = \frac{\sum_k b_k a^{n-k}}{a^n} = 0.$$

This is iff there is a t such that

$$a^t \sum_k b_k a^{n-k} = 0.$$

Defining c_k recursively by $c_0 = -b_0$ and $c_k = ac_{k-1} - b_k$ gives a polynomial such that $p(x) = (ax-1)q(x)$, so $p \in (ax-1)$ (the equality above ensures the recursion terminates). Thus $\ker\varphi = (ax-1)$ and thus

$$A_f \cong \frac{A[x]}{(ax-1)}.$$

Note:

Note that we have two distinct localizations here, which have similar names in some cases. If A is a PID and $p \in A$ is irreducible, then $A_{(p)}$ is localization by $A - (p)$, which A_p is localization by $\{p^n\}_{n<\omega}$. Elements of the former are of the form a/s where p does not divide s, while elements of the former are of the form a/s where $s = p^n$. Quite different!

2.1.12 Definition

For a ring morphism $f\colon A \to B$ and an ideal $I \subseteq A$, define the ideal $IB = (f(I))$ of B. Explicitly, IB is the set of all finite sums $\sum_i f(a_i)b_i$ for $a_i \in I$.

In general we have $f^{-1}(f(I)) \supseteq I$ and $f(f^{-1}(J)) \subseteq J$. These imply $f^{-1}(IB) \supseteq I$ and $f^{-1}(J)B \subseteq J$.

2.1.13 Definition

For two ideals I, J, define $(I : J) = \{x \in A \mid xJ \subseteq I\}$. This is an ideal containing I. For $a \in A$, define $(I : a) = (I : (a)) = \{x \in A \mid xa \subseteq I\}$.

For $S \subseteq R$ multiplicatively closed, define the ideal $S^{-1}I = IS^{-1}A \subseteq S^{-1}A$. This is the ideal generated by $\iota(I)$.

So we have two maps: $S^{-1}\bullet$ mapping ideals of A to ideals of $S^{-1}A$, and ι^{-1} mapping ideals of $S^{-1}A$ to A:

$$\{\text{Ideals of } A\} \underset{\iota^{-1}}{\overset{S^{-1}}{\rightleftarrows}} \{\text{Ideals of } S^{-1}A\}.$$

We wish to relate the two.

2.1.14 Proposition

(1) For an ideal $I \subseteq A$, $S^{-1}I = \{a/s \mid a \in I, s \in S\}$.

(2) For an ideal $J \subseteq S^{-1}A$, we have $S^{-1}\iota^{-1}J = J$.

(3) For an ideal $I \subseteq A$, we have $\iota^{-1}S^{-1}I = \bigcup_{s \in S}(I : s)$.

(4) We have $S^{-1}I = S^{-1}A$ iff $I \cap S \neq \emptyset$.

(5) For $\mathfrak{p} \subseteq A$ prime, if $I \cap S \neq \emptyset$ then $S^{-1}\mathfrak{p} \subseteq S^{-1}A$ is prime (otherwise it is all of $S^{-1}A$).

(6) The correspondence $\mathfrak{q} \mapsto \iota^{-1}\mathfrak{q}$ is a bijection between prime ideals of $S^{-1}A$ and prime ideals of A not intersecting S. Its inverse is S^{-1}. That is, we have a one-to-one correspondence:

$$\{\text{Prime ideals of } A \text{ not intersecting } S\} \underset{\iota^{-1}}{\overset{S^{-1}}{\rightleftarrows}} \{\text{Prime ideals of } S^{-1}A\}.$$

Proof

(1) $\{a/s \mid a \in I, s \in S\}$ is an ideal and contains $\iota(I)$. It is clearly closed under multiplication by $S^{-1}A$ and for $a/s + b/t = (at + bs)/(st)$, notice that $at + bs \in I$. By definition $S^{-1}I$ is the smallest ideal containing $\iota(I)$, and it consists of finite sums $\sum_i \iota(a_i)x_i/s_i$. This clearly contains a/s, and thus they must be equal.

(2) As explained before, in general we have $S^{-1}\iota^{-1}J = \iota^{-1}(J)S^{-1}A \subseteq J$, we must prove the opposite inclusion. As above, we have $S^{-1}\iota^{-1}J = \{a/s \mid \iota(a) \in J, s \in S\}$. For $x/s \in J$, we have that $\iota(x) = x/1 \in J$ and so $x/s \in S^{-1}\iota^{-1}J$ as desired.

(3) If $a \in \iota^{-1}S^{-1}I$ then $\iota(a) \in S^{-1}I$ so by the first point $a/1 = b/s$ for $b \in I$. Then $asu = bu$ for some $u \in S$ and since $bu \in J$ this means that $a \in (I : su)$. Conversely if $a \in (I : s)$ for $s \in S$ then $as \in I$ and so $\iota(a) = a/1 = (as)/s \in S^{-1}I$.

(4) If $S^{-1}I = S^{-1}A$ then $1/s = a/t$ for $a \in I$, so $as = t$ meaning $as \in I \cap S$. Conversely if $a \in I \cap S$ then $1/1 = a/a \in S^{-1}I$.

(5) If $S \cap \mathfrak{p} = \emptyset$ then we need to show that $S^{-1}\mathfrak{p}$ is prime. If $a_1/s_1, a_2/s_2 \notin S^{-1}\mathfrak{p}$ we want to show $(a_1a_2)/(s_1s_2) \notin S^{-1}\mathfrak{p}$. By assumption $a_1, a_2 \notin \mathfrak{p}$.

Now we claim that if $a \notin \mathfrak{p}$, $a/s \notin S^{-1}\mathfrak{p}$, which would complete the proof. Assume the contrary: $a/s = b/t$ for $b \in \mathfrak{p}$ then there is a $u \in S$ such that $atu = bsu$. Then $atu \in \mathfrak{p}$, but $a \notin \mathfrak{p}$ and $t, u \in S$ and thus also are not in $\mathfrak{p}$, a contradiction to $\mathfrak{p}$'s primality.

(6) Preimages of prime ideals are prime, so ι^{-1} maps prime ideals to prime ideals. By the second point $S^{-1}\iota^{-1}\mathfrak{q} = \mathfrak{q}$ for all prime ideals $\mathfrak{q} \subseteq S^{-1}A$. It remains to show that $\iota^{-1}S^{-1}\mathfrak{p} = \mathfrak{p}$ for all prime $\mathfrak{p} \subseteq A$ which don't intersect S. We know that $\mathfrak{p} \subseteq \iota^{-1}S^{-1}\mathfrak{p}$ as is true in general, so we want to show that $\iota^{-1}S^{-1}\mathfrak{p} = \bigcup_{s \in S}(\mathfrak{p} : s)$ is contained in $\mathfrak{p}$. Indeed, if $a \in (\mathfrak{p} : s)$ then $as \in \mathfrak{p}$ and since $s \notin \mathfrak{p}$ we have that $a \in \mathfrak{p}$ as desired. ■

2.1.15 Corollary

Let $\mathfrak{p} \subseteq A$ be prime, and consider its localization $A_\mathfrak{p}$. Then $A_\mathfrak{p}$ is a local ring with maximal ideal $\mathfrak{p}A_\mathfrak{p}$. Moreover, the correspondence $\mathfrak{q} \to \mathfrak{q}A_\mathfrak{p}$ is a bijection between prime ideals of $A_\mathfrak{p}$ and prime ideals $\mathfrak{q} \subseteq \mathfrak{p}$.

Proof

Let $S = A - \mathfrak{p}$, so that $A_\mathfrak{p} = S^{-1}A$, and $\mathfrak{p}A_\mathfrak{p} = S^{-1}\mathfrak{p}$. The last point is clear: $\mathfrak{q} \cap S = \emptyset$ iff $\mathfrak{q} \subseteq \mathfrak{p}$. $A_\mathfrak{p}$ being local is also clear: $S^{-1}\bullet$ is order-preserving, so $S^{-1}\mathfrak{q} \subseteq S^{-1}\mathfrak{p}$ for all prime ideals $\mathfrak{q}$. Thus $S^{-1}\mathfrak{p}$ is maximal. ■

2.1.16 Corollary

If $a \in A$ is not nilpotent, then there is a prime ideal $\mathfrak{p} \subseteq A$ such that $a \notin \mathfrak{p}$.

Proof

As before, consider the multiplicative set $S = \{a^n\}_{n<\omega}$, and $A_a = S^{-1}A$. Since a is not nilpotent, $0 \notin S$, so that $A_a \neq 0$. So take a prime ideal $\mathfrak{p} \subseteq A_a$, and then $\iota^{-1}\mathfrak{p}$ is a prime ideal of A not intersecting S, thus not containing a. ■

Note:
This completes the proof before that

$$\mathrm{nil}(A) = \bigcap_{\mathfrak{p}} \mathfrak{p}$$

but without appealing to Zorn, and thus not relying on the Axiom of Choice.

2.2 Localization of modules

2.2.1 Definition

Let $S \subseteq A$ be multiplicatively closed and M an A-module. Define an equivalence on $M \times S$ by $(m, s) \sim (m', s')$ iff there is a $t \in S$ such that $t(s'm - sm') = 0$. This is an equivalence relation and we denote the equivalence class of (m, s) by m/s. The quotient $S^{-1}M$ forms an A-module by

$$m/s + m'/s' = (s'm + sm')/(ss'), \qquad a \cdot (m/s) = (am)/s.$$

Note that localization at S forms an additive functor $S^{-1}\colon \mathsf{Mod}_A \to \mathsf{Mod}_{S^{-1}A}$. Indeed, $S^{-1}M$ is an $S^{-1}A$ module by

$$a/s \cdot m/s' = (am)/(ss').$$

And if $f\colon M \to N$ is an A-module morphism, then define

$$S^{-1}f\colon S^{-1}M \to S^{-1}N, \qquad S^{-1}f(m/s) = f(m)/s.$$

It is not hard to see that this is functorial and additive.
There is the canonical morphism $\iota\colon A \to S^{-1}A$ and as such it lifts to a "forgetful" functor $\mathsf{Mod}_{S^{-1}A} \to \mathsf{Mod}_A$, where for $M \in \mathsf{Mod}_{S^{-1}A}$ we view it as an A-module by $a \cdot M = \iota(a) \cdot M$.

2.2.2 Proposition

The functor $S^{-1}\colon \mathsf{Mod}_A \to \mathsf{Mod}_{S^{-1}A}$ is left adjoint to the forgetful functor $\mathsf{Mod}_{S^{-1}A} \to \mathsf{Mod}_A$.

Proof

We need to show that there is a canonical bijection

$$\hom_{S^{-1}A}(S^{-1}M, N) \cong \hom_A(M, N),$$

for all A-modules M and $S^{-1}A$-modules N. For $f\colon M \to N$ map it to $\hat{f}\colon S^{-1}M \to N$ by $\hat{f}(m/s) = 1/s \cdot f(m)$. And for $g\colon S^{-1}M \to N$ map it to $\tilde{g}\colon M \to N$ by $\tilde{g}(m) = g(m/1)$. ■

2.2.3 Definition

An additive functor $F\colon \mathsf{Mod}_A \to \mathsf{Mod}_B$ is

(1) **left exact** if it preserves left-exact sequences: if

$$0 \to M \xrightarrow{f} N \xrightarrow{g} P$$

is exact, then so is

$$0 \to FM \xrightarrow{Ff} FN \xrightarrow{Fg} FP.$$

(2) **right exact** if it preserves right-exact sequences: if

$$M \xrightarrow{f} N \xrightarrow{g} P \to 0$$

is exact, then so is

$$FM \xrightarrow{Ff} FN \xrightarrow{Fg} FP \to 0.$$

(3) **exact** if it is both left and right exact.

2.2.4 Proposition

An additive functor is

(1) left exact iff it preserves kernels iff it preserves finite limits iff it preserves the left side of a short exact sequence.

(2) right exact iff it preserves cokernels iff it preserves finite colimits iff it preserves the right side of a short exact sequence.

(3) exact iff it preserves finite limits and colimits iff it preserves short exact sequences iff it preserves all exact sequences.

Proof

Left-exact sequences are kernel diagrams, and right-exact sequences are cokernel diagrams. So by definition, left exact functors are those which preserve kernels, and right exact functors are those which preserve cokernels. Since additive functors preserve finite biproducts, left/right exact functors preserve limits/colimits.

Clearly left-exact functors preserve the left side of a short exact sequence (similarly for right-exact functors). To show the converse, if left-exact sequences can be quotiented to give an appropriate short exact sequence, similar for right-exact sequences. ■

2.2.5 Corollary

Left adjoints are right exact, and right adjoints are left exact.

Proof

Left adjoints preserve all colimits, and right adjoints all limits. ■

2.2.6 Corollary

S^{-1} is an exact functor.

Proof

By the above corollary, it is right-exact. To show that it is exact, it is sufficient to show that it preserves injections. If $f\colon M \to N$ is injective then $m/s \in \ker S^{-1}f$ iff $f(m)/s = 0/1$, so that $tf(m) = 0$ for some $t \in S$. But then $tf(m) = f(tm)$ so $tm \in \ker \varphi$, so $tm = 0$, which means $m/s = 0/1$ as desired. ■

In general if $U\colon \mathsf{Mod}_A \to \mathsf{Mod}_B$ is a restriction functor corresponding to a morphism $\iota\colon B \to A$, then it has a left adjoint via the tensor product. In particular, viewing A as a (A, B)-module, then $A \otimes_B M$ has an A-module structure (given by $a(a' \otimes m) = (aa') \otimes m$). U's left adjoint is then $A \otimes_B M$.

Since left adjoints are unique up to natural isomorphism,

$$S^{-1}M \cong S^{-1}A \otimes_A M.$$

Thus $S^{-1}A \otimes_A \bullet$ is an exact functor, which means that $S^{-1}A$ is a **flat** A-module. This can also be shown directly: $S^{-1}A$ is the filtered colimit of free A-modules, and is thus flat.

2.2.7 Corollary

S^{-1} preserves arbitrary coproducts:

$$S^{-1} \bigoplus_\alpha M_\alpha \cong \bigoplus_\alpha S^{-1} M_\alpha$$

naturally.

Proof

Left adjoints preserve arbitrary colimits. ■

Note:
While S^{-1} is exact, it is not **continuous**: it does not preserve arbitrary limits (equivalently it does not preserve arbitrary products). That is, we do *not* have that

$$S^{-1}\prod_{\alpha} M_\alpha \cong \prod_{\alpha} S^{-1}M_\alpha.$$

This is of course true when the index set is finite, since then it is just a biproduct.

2.2.8 Corollary

(1) If $N \subseteq M$ is a submodule, $S^{-1}N$ can be naturally embedded in $S^{-1}M$.

(2) There is a natural isomorphism $S^{-1}(M/N) \cong (S^{-1}M)/(S^{-1}N)$.

(3) For $f\colon M \to N$ of A-modules, we have natural isomorphisms

$$\ker S^{-1}f \cong S^{-1}\ker f, \qquad \mathrm{im} S^{-1}f \cong S^{-1}\mathrm{im} f, \qquad S^{-1}\mathrm{coker} f \cong \mathrm{coker} S^{-1}f.$$

(4) For every two submodules $M_1, M_2 \subseteq M$ under the natural embeddings,

$$S^{-1}(M_1 \cap M_2) = S^{-1}M_1 \cap S^{-1}M_2 \subseteq S^{-1}M.$$

Proof

The first point is due to S^{-1} being exact and thus preserving injections. The second point is because S^{-1} preserves the short exact sequence $0 \to N \to M \to M/N \to 0$. The third point is clear as well, with only im not being immediately obvious. $\mathrm{im} f$ can be characterized as $\mathrm{im} f = \ker \mathrm{coker} f$, so it follows.

For the last point, note that $M_1 \cap M_2$ is the kernel of $M_1 \oplus M_2 \to M$ by $(x, y) \mapsto x - y$ (via the diagonal embedding), and so this follows from S^{-1}'s exactness. ■

2.2.9 Definition
For a prime ideal $\mathfrak{p} \leq A$ and an A-module M, denote by $M_\mathfrak{p}$ the localization of M by $S = A - \mathfrak{p}$.

2.2.10 Lemma
For an A-module M, consider the natural map

$$\varphi\colon M \to \prod_{\mathfrak{m}} M_\mathfrak{m}, \qquad m \mapsto (m/1)_\mathfrak{m},$$

where the product is over all maximal ideals of A. Then φ is injective.

Proof

Assume there is an $0 \neq x \in M$ such that $\varphi(x) = 0$. Then define

$$I = \operatorname{ann}_A(x) = \{a \in A \mid ax = 0\}.$$

I is a proper ideal of A, and thus is contained in a maximal ideal $\mathfrak{m}$. Then for every $s \in A - \mathfrak{m}$ we have that $s \notin I$ and so $sx \neq 0$, and thus $m/1$ is not zero in $M_{\mathfrak{m}}$, a contradiction. ■

2.2.11 Corollary

For an A-module M, the following are equivalent:

(1) $M = 0$,

(2) $S^{-1}M = 0$ for every multiplicative set $S \subseteq A$,

(3) $M_{\mathfrak{p}} = 0$ for every prime ideal $\mathfrak{p} \leq A$,

(4) $M_{\mathfrak{m}} = 0$ for every maximal ideal $\mathfrak{m} \leq A$.

Proof

Clearly each point implies the next. By above M embeds into $\prod_{\mathfrak{m}} M_{\mathfrak{m}}$, so if $M_{\mathfrak{m}} = 0$ for all $\mathfrak{m}$ then $M = 0$; the last point implies the first. ■

2.2.12 Corollary

For a morphism of A-modules $f \colon M \to N$, the following are equivalent:

(1) f is injective/surjective/an isomorphism,

(2) $S^{-1}f$ is injective/surjective/an isomorphism for every multiplicative $S \subseteq A$,

(3) $f_{\mathfrak{p}}$ is injective/surjective/an isomorphism for every prime $\mathfrak{p} \leq A$,

(4) $f_{\mathfrak{m}}$ is injective/surjective/an isomorphism for every maximal $\mathfrak{m} \leq A$.

Proof

By S^{-1}'s exactness, the first point implies the second, and clearly all the other points imply the next. To show the last implies the first, let $K = \ker f$ so that $0 \to K \to M \xrightarrow{f} N$ is exact. Then $0 \to K_{\mathfrak{m}} \to M_{\mathfrak{m}} \xrightarrow{f_{\mathfrak{m}}} N_{\mathfrak{m}}$ is exact, and so $K_{\mathfrak{m}} = \ker f_{\mathfrak{m}} = 0$ for all $\mathfrak{m}$. By above this means $K = 0$ so that f is injective. Similar for surjectivity. ■

3 Properties of Modules

3.1 Finitely generated, integral, and finite algebras

3.1.1 Definition

For an A-algebra B, and $S \subseteq B$ denote by $A[S]$ the smallest A-subalgebra containing S in B.

Let $\varphi\colon A \to B$ be the algebra map, then $A[S]$ is the subring of B generated by $\varphi(A) \cup S$. Alternatively, for every $s \in S$ define an indeterminant x_s, and let $X = \{x_s\}_{s \in S}$ be the set of all these indeterminants. Consider the polynomial algebra $A[X]$, and define $f\colon A[X] \to B$ by $f(x_s) = s$. Then $A[S]$ is just the image of f. Explicitly, for a function $\alpha\colon S \to \mathbb{Z}_{\geq 0}$ with finite support, define $S^\alpha = \prod_{\alpha(s) \neq 0} s^{\alpha(s)}$. Then

$$A[S] = \left\{ \text{Finite sums } \sum_\alpha \varphi(a_\alpha) S^\alpha \;\middle|\; a_\alpha \in A \right\}.$$

Note that we can consider B as an A-module. Then for $S \subseteq B$, we can consider the *submodule* generated by S: $\langle S \rangle$. Since $A[S]$ is itself a module, $\langle S \rangle \subseteq A[S]$ automatically, and we do not have equality in general.

3.1.2 Definition

An A-algebra B is **finitely generated** if there is a finite $S \subseteq B$ such that $B = A[S]$. B is a **finite algebra** if it is finitely generated as an A-module.

A finite A-algebra is also finitely generated as an A-algebra, but not vice versa in general.

3.1.3 Example

Suppose $S = \{s\}$ is a singleton, then $A[s]$ is the set of polynomials $\sum_k \varphi(a_k) s^k$.

3.1.4 Example

Clearly the polynomial algebra $A[x_1, \ldots, x_n]$ is a finitely generated A-algebra. But it is not a finite A-algebra: we can reduce this to the case of fields, in which x_i^k are all linearly independent and thus there is no finite basis.

3.1.5 Lemma

An A-algebra B is fg iff it is the quotient of some finite polynomial algebra $A[x_1, \ldots, x_n]$ (up to isomorphism of A-algebras).

Proof

This is equivalent to saying B is fg iff there is a surjection $A[x_1, \ldots, x_n] \to B$. This is clearly true: if B is fg then for $s_1, \ldots, s_n$ generating B map $x_i \mapsto s_i$ and extend by freeness. Conversely, quotients of fg algebras are fg. ∎

3.1.6 Definition

Let B be an A-algebra. An element $b \in B$ is **integral** over A if b is the root of a monic polynomial over A. That is, there exist $a_1, \ldots, a_n \in A$ such that $b^n + a_1 b^{n_1} + \cdots + a_n \cdot 1 = 0$.

$b \in B$ is **algebraic** over A if b is the root of any polynomial over A.

Clearly integral elements are algebraic. For fields the converse holds as well, as we can divide a polynomial by its leading coefficient.

3.1.7 Example

The set of integral elements in the $\mathbb{Z}$-algebra $\mathbb{Q}$ is $\mathbb{Z}$. Clearly an integer is integral, as $n \in \mathbb{Z}$ is the root of the monic $x - n$. Conversely if $b = p/q \in \mathbb{Q}$ is integral with p, q coprime then

$$b^n + a_1 b^{n-1} + \cdots + a_n 1 = 0, \qquad a_i \in \mathbb{Z}.$$

Then we have that

$$p^n + a_1 p^{n-1} q + \cdots + a_n q^n = 0$$

and so q must divide p^n, but since p, q are coprime this can only happen if $q = 1$ and $b \in \mathbb{Z}$.

3.1.8 Lemma

If M is a fg A-module, then for every $\varphi \in \mathrm{end}_A(M)$ there exist $a_1, \ldots, a_n \in A$ such that

$$\varphi^n + a_1 \varphi^{n-1} + \cdots + a_n = 0$$

in $\mathrm{end}_A(M)$.

Proof

The proof is by a generalization of Cayley-Hamilton. Let $n \geq 1$ and consider $R = \mathbb{Z}[X_{ij}]$, the polynomial ring over n^2 indeterminants X_{ij}, $1 \leq i, j \leq n$. Let $X = (X_{ij}) \in M_n(R)$, and define its **characteristic polynomial**: $\Phi_X(T) \in R[T]$ given by

$$\Phi_X(T) = \det(T\mathrm{id} - X).$$

By Cayley-Hamilton over R's field of fractions, $\Phi_X(X) = 0$.

Now if A is any ring and $Y \in M_n(A)$, there is a unique ring morphism $\varphi\colon R \to A$ which maps X to Y (coordinate-wise). Indeed, such a map must map X_{ij} to Y_{ij} which uniquely determines φ. Then φ lifts to the ring morphism $R[T] \to A[T]$ (map $\sum_n a_n T^n$ to $\sum_n \varphi(a_n) T^n$), which maps $\Phi_X(T)$ to $\Phi_Y(T)$ (which is also given by the determinant $\det(T\mathrm{id} - Y)$ since the determinant is functorial: $\det(\varphi(B)) = \varphi(\det(B))$). Thus $\Phi_X(X) = 0$ implies that $\Phi_Y(Y) = \varphi(\Phi_X(\varphi(X))) = \varphi(\Phi_X(X)) = 0$, as desired. So Cayley-Hamilton applies to any commutative ring.

Now for our lemma, let $m_1, \ldots, m_n \in M$ be generators. Then for all j we have

$$\varphi(m_j) = \sum_i a_{ij} m_i, \qquad a_{ij} \in A.$$

We associate to φ the representative matrix $[\varphi] = (a_{ij}) \in M_n(A)$ (which is not unique, unlike over fields). This has the usual properties that $[\varphi^n] = [\varphi]^n$ and $[\varphi + \psi] = [\varphi] + [\psi]$ (meaning that the composition or sum of representatives is a representative matrix). Now Cayley-Hamilton gives us that

$$\sum_n b_n [\varphi]^n = 0$$

for some $b_n \in A$, and thus

$$\left[\sum_n b_n \varphi^n\right] = 0,$$

and since only the zero endomorphism has the zero representative, this means $\sum_n b_n \varphi^n = 0$ as desired. ■

3.1.9 Proposition

For an A-algebra B and $b \in B$, the following are equivalent:

(1) b is integral over A,

(2) $A[b] \subseteq B$ is a finite algebra (fg module) over A,

(3) $A[b]$ is contained in a finite algebra over A,

(4) There exists a faithful $A[b]$-module M such that M is finitely generated as an A-module.

Proof

Clearly (2) implies (3) implies (4): take $M = C$, and since $A[b] \subseteq M$ it is faithful (the algebra map is inclusion, which is injective). (1) implies (2) since if $b^n + a_1 b^{n-1} + \cdots + a_n 1 = 0$ then b^n is in $\langle 1, \ldots, b^{n-1} \rangle \subseteq B$ and then by induction so is every higher power of b, so that $A[b] = \langle 1, \ldots, b^{n-1} \rangle$ is finite.
(4) implies (1): we have a ring morphism $\varphi\colon A[b] \to \mathrm{end}_A(M)$, so that every $f \in A[b]$ gives us an endomorphism $\varphi(f) \in \mathrm{end}_A(M)$. In particular by the above lemma there are $a_1, \ldots, a_n \in A$ such that

$$\rho(b)^n + a_1 \rho(b)^{n-1} + \cdots + a_n = 0 \in \mathrm{end}_A(M).$$

For $f(b) = b^n + a_1 b^{n-1} + \cdots + a_n \in A[b]$, we see that $\rho(f(b)) = 0$. But M is a faithful $A[b]$-module, so $f(b) = 0$ and b is integral as desired. ■

3.1.10 Lemma

(1) If B is a finite A-algebra, and M is a fg B-module, then M is a fg A-module.

(2) If B is a finite A-algebra, and C is a finite B-algebra, then C is a finite A-algebra.

Proof

Clearly the second point follows from the first. Now if M is generated by $m_1, \ldots, m_n$ over B, and B is generated by $b_1, \ldots, b_k$ over A then $b_i m_j$ generates M over A. Indeed, for $m \in M$ if $m = \sum_i b_i' m_i$, then $b_i' = \sum_j a_{ij} b_j$ so

$$m = \sum_{i,j} a_{ij} (b_j m_i),$$

as desired. ■

3.1.11 Corollary

Let B be an A-algebra, and suppose that $b_1, \ldots, b_n \in B$ are integral over A. Then $A[b_1, \ldots, b_n]$ is a finite A-algebra.

Proof

By induction: for $n = 1$ this is our above proposition. Then if b_{n+1} is integral over A, it is integral over $A[b_1, \ldots, b_n]$ so that $A[b_1, \ldots, b_{n+1}]$ is a finite $A[b_1, \ldots, b_n]$-algebra, and thus a finite A-algebra by transitivity which we just showed. ■

3.1.12 Corollary

Let B be an A-algebra, then
$$B_0 = \{b \in B \mid b \text{ is integral over } A\}$$
is an A-subalgebra of B

Proof

Clearly $0, 1 \in B_0$. Now if $b_1, b_2 \in B_0$ then $A[b_1 + b_2], A[b_1b_2] \subseteq A[b_1, b_2]$ which is a finite A-algebra and thus $b_1 + b_2, b_1b_2$ are integral over A; B_0 is a subring of B. And if $a \in A, b \in B_0$ then $A[ab] \subseteq A[b]$ is finite, so that $ab \in B_0$. ∎

3.1.13 Definition

(1) For an A-algebra B, we define A's **integral closure** in B to be the set of all integral elements over A in B; which we showed above is an A-subalgebra.

(2) An A-algebra B is **integral** if every element of B is integral over A.

(3) A morphism of rings $f\colon A \to B$ is **integral** if B is an integral A-algebra via f.

(4) For a morphism of rings $f\colon A \to B$, $f(A)$ is **integrally closed** in B if $B_0 = f(A)$ (i.e. no $b \in B - f(A)$ is integral over A).

(5) A domain A is **normal** if it is integrally closed in its field of fractions.

3.1.14 Example

We saw that $\mathbb{Z}$ is normal (its field of fractions is $\mathbb{Q}$); the same proof shows that any UFD is normal. In particular $k[x_1, \ldots, x_n]$ and $\mathbb{Z}[x_1, \ldots, x_n]$ are normal.
Consider the k-subalgebra $A = k[x^2, x^3] \subseteq k[x]$. Then $x \notin A$ is integral over A since it is a root of $t^6 - x^6 \in A[t]$, and $x = x^3/x^2$ is in its field of fractions. Thus A is not normal.

Note:
There is similarity between integral morphisms and algebraic extensions of fields. But while algebraically closed fields exist, there are no "integrally closed ring": every ring can be embedded non-integrally into another ring.

3.1.15 Corollary

An A-alegbra B is finite over A iff it is integral and finitely generated.

Proof

If B if a finite A-algebra, then for every $b \in B$, $A[b] \subseteq B$ is finite so b is integral over A; B is integral and finitely generated. Conversely if B is integral and finitely generated, suppose $B = A[b_1, \ldots, b_n]$; since each b_i is integral we have that B is finite. ∎

3.1.16 Corollary

Let B be an integral A-algebra, and C be a B-algebra.

(1) If $c \in C$ is integral over B, then it is integral over A.

(2) If C is integral over B, then it is integral over A.

Proof

If c is integral over B, then

$$c^n + b_1 c^{n-1} + \cdots + b_n = 0$$

for some $b_i \in B$, which are integral over A. Then $B' = A[b_1, \ldots, b_n] \subseteq B$ is a finite A-algebra, and c is integral over B' and so $B'[c]$ is a finite B'-algebra. Since B' is a finite A-algebra, by transitivity $B'[c]$ is a finite A-algebra as well containing $A[c]$, so c is integral over A.

The second point follows from the first. ■

3.1.17 Corollary

Let B be an A-algebra, and B_0 be the integral closure of A inside B. Then B_0 is integrally closed in B.

Proof

If $b \in B$ is integral over B_0, then by the above corollary, since B_0 is an integral A-algebra, b is integral over A and thus $b \in B_0$. ■

Let B be an A-algebra under $f\colon A \to B$.

(1) For an ideal $J \leq B$, define $J_A = f^{-1}(J) \leq A$. Then f induces an injective homomorphism $A/J_A \to B/J$, so that B/J is an A/J_A-algebra.

(2) For a multiplicative set $S \subseteq A$, $f(S) \subseteq B$ is multiplicative. Then we have a natural morphism $S^{-1}A \to f(S)^{-1}B$ mapping a/s to $f(a)/f(s)$. Thus $f(S)^{-1}B$ is an $S^{-1}A$-algebra.

(3) Viewing B as an A-module, we have already defined $S^{-1}B$, and this clearly coincides with $f(S)^{-1}B$. Thus $S^{-1}B$ is naturally an $S^{-1}A$-algebra.

3.1.18 Proposition

Let B be an integral A-algebra.

(1) Let $J \leq B$ be an ideal, then B/J is an integral A/J_A-algebra.

(2) Let $S \subseteq B$ be multiplicative, then $S^{-1}B$ is an integral $S^{-1}A$-algebra.

Proof

For $\overline{x} \in B/J$, lift it to $x \in B$, which is integral over A. Thus $x^n + a_1 x^{n-1} + \cdots + a_n = 0$ for $a_i \in A$, and then

$$\overline{x}^n + \overline{a}_1 \overline{x}^{n-1} + \cdots + \overline{a}_n = 0,$$

where $\overline{a}_i = a_i + J_A \in A/J_A$, so that $\overline{x}$ is integral over A/J_A.

An element of $S^{-1}B$ is of the form $\tilde{x} = x/f(s)$ with $x \in B, s \in S$. By assumption, $x^n + a_1 x^{n-1} + \cdots + a_n = 0$ and thus

$$\left(\frac{x}{f(s)}\right)^n + \frac{a_1}{s}\left(\frac{x}{f(s)}\right)^{n-1} + \cdots + \frac{a_n}{s^n} = \frac{x^n + a_1 x^{n-1} + \cdots + a_0}{f(s)^n} = 0,$$

so that $\tilde{x}$ is integral over $S^{-1}A$ as desired. ■

3.1.19 Proposition

Let $A \subseteq B$ be integral domains so that B is integral over A. Then A is a field iff B is a field.

Proof

If A is a field, then let $0 \neq b \in B$. It is integral over A so $b^n + a_1 b^{n-1} + \cdots + a_n = 0$ for $a_i \in A$, and assume n is minimal. Then

$$b(b^{n-1} + a_1 b^{n-2} + \cdots + a_{n-1}) = -a_n.$$

Since by minimality $b^{n-1} + a_1 b^{n-1} + \cdots + a_{n-1} \neq 0$, we have $a_n \neq 0$. Thus $-a_n$ is a unit, and thus b must be a unit as well (if $xy = u$ is a unit, then x is a unit since $x(yu^{-1}) = 1$). So B is a field.

Conversely if B is a field, then let $0 \neq x \in A$. Then $x^{-1} \in B$ since it is a field. Since B is integral over A, there are $a_i \in A$ such that

$$x^{-n} + a_1 x^{-(n-1)} + \cdots + a_n = 0,$$

and multiplying by x^{n-1} we get

$$x^{-1} + a_1 + \cdots + a_n x^{n-1},$$

so that x^{-1} is the sum of elements from A, and thus is in A. ■

3.1.20 Corollary

Let B be an integral A-algebra, and $\mathfrak{q} \leq B$ a prime ideal. Let $\mathfrak{p} = \mathfrak{q}_A \leq A$ be its preimage, then $\mathfrak{q}$ is maximal iff $\mathfrak{p}$ is maximal.

Proof

Both $\mathfrak{p}, \mathfrak{q}$ are prime so that $B/\mathfrak{q}$ and $A/\mathfrak{p}$ are integral domains. As we showed, $B/\mathfrak{q}$ is integral over $A/\mathfrak{p}$. Thus $B/\mathfrak{q}$ is a field iff $A/\mathfrak{p}$ is a field; so $\mathfrak{q}$ is maximal iff $\mathfrak{p}$ is maximal. ■

3.1.21 Corollary

Let B be an integral A-algebra, $\mathfrak{q} \leq \mathfrak{q}' \leq B$ be prime ideals such that $\mathfrak{q}_A = \mathfrak{q}'_A$. Then $\mathfrak{q} = \mathfrak{q}'$.

Proof

Let $\mathfrak{p} = \mathfrak{q}_A = \mathfrak{q}'_A$; it is prime over A. Consider the following commutative square

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "\iota_A"'] & B \arrow[d, "\iota_B"] \\ A_{\mathfrak{p}} \arrow[r, "f_{\mathfrak{p}}"] & B_{\mathfrak{p}} \end{tikzcd}$$

which commutes since the structure morphism $A \to S^{-1}A$ is a natural transformation (literally what the commutative square says).

Since $f^{-1}\mathfrak{q} = \mathfrak{p}$ we see that $\mathfrak{q} \cap f(A - \mathfrak{p}) = \emptyset$ and similarly for $\mathfrak{q}'$. Thus $\mathfrak{q}B_{\mathfrak{p}} \subseteq \mathfrak{q}'B_{\mathfrak{p}} \subseteq B_{\mathfrak{p}}$ are prime ideals (since they don't intersect the localizing set $f(A - \mathfrak{p})$).

We are going to show that $\mathfrak{q}B_{\mathfrak{p}} \subseteq B_{\mathfrak{p}}$ is a maximal ideal so that $\mathfrak{q}B_{\mathfrak{p}} = \mathfrak{q}'B_{\mathfrak{p}}$. And then since $\mathfrak{q} \mapsto \mathfrak{q}B_{\mathfrak{p}}$ is bijective with inverse ι_B^{-1}, we will have that $\mathfrak{q} = \mathfrak{q}'$.

Since B is integral over A, $B_{\mathfrak{p}}$ is integral over $A_{\mathfrak{p}}$. Thus, by the above corollary, it is sufficient to show that $\tilde{\mathfrak{p}} = f_{\mathfrak{p}}^{-1}(\mathfrak{q}B_{\mathfrak{p}}) \subseteq A_{\mathfrak{p}}$ is maximal. By assumption,

$$\iota_A^{-1}(\tilde{\mathfrak{p}}) = \iota_A^{-1}(f_{\mathfrak{p}}^{-1}(\mathfrak{q}B_{\mathfrak{p}})) = f^{-1} \circ \iota_B^{-1}(\mathfrak{q}B_{\mathfrak{p}}) = f^{-1}(\mathfrak{q}) = \mathfrak{p}.$$

Thus we have that $\tilde{\mathfrak{p}} = \mathfrak{p}A_{\mathfrak{p}}$ (since ι_A^{-1} is bijective) which is maximal, as desired. ■

The main result we want is the following.

3.1.22 Theorem

Let $f\colon A \to B$ be an injective integral morphism. Then for every prime $\mathfrak{p} \leq A$, there exists a prime ideal $\mathfrak{q} \leq B$ such that $\mathfrak{p} = \mathfrak{q}_A$.

Proof

Since $B_{\mathfrak{p}} = f(A - \mathfrak{p})^{-1}B$, and $0 \notin f(A - \mathfrak{p})$ because f is injective, $B_{\mathfrak{p}} \neq 0$. As before, $B_{\mathfrak{p}}$ is integral over $A_{\mathfrak{p}}$ and we have the same commutative square

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "\iota_A"'] & B \arrow[d, "\iota_B"] \\ A_{\mathfrak{p}} \arrow[r, "f_{\mathfrak{p}}"] & B_{\mathfrak{p}} \end{tikzcd}$$

Let $\mathfrak{m} \subseteq B_{\mathfrak{p}}$ be maximal, so that $f_{\mathfrak{p}}^{-1}(\mathfrak{m}) \subseteq A_{\mathfrak{p}}$ is maximal (as we showed $\mathfrak{q}_A$ is maximal iff $\mathfrak{q}$ is). Thus $f_{\mathfrak{p}}^{-1}(\mathfrak{m}) = \mathfrak{p}A_{\mathfrak{p}}$. Let $\mathfrak{q} = \iota_B^{-1}(\mathfrak{m})$, which is prime in B, and

$$f^{-1}(\mathfrak{q}) = f^{-1} \circ \iota_B^{-1}(\mathfrak{m}) = \iota_A^{-1} \circ f_{\mathfrak{p}}^{-1}(\mathfrak{m}) = \iota_A^{-1}(\mathfrak{p}A_{\mathfrak{p}}) = \mathfrak{p},$$

as desired. ■

Note:

Equivalently, this theorem says that if $f\colon A \to B$ is integral injective, then $f^*\colon \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is

surjective.
Note that injectivity is necessary here. This is because $\pi\colon A \to A/I$ is integral for any ideal $I \leq A$, but $\pi^*\colon \mathrm{Spec}(A/I) \to \mathrm{Spec}(A)$ is not surjective: its image is $V(I)$, the set of prime ideals containing I.

3.1.23 Corollary

Let B be an integral A-algebra, then for every prime $\mathfrak{q} \leq B$ and every prime ideal $\mathfrak{p}' \supseteq \mathfrak{p} = \mathfrak{q}_A$, there exists a prime ideal $\mathfrak{q}' \supseteq \mathfrak{q}$ of B such that $\mathfrak{q}'_A = \mathfrak{p}'$.

Proof

Consider the commutative square

$$\begin{tikzcd} A \arrow[r, "f"'] \arrow[d, "\pi_A"'] & B \arrow[d, "\pi_B"'] \\ A/\mathfrak{p} \arrow[r, "\overline{f}"] & B/\mathfrak{q} \end{tikzcd}$$

Then $\mathfrak{p}' \supseteq \mathfrak{p}$ gives rise to a prime ideal $\overline{\mathfrak{p}}' = \mathfrak{p}'/\mathfrak{p} \subseteq A/\mathfrak{p}$. Since $B/\mathfrak{q}$ is an integral $A/\mathfrak{p}$-algebra, there is a prime ideal $\overline{\mathfrak{q}}' \subseteq B/\mathfrak{q}$ such that $\overline{f}^{-1}(\overline{\mathfrak{q}}') = \overline{\mathfrak{p}}'$. Then $\mathfrak{q}' = \pi_B^{-1}(\overline{\mathfrak{q}}')$ is a prime ideal of B containing $\mathfrak{q}$ and

$$f^{-1}(\mathfrak{q}') = f^{-1} \circ \pi_B^{-1}(\overline{\mathfrak{q}}') = \pi_A^{-1} \circ \overline{f}^{-1}(\overline{\mathfrak{q}}') = \pi_A^{-1}(\overline{\mathfrak{p}}') = \mathfrak{p}',$$

as desired. ∎

3.1.24 Proposition

Let B be an A-algebra, $C \subseteq B$ be the integral closure of A inside of B. Then for any multiplicative $S \subseteq A$, $S^{-1}C$ is the integral closure of $S^{-1}A$ inside of $S^{-1}B$.

Proof

Since C is integral over A, $S^{-1}C$ is integral over $S^{-1}C$. Now if b/s is integral over $S^{-1}A$, then $b/1 = s \cdot b/s$ is integral over $S^{-1}A$ as well, and thus

$$\left(\frac{b}{1}\right)^n + \frac{a_1}{s_1}\left(\frac{b}{1}\right)^{n-1} + \cdots + \frac{a_n}{s_n} = 0$$

for $a_i/s_i \in S^{-1}A$. Let $t = s_1 \cdots s_n$ and $u_i = t/s_i \in S$, and multiplying this equation by t^n gives us

$$\left(\frac{tb}{1}\right)^n + a_1u_1\left(\frac{bt}{1}\right)^{n-1} + \cdots + a_nu_nt^{n-1} = 0.$$

Thus there is a $v \in S$ such that

$$v((bt)^n + a_1u_1(bt)^{n-1} + a_2u_2t(bt)^{n-2} + \cdots + a_nu_nt^{n-1}) = 0.$$

Then btv is integral over A (multiply by v^{n-1}), so that $btv \in C$. Then $b/s = (btv)/(stv) \in S^{-1}C$, as desired. ∎

3.1.25 Corollary

Let A be a domain, then the following are equivalent:

(1) A is normal,

(2) $S^{-1}A$ is normal for every multiplicative $S \subseteq A$ not containing 0,

(3) $A_{\mathfrak{p}}$ is normal for every prime $\mathfrak{p}$,

(4) $A_{\mathfrak{m}}$ is normal for every maximal $\mathfrak{m}$.

Proof

Let K be A's field of fractions, and let $C \subseteq K$ be A's integral closure in K. Then $S^{-1}C$ is $S^{-1}A$'s integral closure in $S^{-1}K = K$ for all multiplicative S not containing 0. If A is normal, then $C = A$ so that $S^{-1}A$ is integrally closed in K, and thus normal. So (1) implies (2), which implies (3) which implies (4).

Now if $A_{\mathfrak{m}}$ is normal, then $A_{\mathfrak{m}} \to C_{\mathfrak{m}}$ is an isomorphism for all $\mathfrak{m}$. We showed that this implies $A \to C$ is an isomorphism, so that A is normal. ∎

3.2 Noetherian rings and modules

3.2.1 Lemma

Let Σ be a poset, then the following conditions are equivalent:

(1) every nonempty subset of Σ has a maximal element,

(2) every increasing chain $x_1 \leq x_2 \leq \cdots$ in Σ eventually stabilizes: at some point $x_n = x_{n+1}$ (equivalently, there are no infinite strictly increasing chains).

A poset satisfying these equivalent conditions is said to have the **ascending chain condition** (**ACC**).

Proof

The first implies the second as we can consider the subset $\{x_i\}$. The second implies the first since if $\Sigma' \subseteq \Sigma$ doesn't have a maximal element we can form a strictly increasing chain inductively. ∎

3.2.2 Definition

An A-module M is **Noetherian** if its poset of A-submodules has the ACC. A ring A is **Noetherian** if it is Noetherian as a module over itself (equivalently its ideals satisfy the ACC).

3.2.3 Example

Any field is a Noetherian ring, since it only has two ideals.

$\mathbb{Q}$ is not a Noetherian $\mathbb{Z}$-module, since $\frac{1}{n!}\mathbb{Z}$ does not stabilize (the reason for the factorial is so that it forms an increasing chain).

The ring $k[x_1, x_2, \ldots]$ with infinitely many indeterminants is not Noetherian, since $(x_1) \subsetneq (x_1, x_2) \subsetneq (x_1, x_2, x_3) \subsetneq \cdots$ forms a strictly increasing chain.

3.2.4 Proposition

An A-module is Noetherian iff every submodule is finitely generated. Thus a ring is Noetherian iff every ideal is finitely generated.

Proof

If M is Noetherian and $N \subseteq M$ is a submodule, suppose it is not finitely generated. Then inductively we will form an infinitely increasing sequence of submodules. Let $x_1 \in N$ arbitrary, then since N is not finitely generated $x_1, \ldots, x_n$ does not generate N so that there is an $x_{n+1} \in N - \langle x_1, \ldots, x_n \rangle$. Then $N_n = \langle x_1, \ldots, x_n \rangle$ is a strictly increasing infinite chain of submodules of M, a contradiction.

Conversely if every submodule of M is finitely generated, then M is Noetherian. Otherwise, let $M_1 \subsetneq M_2 \subseteq \cdots$ be an increasing chain of submodules. Let $N = \bigcup_n M_n$ be their union, which is also a submodule of M, and thus is finitely generated. Suppose $x_1, \ldots, x_n \in N$ generate N, and since M_i form a chain there is an m such that $x_1, \ldots, x_n \in M_m$ and thus $M_m = N$ so that M_i eventually stabilizes to M_m. ∎

3.2.5 Corollary

Every PID is Noetherian.

3.2.6 Lemma

Let $0 \to M' \xrightarrow{f} M \xrightarrow{g} M'' \to 0$ be a short exact sequence of A-modules. Then M is Noetherian iff M', M'' are.

Equivalently, if M is an A-module and $N \subseteq M$ a submodule, M is Noetherian iff N and M/N are.

Proof

If M is Notherian, let $M'_n \subseteq M', M''_n \subseteq M''$ be increasing sequences of submodules. Then $f(M'_n)$ and $g^{-1}(M''_n)$ are submodules of M and thus stabilize: $f(M'_n) = f(M'_{n+1})$ and $g^{-1}(M'_n) = g^{-1}(M'_{n+1})$ eventually. By the injectivity of f and surjectivity of g, this means M'_n and M''_n both eventually stabilize.

Conversely if M', M'' are Noetherian, let $M_n \subseteq M$ be an increasing sequence of submodules. Then $f^{-1}(M_n) = M_n \cap M'$ and $g(M_n) \subseteq M''_n$ are increasing sequences of submodules of M', M'' respectively. Thus eventually they stabilize: $f^{-1}(M_n) = f^{-1}(M_{n+1})$ and $g(M_n) = g(M_{n+1})$ eventually. We claim that this means $M_n = M_{n+1}$.

Indeed, for $x \in M_{n+1}$ consider $g(x) \in g(M_{n+1})$ so there is a $y \in M_n$ such that $g(x) = g(y)$. Then $x - y \in \ker g = \operatorname{im} f$ so that $x - y = f(z)$ for $z \in M'$, and $x - y \in M_{n+1}$ so that $z \in f^{-1}(M_{n+1})$ and thus $z \in f^{-1}(M_n)$. This means that $x - y \in M_n$ and thus $x \in M_n$ as desired. ∎

3.2.7 Corollary

(1) The finite direct sums of Noetherian modules is Noetherian.

(2) If A is a Noetherian ring and M is a fg A-module, then M is a Notherian module.

(3) If A is a Notherian ring and B a finite A-algebra, then B is a Noetherian ring. In particular, A/I is Notherian for any ideal $I \leq A$.

Proof

For the first point it is sufficient to show this for binary direct sums. If M, N are Noetherian, then $0 \to M \to M \oplus N \to N \to 0$ is a short exact sequence and thus $M \oplus N$ is Notherian by above.

Since M is fg, $M \cong A^n/N$ for some submodule $N \leq A^n$ (consider the surjection $A^n \to M$ mapping to generators). By the first point, A^n is Noetherian and thus M is as well.

B is a fg A-module, and thus a Noetherian A-module. Since ideals of B are A-submodules, the ACC on A-submodules implies ACC on B's ideals. ■

3.2.8 Lemma

Let A be a Noetherian ring, then all of its localizations are Noetherian rings as well.

Proof

Let $S \subseteq A$ be multiplicative, and consider an increasing sequence of ideals $\tilde{I}_1 \subseteq \tilde{I}_2 \subseteq \cdots$ in $S^{-1}A$. Then $I_n = \iota^{-1}(\tilde{I}_n)$ forms an increasing sequence of ideals of A, and thus stabilizes. Since $\tilde{I}_n = S^{-1}I_n$, we get that $\tilde{I}_n$ stabilizes. ■

3.2.9 Theorem (Hilbert basis theorem)

If A is Noetherian, then $A[x]$ is a Noetherian ring.

Proof

Let $I \subseteq A[x]$ be an ideal, we wish to show it is finitely generated. Let us adopt the notation that for $f \in A[x]$, $a_f \in A$ denotes the leading term of f (and $a_f = 0$ only when $f = 0$).

We claim that $J = \{a_f \mid f \in I\}$ is an ideal in A. If $f \in I$ and $b \in A$ then either $ba_f = 0$ or $ba_f = a_{bf}$; in both cases $ba_f \in J$. Now if $f, g \neq 0$ are in I then $a_f + a_g$ is either zero or the leading term of $x^{\deg g} f + x^{\deg f} g \in J$ and thus $a_f + a_g \in J$.

Since J is Noetherian, there are $f_1, \ldots, f_n \in I$ such that $J = (a_{f_1}, \ldots, a_{f_n})$. Let $d_i = \deg f_i$, and d be the maximum d_i. Consider the fg A-submodule

$$M = \sum_{i=0}^{d-1} Ax^i \subseteq A[x].$$

Then $M \cap I$ is an A-submodule of M, which is Noetherian (since M is fg and A is Noetherian so M is Noetherian, and thus its submodules are Noetherian).

Now define $I' = (f_1, \ldots, f_n) \subseteq I$. We claim that

$$I = I' + (M \cap I) \text{ as } A\text{-submodules.}$$

Let $f \in I$, then we claim there is an $f' \in I'$ such that $\deg(f - f') < d$ and thus $f - f' \in M \cap I$. If $\deg f < d$ then we are done, so assume $\deg f \geq d$. By induction, it is sufficient to show that there is an $f' \in I'$ such that $\deg(f - f') < \deg f$.

Now, $a_f \in J = (a_{f_1}, \ldots, a_{f_n})$ so that $a_f = \sum_i b_i a_{f_i}$ for $b_i \in A$. Then define

$$f' = \sum_{i=1}^{n} b_i x^{(\deg f) - d_i} f_i \in I'.$$

By assumption, $\deg f \geq d \geq d_i$ so that $\deg(x^{(\deg f) - d_i} f_i) = d_i$, and so $\deg f' \leq \deg f$. Moreover, the leading coefficient of f' is

$$a_{f'} = \sum_{i=1}^{n} b_i a_{f_i} = a_f,$$

so that $\deg(f - f') < \deg f$ as desired.
Now since $I = I' + (M \cap I)$, I' is a fg ideal and $M \cap I$ is a fg A-module (and thus fg by $A[x]$), we have that I is a fg ideal, as desired. ■

3.2.10 Corollary

(1) If A is a Noetherian ring, so is $A[x_1, \dots, x_n]$.

(2) If A is a Noetherian ring, and B is a fg A-algebra, B is Noetherian.

(3) Any fg algebra over a field is Noetherian.

(4) Any fg algebra over a PID (in particular $\mathbb{Z}$) is Noetherian.

Proof

The first point follows by induction and Hilbert's basis theorem. The second point is because if B is a fg A-algebra, it is isomorphic to $A[x_1, \dots, x_n]/I$ for some ideal $I \leq A[x_1, \dots, x_n]$, and is thus Noetherian. The last two points are due to the second (fields and PIDs are Noetherian). ■

We now utilize these results to reprove a lemma we showed above.

3.2.11 Lemma

Let A be a ring and $T \in M_n(A)$ a matrix. Then there are $a_1, \dots, a_n \in A$ such that

$$T^n + a_1 T^{n-1} + \cdots + a_m = 0.$$

Proof

Denote $T = (a_{ij})$, and consider the subalgebra $A' = \mathbb{Z}[\{a_{ij}\}] \subseteq A$. This is a fg $\mathbb{Z}$-algebra and thus Noetherian. Since $T \in M_n(A')$, we can assume from the getgo that A is Noetherian.
Since $M_n(A) \cong A^{n^2}$ as A-modules, $M_n(A)$ is a Noetherian A-module. Thus the increasing sequence of A-modules

$$N_m = \langle 1, T, \dots, T^m \rangle \subseteq M_n(A)$$

stabilizes, so eventually $N_n = N_{n-1}$ so that $T^n \in N_{n-1}$ and we get the desired result. ■

3.2.12 Lemma

If M is a fg A-module, then for $\varphi \in \mathrm{end}_A(M)$, there are $a_1, \dots, a_n \in A$ such that

$$\varphi^n + a_1 \varphi^{n-1} + \cdots + a_n = 0.$$

Proof

When $M = A^n$ there is a natural isomorphism of A-algebras $M_n(A) \cong \mathrm{end}_A(M)$, so the result follow from the above lemma. Otherwise, consider a surjective map $\pi\colon A^n \to M$, then we claim there is a $\tilde{\varphi}\colon A^n \to A^n$

such that the below diagram commutes

$$\begin{tikzcd}
A^n \arrow[r, "\tilde{\varphi}"'] \arrow[d, "\pi"'] & A^n \arrow[d, "\pi"'] \\
M \arrow[r, "\varphi"] & M
\end{tikzcd}$$

Meaning $\pi \circ \tilde{\varphi} = \varphi \circ \pi$. For every i choose a preimage $V_i \in \pi^{-1}(\varphi(\pi(e_i))) \in A^n$ and define

$$\tilde{\varphi}(x_1, \ldots, x_n) = \sum_{i=1}^{n} x_i v_i.$$

This satisfies the condition.
Now there is a monic polynomial $f \in A[x]$ such that $f(\tilde{\varphi}) = 0$ by the $M = A^n$ case. Then

$$f(\varphi) \circ \pi = \pi \circ f(\tilde{\varphi}) = 0,$$

which implies $f(\varphi) = 0$ because π is surjective. ∎

3.3 Simple modules and length

3.3.1 Definition

A poset Σ has the **descending chain condition** (**DCC**) if it satisfies the dual of the ACC (equivalently, Σ^{op} has ACC). This means that every subset has a minimal element, or every descending chain stabilizes.

3.3.2 Definition

An A-module is **Artinian** if it has DCC on its submodules. A ring A is **Artinian** if it is Artinian as a module over itself, equivalently it has DCC on its ideals.

Note that the short exact sequence result on Noetherian rings holds for Artinian rings as well. Thus finite direct sums of Artinian modules are Artinian, fg modules over Artinian rings are Artinian, and thus finite algebras over Artinian rings are Artinian.
Note that it is not the case that a module is Artinian iff its submodules are all fg. That is, Artinian is not equivalent to Noetherian (and for modules they are distinct conditions, for rings we will see that Artinian implies Noetherian).

3.3.3 Example

$\mathbb{Z}$ is a Noetherian but not an Artinian ring. It is not Artinian because $\mathbb{Z} \supsetneq (n) \supsetneq (n^2) \supsetneq (n^3) \supsetneq \cdots$ is an infinite decreasing sequence of ideals.
Similarly $k[x]$ is Noetherian but not Artinian: (x^k) forms a descending sequence of ideals.

3.3.4 Definition

A nonzero module M is **simple** if it has no non-trivial submodules.

3.3.5 Lemma

An A-module M is simple iff it is isomorphic (as A-modules) to $A/\mathfrak{m}$ for a maximal ideal $\mathfrak{m} \leq A$.

Proof

If M is simple, then for $0 \neq m \in M$ the submodule $\langle m \rangle \subseteq M$ is isomorphic to $A/\mathrm{ann}(m)$ (the map $a \mapsto am$ has kernel $\mathrm{ann}(m)$). Since M is simple, $\langle m \rangle = M$ and so $A/\mathrm{ann}(m) \cong M$. There is then a bijection between submodules of M and ideals of A containing $\mathrm{ann}(m)$. Since M is simple, there cannot be any nontrivial such ideals, so $\mathrm{ann}(m)$ is maximal.

And if $M \cong A/\mathfrak{m}$ then again there is a bijection between submodules of M and ideals containing $\mathfrak{m}$. So M is simple, as $\mathfrak{m}$ is maximal. ■

3.3.6 Definition

A **composition series** of M is a chain of submodules

$$0 = M_0 \subsetneq M_1 \subsetneq M_2 \subsetneq \cdots \subsetneq M_n = M$$

which is maximal. This is equivalent to M_{i+1}/M_i being simple for all i (no modules between M_i and M_{i+1}). The **length** of such a series is n (not $n+1$).

Denote by $\ell_A(M)$ the minimal length of a composition series of M (or infinity if no composition series exists).

3.3.7 Lemma

If M has finite length $\ell(M) = n$, then all of its composition series have the same length n, and every chain of submodules can be extended to a composition series.

Proof

First we claim that if $N \subsetneq M$ is a submodule, then $\ell(N) < \ell(M)$. Indeed, if $\{M_i\}$ is a composition series for N, then $N_i = M_i \cap N$ is a sequence of submodules of N. Then $N_i = N_{i+1} \cap M_i$ so

$$N_{i+1}/N_i = N_{i+1}/(N_{i+1} \cap M_i) \cong (N_{i+1} + M_i)/M_i \subseteq M_{i+1}/M_i$$

which is either simple or zero. Thus removing repeated values, we have formed a composition series for N of shorter length, so $\ell(N) \leq \ell(M)$.

Now if $\ell(N) = \ell(M)$ then $N_{i+1} + M_i = M_{i+1}$ for all i, and thus we claim $N_i = M_i$ for all i so that $N = M$. This is by induction, since $M_0 = N_0 = 0$, and $N_{i+1} = N_{i+1} + N_i = N_{i+1} + M_i = M_{i+1}$, as desired.

Now if $\{M_i\}_0^n$ is any chain in M, then

$$\ell(M) = \ell(M_n) > \ell(M_{n-1}) > \cdots > \ell(M_0) = 0,$$

so $\ell(M) \geq n$.

Now if $\{M_i\}$ is any composition series of M of length n, then $\ell(M) \geq n$ by what we just showed. But by definition $\ell(M) \leq n$ so that $\ell(M) = n$ and thus all composition series have length $\ell(M)$.

And for any general chain of submodules of M, if it is not a composition series then new terms can be inserted. This process can be repeated, and must terminate (since once it reaches a length of $\ell(M)$ it must be a composition series). ■

Note:
Note that we can prove something stronger, the **Jordan-Holder theorem**: for any two composition series M_i, N_j there is a permutation σ between i and j such that $M_{i+1}/M_i \cong N_{\sigma(i)+1}/N_{\sigma(i)}$. A proof of this can be found in my representation theory homework somewhere.

3.3.8 Lemma
An A-module M has finite length iff it is both Noetherian and Artinian.

Proof

If M has finite length, then any increasing/decreasing sequence has length bound by $\ell(M)$ so that M is both Noetherian and Artinian.
Converslеy if M is Artinian and Noetherian, then consider the set of submodules $N \subseteq M$ of finite length. It is nonempty since it contains 0, and because M is Noetherian contains a maximal element M'. If $M = M'$ then we are done. Otherwise, the set of submodules $M' \subsetneq N \subseteq M$ is nonempty (containing M) and has a minimal element N since M is Artinian. But then N/M' is simple, and so composing a composition series of M' with $\subsetneq N$ forms a composition series of N, so N has finite length, contradicting M''s maximality. ■

3.3.9 Proposition
Let M be fg over a Notherian ring A.

(1) If $M \neq 0$, then M has a submodules $N \subseteq M$ such that $N \cong A/\mathfrak{p}$ for some prime $\mathfrak{p} \leq A$.

(2) There exists a chain $0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M$ of submodules such that for each i, there is an isomorphism of A-modules $M_{i+1}/M_i \cong A/\mathfrak{p}_i$ for prime ideals $\mathfrak{p}_i \leq A$.

(3) The chain in the previous point is not necessarily unique, but

- **(i)** for a prime ideal $\mathfrak{p} \leq A$ we have $\text{ann}M \subseteq \mathfrak{p}$ iff $\mathfrak{p}_i \subseteq \mathfrak{p}$ for some i. In particular, $\text{ann}M \subseteq \mathfrak{p}_i$ for all i, and the minimal ideals between $\mathfrak{p}_1, \ldots, \mathfrak{p}_n$ are precisely minimal prime ideals $\text{ann}(M) \subseteq \mathfrak{p}$.

- **(ii)** for every minimal prime ideal $\text{ann}(M) \subseteq \mathfrak{p}$, the $\mathcal{A}_\mathfrak{p}$ module $M_\mathfrak{p}$ has finite length equal to the number of times $\mathfrak{p}$ occurs in $\{\mathfrak{p}_1, \ldots, \mathfrak{p}_n\}$.

Proof

Let Θ be the set of all ideal $I \leq A$ of the form $I = \text{ann}(m)$ for $0 \neq m \in M$. Since A is Noetherian, there is an m such that $\text{ann}(m)$ is maximal. Since $\langle m \rangle \subseteq M$ is isomorphic to $A/\text{ann}(m)$, it is sufficient to show $\text{ann}(m)$ is prime. So suppose $a, b \notin \text{ann}(m)$, then $bm \neq 0$ and $\text{ann}(m) \subseteq \text{ann}(bm)$. Thus by maximality $\text{ann}(m) = \text{ann}(bm)$ and then $a \notin \text{ann}(bm)$ so $abm \neq 0$ meaning $ab \notin \text{ann}(m)$, showing it is prime. This proves the first point.
Let Σ be the set of all submodules of M which possess such a chain. It is nonempty since the zero module is in it. Since M is Noetherian, let M' be a maximal element of Σ.
Now consider $M'' = M/M'$; if $M'' = 0$ we are finished. Otherwise by the first point it has a submodule $N'' = N'/M'$ isomorphic to $A/\mathfrak{p}$. But then composing the chain of M' with $\subsetneq N'$ gives such a chain, contradicting M''s maximality.
For the final point, notice that for $M' \subsetneq M$,

$$\text{ann}(M') \cdot \text{ann}(M/M') \subseteq \text{ann}(M) \subseteq \text{ann}(M') \cap \text{ann}(M/M').$$

The second inclusion is clear, the first is because for $a \in \operatorname{ann}(M')$ and $b \in \operatorname{ann}(M/M')$ and $m \in M$ then $bm \in M'$ and $abm \in aM' = 0$. Thus for a prime ideal $\mathfrak{p} \leq A$, $\operatorname{ann}(M) \subseteq \mathfrak{p}$ iff $\operatorname{ann}(M') \subseteq \mathfrak{p}$ or $\operatorname{ann}(M/M') \subseteq \mathfrak{p}$. Thus by induction $\operatorname{ann}(M) \subseteq \mathfrak{p}$ iff $\operatorname{ann}(M_{i+1}/M_i) \subseteq \mathfrak{p}$ for some i. Since $\operatorname{ann}(A/\mathfrak{p}_i) = \mathfrak{p}_i$, the first point follows.

For the second point, the chain $\{M_i\}$ gives the chain $\left\{S^{-1}M_i\right\}$ and

$$S^{-1}M_{i+1}/S^{-1}M_i \cong S^{-1}(M_{i+1}/M_i) \cong S^{-1}(A/\mathfrak{p}_i) \cong S^{-1}A/S^{-1}\mathfrak{p}_i.$$

Moreover $S^{-1}\mathfrak{p}_i = S^{-1}A$ iff $S \cap \mathfrak{p}_i \neq \emptyset$, and otherwise is a prime ideal. For $S = A - \mathfrak{p}$ for a minimal prime ideal $\operatorname{ann}(M) \subseteq \mathfrak{p}$, then $S^{-1}\mathfrak{p}_i = A_\mathfrak{p}$ if $\mathfrak{p} \neq \mathfrak{p}_i$ and otherwise $S^{-1}\mathfrak{p}_i = \mathfrak{p}A_\mathfrak{p}$ is a maximal ideal. Thus the sequence $(M_i)_\mathfrak{p}$ forms a composition series after deleting the repeated entries (when $\mathfrak{p} \neq \mathfrak{p}_i$), and we are left with a composition series (since the quotients are quotients by a maximal ideal) whose length is equal to the number of $\mathfrak{p}_i$ equal to $\mathfrak{p}$. ■

3.3.10 Definition

A minimal prime ideal is a prime ideal not containing any other prime ideal.

3.3.11 Lemma

A Noetherian ring A has finitely many minimal prime ideals.

Proof

The minimal prime ideals of A are the minimal primes containing $\operatorname{ann}(A) = 0$. Then for any chain as in the above lemma (whose components are $A/\mathfrak{p}$ for some prime), the minimal prime ideals are the minimal primes from $\{\mathfrak{p}_1, \dots, \mathfrak{p}_n\}$ which is finite. ■

3.3.12 Proposition

A ring is Artinian iff it is Noetherian and every prime ideal is maximal.

Proof

If A is Noetherian and every prime ideal is maximal, then every prime ideal is also minimal. Thus A has finitely many prime/maximal ideals, $\mathfrak{m}_1, \dots, \mathfrak{m}_k$. Then their product $I = \mathfrak{m}_1 \cdots \mathfrak{m}_k$ is contained in the intersection of all prime ideals, $\operatorname{nil}(A)$. Because A is Noetherian I is finitely generated, so $I^n = 0$ for some n (since every element of I is nilpotent, and it is finitely generated).

We have a chain of submodules

$$0 = M_0 \subseteq M_1 \subseteq \cdots \subseteq M_k = A,$$

where $M_{i+1}/M_i \cong A/\mathfrak{m}_i$ for some i (by the above proposition). As a subquotient, M_{i+1}/M_i is a Noetherian A-module, and thus a Notherian module over the field $K = A/\mathfrak{m}_i$. A Noetherian vector space is finite-dimensional (it is fg), so that M_{i+1}/M_i is of finite length over K (its length is equal to its dimension). Then M_{i+1}/M_i has finite length over A as well, and then inductively we see that each M_i has finite length, and in particular A is Artinian.

Conversely if A is Artinian, then all of its prime ideals are maximal (homework), so we need only show it is Noetherian. If $\mathfrak{m}_1, \dots, \mathfrak{m}_{n+1}$ are distinct maximal ideals then $\mathfrak{m}_1 \cdots \mathfrak{m}_n \supset \mathfrak{m}_1 \cdots \mathfrak{m}_{n+1}$ is strict, so by Artinianness there must be finitely many maximal ideals. If the inclusion were not strict then $\mathfrak{m}_1 \cdots \mathfrak{m}_n \subseteq \mathfrak{m}_{n+1}$, and then by $\mathfrak{m}_{n+1}$'s primality, $\mathfrak{m}_i \subseteq \mathfrak{m}_{n+1}$ for some i. But then $\mathfrak{m}_i = \mathfrak{m}_{n+1}$ by maximality, a contradiction to distinctness.

So let $I = \mathfrak{m}_1 \cdots \mathfrak{m}_n$ be the product of all A's prime/maximal ideals. Consider the descending chain $I \supseteq I^2 \supseteq I^3 \supseteq \cdots$ which must terminate, so $I^k = I^{k+1}$ eventually. We claim that $I^k = 0$.

If not, let $x_1, \ldots, x_{k+1} \in I$ such that $x_1 \cdots x_{k+1} \neq 0$. Because $I^k = I^{k+1}$, $\prod_{i=2}^{k+1} x_i = \sum_\alpha y_{\alpha,1} \cdots y_{\alpha,k+1}$ where the sum is over a finite set and each $y_{\alpha,i} \in I$. Thus $x_1 y_{\alpha_0,1} \cdots y_{\alpha_0,k+1} \neq 0$ for some α_0. Then from the original tuple $x_1, \ldots, x_{k+1} \in I$ we obtain a new tuple $x_1' = x_1, x_2', \ldots, x_{k+2}'$ ($x_i' = y_{\alpha_0,i-1}$) with nonzero product, and the first element unchanged.

We similarly may write the product of the final k elements $x_3' \cdots x_{k+2}'$ as an element of I^{k+1} and obtain $x_1, x_2', x_3', x_4'', \ldots, x_{k+3}''$ with nonzero product. Continuing this inductively, we produce an infinite sequence $y_1, y_2, \ldots$ such that $y_1 \cdots y_m \neq 0$ for all $m > 0$. Now consider the descending chain $(y_1) \supseteq (y_1 y_2) \supseteq \cdots$, which stabilizes since A is Artinian. Thus for some $m > 0$ and $a \in A$,

$$\prod_{i=1}^{m} y_i = a \prod_{i=1}^{m+1} y_i \implies \left(\prod_{i=1}^{m} y_i \right) (1 - a y_{m+1}) = 0.$$

Since $y_{m+1} \in I$ and thus $a y_{m+1} \in I$ is in the Jacobson radical (I is the product of all maximal ideals, and thus contained in their intersection), we have that $1 - a y_{m+1} \in A^\times$. Thus $\prod_{i=1}^m y_i = 0$, a contradiction.

So $I^k = 0$, and so there is a chain (taking $M_j = \mathfrak{m}_1 \cdots \mathfrak{m}_{n-j}$),

$$0 = M_0 \subseteq M_1 \subseteq \cdots \subseteq M_m = A$$

where M_{i+1}/M_i is annihlated by some $\mathfrak{m}_i$, and thus is Artinian over the field $K = A/\mathfrak{m}_i$ so finite-dimensional over A. Thus M_{i+1}/M_i has finite length over $A/\mathfrak{m}_i$, and thus over A, and then by induction we see that each M_i has finite length, in particular so does A and thus is Noetherian. ■

Note:

Recall that $\mathbb{Z}$ is Notherian but not Artinian. This is because while all of its non-zero prime ideals are maximal, (0) is a prime ideal which is not maximal. We see that in general an integral domain which is not a field cannot be Artinian.

4 Tensoring and Flatness

4.1 Tensor products

I assume some familiarity with the tensor product (see e.g. my summary on representation theory otherwise). The tensor product is a bifunctor

$$\bullet \otimes_B \bullet \colon {}_A\mathsf{Mod}_B \times {}_B\mathsf{Mod}_C \to {}_A\mathsf{Mod}_C,$$

with the universal property that it represents the balanced map functor:

$$\hom(M \otimes_B N, P) \cong \mathrm{Bal}(M, N;\, P).$$

Balanced maps are maps $M \times N \to P$ which are additive and multiplicative in each coordinate, and $f(mb, n) = f(m, bn)$ for all $b \in B$.

The tensor product also has a right-adjoint. For an (A, B)-bimodule M, $\bullet \otimes_A M$ is a functor $\mathsf{Mod}_A \to \mathsf{Mod}_B$ with right adjoint $\hom_B(M, \bullet)$.

In the case of commutative rings (this course), all modules are bimodules, so the tensor product is a bifunctor

$$\bullet \otimes_A \bullet \colon \mathsf{Mod}_A \to \mathsf{Mod}_A,$$

which represents the bilinear map functor (bilinear in the usual sense).

Note still that we use tensor products between module categories. In particular if B is an A-algebra, then $B \otimes_A M$ is a B-module ($b(c \otimes m) = (bc) \otimes m$). This defines a map $\mathsf{Mod}_A \to \mathsf{Mod}_B$, and its right-adjoint is the restriction map (restrict a B-module to an A-module). Note that this is a special case of the tensor-hom adjunction: $\bullet \otimes_A B$ is left-adjoint to $\hom_B(B, \bullet) \cong B$.

So we have a natural isomorphism

$$\hom_B(B \otimes_A M, P) \cong \hom_A(M, P|_A), \qquad M \in \mathsf{Mod}_A, P \in \mathsf{Mod}_B.$$

The correspondence is given by mapping $f \colon B \otimes_A M \to P$ to $\tilde{f} \colon M \to P$ by $\tilde{f}(m) = f(1 \otimes m)$. And $g \colon M \to P|_A$ to $\hat{g} \colon B \otimes_A M \to P$ by $\hat{g}(1 \otimes m) = g(m)$.

4.1.1 Proposition

Let M be an A-module.

(1) For a multiplicative set $S \subseteq A$, $S^{-1}A \otimes_A M$ is naturally isomorphic to $S^{-1}M$.

(2) For an ideal $I \leq A$, $A/I \otimes_A M$ is naturally isomorphic to M/IM.

Proof

The canonical map $M \to S^{-1}M$ mapping $m \mapsto m/1$ defines a map $f\colon S^{-1}A \otimes_A M \to S^{-1}M$ by $f(1 \otimes m) = m/1$. Then

$$f(a/s \otimes m) = a/sf(1 \otimes m) = am/s.$$

We define its inverse $g\colon S^{-1}M \to S^{-1}A \otimes_A M$ by

$$g(m/s) = 1/s \otimes m.$$

We claim this is well-defined: if $m/s = m'/s'$ then $1/s \otimes m = 1/s' \otimes m'$. If $t(ms' - m's)$ then

$$\frac{1}{s} \otimes m = \frac{ts'}{ts's} \otimes m = \frac{1}{ts's} \otimes ts'm = \frac{1}{ts's} \otimes tsm' = \frac{1}{s'} \otimes m'$$

as desired, and clearly g is a module morphism. f and g are also clearly morphisms.
The proof for the second point is similar: projection $M \to M/IM$ is an A-module morphism. Thus this gives rise to an A/IA-module morphism $f\colon (A/IA) \otimes_A M \to M/IM$ by $f(1 \otimes m) = \overline{m}$. Then

$$f(\overline{a} \otimes m) = af(1 \otimes m) = a\overline{m} = \overline{am}.$$

Its inverse is $g\colon M/IM \to (A/IA) \otimes_A M$ given by $g(\overline{m}) = 1 \otimes m$; this is easily seen to be well-defined and an inverse. ∎

4.1.2 Corollary

Let B be an A-algebra and M a B-module. Then there is a natural map

$$B \otimes_A M|_A \to M, \qquad b \otimes m \to bm.$$

In particular this is a natural map $m_B\colon B \otimes_A B \to B$ by $b \otimes b' \mapsto bb'$.

Proof

This map is the map corresponding to the identity of $\hom_A(M|_A, M|_A)$. The second point is for $M = B$. ∎

4.1.3 Lemma

If B, C are A-algebras, then their tensor product $B \otimes_A C$ is naturally an A-algebra and their coproduct in the category of A-algebras.

Proof

Let $D = B \otimes_A C$. The natural algebra structure is given by $(b \otimes c)(b' \otimes c') = (bb') \otimes (cc')$, and $A \to D$ by $a \mapsto a(1 \otimes 1)$. This is the composition of

$$D \times D \to D \otimes_A D, \qquad (d, d') \mapsto d \otimes d',$$

and

$$D \otimes_A D = (B \otimes_A C) \otimes_A (B \otimes_A C) \cong (B \otimes_A B) \otimes_A (C \otimes_A C) \to B \otimes_A C$$

where the final map is the tensor of $m_B \otimes m_C$.

The inclusion maps $i_B : B \to D$ and $i_C : C \to D$ are given by $b \mapsto b \otimes 1$ and $c \mapsto 1 \otimes c$.

Now if E is another A-algebra and there are maps $f : B \to E, g : C \to E$, then if $h : D \to E$ commutes with the rest of the diagram, i.e.

$$g = h \circ i_C, \qquad f = h \circ i_B,$$

then we must have

$$h(b \otimes c) = h((b \otimes 1)(1 \otimes c)) = h(b \otimes 1)h(1 \otimes c) = hi_B(b)hi_C(c) = f(b)g(c).$$

Thus h is unique, and this is well-defined (it is bilinear). ■

4.2 Flatness

4.2.1 Definition

A functor is **left-exact** if it preserves all finite limits, and is **right-exact** if it preserves all finite colimits. A functor is **exact** if it is both left- and right-exact (preserves all finite limits and colimits).

Clearly the composition of (left/right-) exact functors are (left/right-) exact.

We quickly note a few equivalences which are useful to keep in mind. The proof provided is not complete, but is not too complicated to complete.

4.2.2 Proposition

For an additive functor F between Abelian categories (in particular between module categories), the following are equivalent:

(1) F is left-exact,

(2) If $0 \to A \to B \to C$ is an exact sequence, then the induced sequence $0 \to FA \to FB \to FC$ is exact as well,

(3) If $0 \to A \to B \to C \to 0$ is a short exact sequence, then the induced sequence $0 \to FA \to FB \to FC$ is exact.

In particular from (2) we see that left-exact functors preserve injections.

Similarly, the following are equivalent:

(1) F is right-exact,

(2) If $A \to B \to C \to 0$ is an exact sequence, then the induced sequence $FA \to FB \to FC \to 0$ is exact as well,

(3) If $0 \to A \to B \to C \to 0$ is a short exact sequence, then the induced sequence $FA \to FB \to FC \to 0$ is exact.

In particular right-exact functors preserve surjections.

Consequently, the following are equivalent:

(1) F is exact,

(2) F preserves short exact sequences,

(3) F preserves long exact sequences,

(4) F is left-exact and preserves surjections,

(5) F is right-exact and preserves injections.

Proof

First we note that an additive functor immediately preserves all finite biproducts. And $0 \to A \to B \to C$ being exact is equivalent to it being a kernel diagram. Thus F is left-exact iff it preserves it. And F preserves left-exact sequences (i.e. ones in point (2)) iff it maps short exact sequences to left-exact ones. Clearly if it preserves left-exact sequences it maps short exact ones to left-exact ones (SES are in particular left-exact), and the converse is shown by restricting the rightmost map to its image.

Since all injections are kernels, we see that left-exact functors preserve injections.

Similarly we show the equivalences for right-exact functors.

By the previous two equivalences, we see F being exact is equivalent to preserving SES sequences. And preserving long-exact sequences is done by properly restricting and quotienting. If F is left-exact and preserves surjections, then if $0 \to A \to B \to C \to 0$ is exact, we know that $0 \to FA \to FB \to FC$ is exact, and $FB \to FC \to 0$ is exact (as it preserves surjections). Putting this together, if F is left-exact and preserves surjections it is exact. ∎

4.2.3 Proposition

Left-adjoint functors are right-exact, and right-adjoint functors are left-exact.

Proof

Left-adjoint functors preserve *all* colimits, and are thus in particular right-exact. Similar for right-adjoint functors. ∎

Tensor products are left-adjoints and thus right-exact:

4.2.4 Proposition

If $M' \to M \to M'' \to 0$ is an exact sequence of A-modules, then for any A-module N, the induced sequence

$$M' \otimes_A N \to M \otimes_A N \to M'' \otimes_A N \to 0$$

is exact.

Note:

Tensor products are not exact; they do not preserve injections. Indeed, if $f\colon A \to A$ is the map $f(x) = ax$ for $a \in A$, then $f \otimes M = f \otimes \mathrm{id}_M$ is the endomorphism of $A \otimes_A M \cong M$ which maps $m \mapsto am$. When a is not a zero divisor (e.g. A is a domain), f is injective, but $f \otimes M$ need not be (e.g. $M = A/(a)$).

Since $\hom_A(M, \bullet)$ is right-adjoint, it is left exact so it preserves finite limits. It is not hard to show that $\hom_A(\bullet, M)$ is left-exact as well (but contravariantly). Furthermore the Yoneda embedding reflects exactness:

4.2.5 Proposition

A sequence of A-modules $M_\bullet = M' \to M \to M'' \to 0$ is exact iff

$$\hom(M_\bullet, N) = 0 \to \hom(M'', N) \to \hom(M, N) \to \hom(M', N)$$

is exact for all $N \in \mathsf{Mod}_A$. Similarly $N_\bullet = 0 \to N' \to N \to N''$ is exact iff

$$\hom(M, N_\bullet) = 0 \to \hom(M, N') \to \hom(M, N) \to \hom(M, N'')$$

is exact for all M.

Proof

The "only if" part is due to left exactness. The converse (second point) is done by choosing $M = N'$ and looking at where id goes (this shows that the composition of maps is zero), then M to be the kernel of the second map, and showing that this implies exactness at M, and then you just need to show the first map is injective. ■

While not all tensor products are exact, some are. We give this case a special term.

4.2.6 Definition

An A-module M is **flat** if the functor $\bullet \otimes_A M$ is exact.

Note:

Since $\bullet \otimes_A M$ is right-exact for any module M, M is flat iff $\bullet \otimes_A M$ preserves injections.

4.2.7 Example

If $S \subseteq A$ is multiplicative, then $\bullet \otimes_A S^{-1}A$ is naturally isomorphic to $S^{-1}\bullet$, which is exact. Thus $S^{-1}A$ is a flat A-module.
If M is free, so $M \cong A^{\oplus I}$, then

$$\bullet \otimes_A M \cong \bullet \otimes_A A^{\oplus I} (\bullet \otimes_A A)^{\oplus I} \cong \bullet^{\oplus I}.$$

The second isomorphism is due to left adjoints preserving colimits. And $\bullet^{\oplus I}$ is exact (it maps injective maps to injective maps, more generally direct sums in module categories are exact, satisfying Grothendieck's axiom (AB4)).

4.2.8 Lemma

Let M be an A-module and $S \subseteq A$ a multiplicative set.

(1) If N is an $S^{-1}A$-module, it is an A-module via the canonical map $A \to S^{-1}A$, and there is a natural isomorphism

$$M \otimes_A N \to S^{-1}M \otimes_{S^{-1}A} N, \qquad m \otimes n \mapsto m/1 \otimes n.$$

(2) For an A-module N, there is a natural isomorphism

$$S^{-1}M \otimes_{S^{-1}A} S^{-1}N \to S^{-1}(M \otimes_A N), \qquad \frac{m}{s} \otimes \frac{n}{t} \mapsto \frac{m \otimes n}{st}.$$

In particular for every prime $\mathfrak{p} \leq A$, there is a natural isomorphism

$$M_\mathfrak{p} \otimes_{A_\mathfrak{p}} N_\mathfrak{p} \cong (M \otimes_A N)_\mathfrak{p}.$$

Proof

There is a natural isomorphism (by associativity of the tensor)

$$S^{-1}M \otimes_{S^{-1}A} N \cong (M \otimes_A S^{-1}A) \otimes_{S^{-1}A} N \cong M \otimes_A N$$

which maps $m/s \otimes n$ to $m \otimes 1/s \otimes n$ to $m \otimes n/s$. So in reverse it maps $m \otimes n$ to $m/1 \otimes n$ as desired.
There is a natural isomorphism

$$S^{-1}M \otimes_{S^{-1}A} S^{-1}N \cong M \otimes_A S^{-1}N \cong M \otimes_A (N \otimes_A S^{-1}A) \cong (M \otimes_A N) \otimes_A S^{-1}A,$$

and this is naturally isomorphic to $S^{-1}(M \otimes_A N)$. ■

4.2.9 Corollary

Flatnes is local, i.e. the following are equivalent:

(1) M is a flat A-module,

(2) $S^{-1}M$ is a flat $S^{-1}A$-module for all multiplicative $S \subseteq A$,

(3) $M_\mathfrak{p}$ is a flat $A_\mathfrak{p}$-module for all prime $\mathfrak{p} \leq A$,

(4) $M_\mathfrak{m}$ is a flat $A_\mathfrak{m}$-module for all maximal $\mathfrak{m} \leq A$.

Proof

If M is flat then by the above lemma

$$\bullet \otimes_{S^{-1}A} S^{-1}M \cong S^{-1}(\bullet \otimes_A M),$$

which is a composition of exact functors and thus exact.
Now if $M_\mathfrak{m}$ is flat for all maximal $\mathfrak{m} \leq A$, we show M is flat. Recall we need only show that $\bullet \otimes_A M$ preserves injectives. Recall that f is injective iff $f_\mathfrak{m}$ is injective for all maximal $\mathfrak{m} \leq A$. Now, $f_\mathfrak{m}$ is isomorphic to $f \otimes M_\mathfrak{m}$, which is injective since $M_\mathfrak{m}$ is flat, we see $f_\mathfrak{m}$ is injective for all $\mathfrak{m}$ as desired. ■

4.3 The Tor functor

We now quickly state a few definitions and results from homological algebra.

4.3.1 Definition

Let $\mathcal{A}, \mathcal{B}$ be Abelian categories (e.g. module categories). Then a **(homological) δ-functor** is a collection of functors $\{F_n\colon \mathcal{A} \to \mathcal{B}\}_{n \geq 0}$ satisfying the following two properties:

(1) If $0 \to A \to B \to C \to 0$ is an SES in $\mathcal{A}$, then there exist maps $\delta_n\colon F_nC \to F_{n-1}A$ inducing a long exact sequence:

$$\cdots \to F_{n+1}C \xrightarrow{\delta} F_nA \to F_nB \to F_nC \xrightarrow{\delta} F_{n-1}A \to \cdots.$$

(2) The maps δ are natural: if we have a map of SES $0 \to A \to B \to C \to 0$ to $0 \to A' \to B' \to C' \to 0$, the following diagram commutes:

$$\begin{tikzcd} F_nC \arrow[r, "\delta"'] \arrow[d] & F_{n-1}A \arrow[d] \\ F_nC' \arrow[r, "\delta"] & F_{n-1}A' \end{tikzcd}$$

The functor F_0 is called the **base** of the δ-functor. Cohomological δ-functors are defined similarly but with the arrows inverted.

If F and G are δ-functors, then a morphism of δ-functors is a sequence of natural transformations $f_n\colon F_n \Rightarrow G_n$ which commute with the δ-maps.

A **universal** δ-functor is a δ-functor F such that for any other δ-functor G and a natural transformation $f_0\colon G_0 \to T_0$, f_0 extends uniquely to a morphism of δ-functors $G \to T$.

Clearly universal δ-functors are unique up to isomorphism. A functor F is called **coeffaceable** if for every object A there is a surjective $g\colon B \to A$ such that $F(g) = 0$. A δ-functor F is **coeffaceable** if every F_n is coeffaceable for $n \geq 1$. An important due to Grothendieck is that a coeffaceable δ-functor is universal.

Notice that if F is a δ-functor then its base F_0 is right-exact. This is because if $0 \to A \to B \to C \to 0$ is exact, then the end of the long exact sequence induced by F is

$$\cdots \to F_0A \to F_0B \to F_0C \to 0.$$

Another result, due to Grothendieck again, is that every right-exact functor is the base of a (necessarily unique up to isomorphism) universal δ-functor (under some basic assumptions on the underlying category). This is called the **derived functor**.

In particular, $\bullet \otimes_A M$ is right-exact for any A-module M. Thus it has a derived functor, called the **Tor-functor**, denoted by $\mathrm{Tor}_n^A(\bullet, M)$. We can also compute the derived functor of $N \otimes_A \bullet$ giving us another functor $\mathrm{Tor}_n^A(N, \bullet)$, and a non-trivial result tells us that these are two parts of the same bifunctor $\mathrm{Tor}_n^A(\bullet, \bullet)\colon \mathsf{Mod}_A \times \mathsf{Mod}_A \to \mathsf{Mod}_A$. Since the tensor product is commutative (up to isomorphism), the derived functors $\mathrm{Tor}_n^A(M, \bullet)$ and $\mathrm{Tor}_n^A(\bullet, M)$ are isomorphic (by uniqueness of universal δ-functors).

We summarize these results in the following theorem:

4.3.2 Theorem (The Tor functor)

There is a unique (up to unique isomorphism) collection of additive bi-functors $\mathrm{Tor}_n^A\colon \mathsf{Mod}_A \times \mathsf{Mod}_A \to \mathsf{Mod}_A$ such that

(1) $\mathrm{Tor}_0^A = \bullet \otimes_A \bullet$,

(2) $\mathrm{Tor}_n^A(\bullet, M)$ is a δ-functor for each A-module M.

(3) $\mathrm{Tor}_n^A(N, M) \cong \mathrm{Tor}_n^A(M, N)$ for all A-modules N, M and all $n \geq 0$.

The construction of derived functors is relatively straight-forward, but proving that they are (1) functors, (2) δ-functors, (3) universal δ-functors is less trivial. (1) and (2) is done pretty much directly as follows. Suppose F is right-exact, and let A be an object (a module in our case). Then we define a **projective resolution** of A to be a long exact sequence

$$P_\bullet \to A = \cdots \xrightarrow{d_1} P_1 \xrightarrow{d_0} P_0 \to A \to 0,$$

where each P_n is a **projective object**. In category of modules, free modules are a type of projective object (but not in general the only type), so we can instead consider only resolutions by free modules.

We then consider the induced chain complex

$$F(P_\bullet) = \cdots \xrightarrow{Fd_2} F(P_2) \xrightarrow{Fd_1} F(P_1) \xrightarrow{Fd_0} F(P_0) \to 0.$$

And then we define F_n to be the nth homology modules of this sequence:

$$F_n(A) = H_n(F(P_\bullet)) = \frac{\ker Fd_n}{\mathrm{im} Fd_{n+1}}.$$

It can be shown that any map $f\colon A \to B$ there is a unique lift of f to a sequence of maps between projective resolutions of A and B (unique up to homotopy), and we define $F_n(f)$ to be the induced map on the homology modules.

It can be shown that this construction is not dependent on the choice of projective resolution: all projective resolutions give naturally isomorphic functors. Furthermore, this is a δ-functor. And it is not hard to see that $F_0 \cong F$.

Now notice that if P is a projective object (whatever that means), certainly $0 \to P \to P \to 0$ is a projective resolution (the non-trivial map is the identity). Thus $F_\bullet$ are the homology modules of the chain complex $0 \to FP \to 0$, which vanish for $n \geq 1$. In particular, as we said that free modules are projective, if N is a free A-module then $\mathrm{Tor}_n^A(N, M) = 0$ for all $M \in \mathsf{Mod}_A$.

Now we call an object P **F-acylic** if $F_n P = 0$ for all $n \geq 1$, so clearly projectives are F-acylic for any F. A simple result is that derived functors can be computed using F-acylic resolutions. That is, if $P_\bullet \to A \to 0$ is a resolution of A using F-acylic objects, then $F_n(A) \cong H_n(F(P))$. Of course to know what objects are F-acylic we need to compute F_n via usual projective resolutions.

But in our case, if $N \in \mathsf{Mod}_A$ is flat, then it is $\bullet \otimes_A M$-acylic for all $M \in \mathsf{Mod}_A$ and vice versa.

4.3.3 Proposition

Let $N \in \mathsf{Mod}_A$, then the following are equivalent:

(1) N is flat,

(2) $\mathrm{Tor}_n^A(N, M) = 0$ for all $M \in \mathsf{Mod}_A$ and all $n \geq 1$,

(3) $\mathrm{Tor}_1^A(N, M) = 0$ for all $M \in \mathsf{Mod}_A$

In particular free modules are flat.

Proof

Consider a projective resolution of M, $P_\bullet \to M \to 0$. Then $\mathrm{Tor}_n^A(N, M)$ are the homology modules of the chain complex

$$\cdots \to P_1 \otimes_A N \to P_0 \otimes_A N \to 0.$$

But N is flat, and thus tensoring with it is exact so it preserves exact sequences. So the above chain complex is exact at every index $n \geq 1$ (at $n = 0$ we have taken away $M \otimes_A N$ so it is no longer exact, this is because we want $\mathrm{Tor}_0^A \cong \bullet \otimes_A \bullet$). Thus the homology modules are all trivial, so $\mathrm{Tor}_n^A(N, M) = 0$ for all $n \geq 1$; so we have shown that (1) implies (2).

And (2) clearly implies (3).

Now suppose (3) then let $0 \to X \to Y \to Z \to 0$ be an SES. Then by δ-functorality there is a long exact sequence

$$\cdots \to \mathrm{Tor}_1^A(N, Z) \to N \otimes_A X \to N \otimes_A Y \to N \otimes_A Z \to 0.$$

Since $\mathrm{Tor}_1^A(N, Z) = 0$ by assumption, this shows that

$$0 \to N \otimes_A X \to N \otimes_A Y \to N \otimes_A Z \to 0$$

is exact, and thus N is flat.
And as explained, $\mathrm{Tor}_n^A(P,\bullet) = 0$ for any projective, and thus free, module P. So by the equivalence, free modules are flat. ∎

Notice that free modules are flat for a simpler reason, which we show in the following lemma.

4.3.4 Lemma
The direct sums and filtered colimits of flat modules are flat.

Proof

Since the tensor product is a left-adjoint it commutes with (preserves) colimits. That is,

$$\bullet \otimes_A \bigoplus_i M_i \cong \bigoplus_i \bullet \otimes_A M_i, \qquad \bullet \otimes_A \varinjlim M_i \cong \varinjlim(\bullet \otimes_A M_i).$$

Since direct sums and filtered colimits are exact (Grothendieck's axioms (AB4) and (AB5)), if each M_i is flat and thus $\bullet \otimes_A M_i$ are all exact, then the above are exact as well. ∎

Since A is a flat A-module (since $\bullet \otimes_A A$ is isomorphic to the identity functor), direct sums of A are flat. Direct sums of A are just free modules.

4.3.5 Lemma
Homology commutes with exact functors: if $C_\bullet$ is a chain complex and F an exact functor, then

$$H_n(F(C_\bullet)) \cong F(H_n(C_\bullet)) \text{ for all } n.$$

Proof

Notice that a quotient M/N is the cokernel of the inclusion $N \hookrightarrow M$, and the image of f is just $\mathrm{im} f = \ker \mathrm{coker} f$. Since F is exact, it preserves injections and cokernels, so $F(N)$ remains a subobject of $F(M)$ and $F(M)/F(N) \cong F(M/N)$ since F preserves cokernels. And it preserves images as well since $F \mathrm{im} f = F \ker \mathrm{coker} f \cong \ker \mathrm{coker} F f = \mathrm{im} F f$.
In particular,

$$F(H_n(C_\bullet)) = F(\ker f_n / \mathrm{im} f_{n+1}) \cong (F \ker f_n)/(F \mathrm{im} f_{n+1}) \cong (\ker F f_n)/(\mathrm{im} F f_{n+1}) = H_n(F(C_\bullet))$$

as desired. ∎

4.3.6 Theorem
The Tor functors commute with direct sums and filtered colimits:

$$\mathrm{Tor}_n^A\left(N, \bigoplus_i M_i\right) \cong \bigoplus_i \mathrm{Tor}_n^A(N, M_i), \qquad \mathrm{Tor}_n^A(N, \varinjlim M_i) \cong \varinjlim \mathrm{Tor}_n^A(N, M_i).$$

Proof

We explained that Tor can be computed via acylic resolutions, and that flat modules are acylic. So if

$$P^i_\bullet \to M_i \to 0$$

are flat resolutions of M_i for each i, then by exactness of direct sums and filtered colimits,

$$\bigoplus_i P^i_\bullet \to \bigoplus_i M_i \to 0, \qquad \varinjlim P^i_\bullet \to \varinjlim M_i \to 0$$

are exact sequences. Furthermore we just showed that these are flat, and thus Tor can be computed using their homology. And again since direct sums and filtered colimits are exact, homology commutes with them. Putting everything together,

$$\begin{aligned}\mathrm{Tor}_n^A\left(N, \bigoplus_i M_i\right) = H_n\left(\bigoplus_i P^i_\bullet\right) = \bigoplus_i H_n(P^i_\bullet) = \bigoplus_i \mathrm{Tor}_n^A(M_i),\\ \mathrm{Tor}_n^A(N, \varinjlim M_i) = H_n\left(\varinjlim P^i_\bullet\right) = \varinjlim H_n(P^i_\bullet) = \varinjlim \mathrm{Tor}_n^A(M_i).\end{aligned}$$

■

Now that we have the basic properties of the Tor functor down, we will use it to prove properties about flatness.

4.3.7 Lemma

An A-module M is flat iff $\mathrm{Tor}_1^A(M, A/I) = 0$ for all ideals $I \leq A$.

Proof

The "only if" case is clear, as the first torsion module of a flat module with any other module is zero. Conversely, we need to show that $\mathrm{Tor}_1^A(M, N) = 0$ for any $N \in \mathsf{Mod}_A$.

First suppose that N is fg, and we induct on the number of generators $a_1, \dots, a_n$. For $n = 0$ it is clear since then $N = 0$. And for $n = 1$ we have that $N = (a_1) \cong A/\mathrm{ann}(a_1)$ and thus the result follows from the assumption. Now inductively consider the SES

$$0 \to (a_n) \to N \to N/(a_n) \to 0.$$

Thus we have an exact sequence

$$\cdots \to \mathrm{Tor}_1^A(M, (a_n)) \to \mathrm{Tor}_1^A(M, N) \to \mathrm{Tor}_1^A(M, N/(a_n)) \to \cdots.$$

Now $N/(a_n)$ is generated by $n-1$ elements, and thus $\mathrm{Tor}_1^A(M, N/(a_n)) = 0$ by induction, and by our base case $\mathrm{Tor}_1^A(M, (a_n)) = 0$. Thus $\mathrm{Tor}_1^A(M, N) = 0$ by exactness.

And if N is not fg, then $N = \varinjlim N_i$ where N_i ranges over all the fg submodules of N. Since Tor commutes with filtered colimits, we have

$$\mathrm{Tor}_1^A(M, \varinjlim N_i) \cong \varinjlim \mathrm{Tor}_1^A(M, N_i) = \varinjlim 0 = 0,$$

as desired. ■

4.3.8 Theorem

An A-module M is flat iff for every ideal $I \leq A$, the map $I \otimes_A M \to IM$ given by $a \otimes m \mapsto am$ is an isomorphism.

Proof

First note that the given map is induced by an A-bilinear map and is thus well-defined.
If M is flat and $I \leq A$ is an ideal, then we have a short exact sequence

$$0 \to I \to A \to A/I \to 0,$$

where the first map is given by inclusion and the second by projection. By flatness, this induces an SES, recalling that $A \otimes_A M \cong M$,

$$0 \to I \otimes_A M \to M \to M \otimes_A (A/I) \to 0.$$

The first map is then the map $M \otimes_A I \to M$ given by $m \otimes a \mapsto am$, the map we are investigating, which is injective by exactness.
Recall that $M \otimes_A (A/I) \cong M/IM$ by $m \otimes (a + I) \mapsto am + IM$. Thus the second map corresponds to the map $M \to M/IM$ given by projection. By exactness, the image of the first map is the kernel of the second, which is IM. So we have that our map is injective onto its image, IM, and thus is an isomorphism $I \otimes_A M \to IM$ as desired.
Conversely by the above lemma, it is sufficient to show that $\mathrm{Tor}_1^A(M, R/I) = 0$. From the SES $0 \to I \to A \to A/I \to 0$, we have an exact sequence

$$\mathrm{Tor}_1^A(M, I) \to \mathrm{Tor}_1^A(M, A) \to \mathrm{Tor}_1^A(M, A/I) \to M \otimes_A I \to M \to M \otimes_A (A/I) \to 0.$$

Since A is free, $\mathrm{Tor}_1^A(M, A) = 0$. And the map $M \otimes_A I \to M$ is precisely the map $m \otimes a \mapsto am$, which is injective by assumption. So we have an exact sequence

$$0 \to \mathrm{Tor}_1^A(M, A/I) \to M \otimes_A I \xrightarrow{f} M,$$

where f is injective. This requires that $\mathrm{Tor}_1^A(M, A/I) = 0$ as desired. ■

4.3.9 Example

Let $R = \mathbb{C}[x]$ and M be the complex algebra of functions $\mathbb{C} \to \mathbb{C}$. Then we define $R \to M$ by mapping x to the zero map. This turns M into an R-algebra, where $p \in R$ is mapped to $p(0)$.
We claim that M is not a flat R-module. Indeed, let $I = (x)$ so that $IM = 0$. But $I \cong R$ as R-modules (since they are both free R-modules on a single generator), so $I \otimes_R M \cong M \neq 0$. Thus IM and $I \otimes_R M$ are not isomorphic and in particular M is not flat.

While free modules are always flat, the converse is not always true. But it is in an important special case, whose proof is homework.

4.3.10 Proposition

Suppose A is a local ring and M a fg A-module. Then M is free iff it is flat.

4.3.11 Corollary

Let M be an A-module, then the following are equivalent:

(1) M is flat,

(2) $M_{\mathfrak{p}}$ is a free $A_{\mathfrak{p}}$-module for all prime $\mathfrak{p} \leq M$,

(3) $M_{\mathfrak{m}}$ is a free $A_{\mathfrak{m}}$-module for all maximal $\mathfrak{m} \leq M$.

Proof

Flatness is local: M is flat iff $M_\mathfrak{p}$ is a flat $A_\mathfrak{p}$-module for all prime $\mathfrak{p} \leq M$ or equivalently all maximal $\mathfrak{m} \leq M$. And $A_\mathfrak{p}$ is a local ring, so by the previous proposition this is iff $M_\mathfrak{p}$ is free. ■

We can also give a nicer criteria in the case that our ring is a domain.

4.3.12 Lemma

Let A be a local domain with maximal ideal $\mathfrak{m}$, and let $k = A/\mathfrak{m}$ be its residue field and $K = \operatorname{Frac}(A)$ its field of fractions. Let M be a fg A-module, and suppose $\dim_k(k \otimes_A M) = \dim_K(K \otimes_A M) = n$. Then M is free of rank n.

Proof

Nakayama's lemma implies that there is a generating set $x_1, \ldots, x_n \in M$ such that their projections form a basis in $k \otimes_A M \cong M/\mathfrak{m}M$ (the final formulation of Nakayama's lemma). Define $\varphi\colon A^n \to M$ which maps the standard basis vectors e_i to x_i. This gives an SES

$$0 \to \ker\varphi \to A^n \to M \to 0.$$

Recall that localization of flat modules are flat, and since K is the localization of A by $A - 0$, it is a flat A-module. Thus tensoring with K preserves the SES,

$$0 \to K \otimes_A \ker\varphi \to K \otimes_A A^n \to K \otimes_A M \to 0.$$

Recall that tensoring commutes with direct sums, so this is isomorphic to

$$0 \to K \otimes_A \ker\varphi \to K^n \to K \otimes_A M \to 0.$$

By assumption $K \otimes_A M$ has dimension n over K, and since the final map is a surjection it must be an isomorphism by considering dimensions. Thus $K \otimes_A \ker\varphi = 0$, and since localization commutes with tensoring,

$$0 = K \otimes_A \ker\varphi \cong \ker\varphi_{A-0}.$$

So for every $x \in \ker\varphi$ we must have that $x/1 = 0$, i.e. there is an $a \in A - 0$ such that $ax = 0$, so $\ker\varphi$ is a torsion module. But $\ker\varphi \subseteq A^n$ which is not torsion as A is a domain, so $\ker\varphi = 0$ and thus φ is an isomorphism and $M \cong A^n$ as desired. ■

4.3.13 Theorem (Fiber dimension criterion for flatness)

Let A be a domain, and M a fg A-module. Let $S = \operatorname{Spec}(A)$, then M is flat iff the map $S \to \mathbb{N}$ given by $\mathfrak{p} \mapsto \dim_{k(\mathfrak{p})}(k(\mathfrak{p}) \otimes_A M)$ is constant.

Proof

As we showed by locality of flatness, M is flat iff $M_\mathfrak{p}$ is a free $A_\mathfrak{p}$-module for all prime $\mathfrak{p} \leq A$. So first suppose $M_\mathfrak{p}$ is free for all prime $\mathfrak{p} \leq A$, say $M_\mathfrak{p} \cong A_\mathfrak{p}^{n_\mathfrak{p}}$ ($n_\mathfrak{p}$ is finite since M is fg). Then

$$k(\mathfrak{p}) \otimes_A M = (A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}) \otimes_A M \cong M_\mathfrak{p}/\mathfrak{p}M_\mathfrak{p} \cong A_\mathfrak{p}^{n_\mathfrak{p}}/(\mathfrak{p}A_\mathfrak{p}^{n_\mathfrak{p}}) \cong k(\mathfrak{p})^{n_\mathfrak{p}},$$

so $\dim_{k(\mathfrak{p})}(k(\mathfrak{p}) \otimes_A M) = n_\mathfrak{p}$.
Now in general notice that if $\mathfrak{p} \leq \mathfrak{q} \leq A$ are prime, and M is an A-module, then $M_\mathfrak{q}$ remains an A-module and we can localize it by $\mathfrak{p}$. It is easy to see that

$$(M_\mathfrak{q})_\mathfrak{p} \cong M_\mathfrak{p}.$$

Thus,

$$M_{(0)} \cong (M_{\mathfrak{p}})_(0) \cong (A_{\mathfrak{p}}^{n_{\mathfrak{p}}})_{(0)} \cong (A_{\mathfrak{p}})_{(0)}^{n_{\mathfrak{p}}} \cong A_{(0)}^{n_{\mathfrak{p}}} = \mathrm{Frac}(A)^{n_{\mathfrak{p}}}.$$

And so we have that

$$\mathrm{Frac}(A)^{n_{\mathfrak{p}}} \cong M_{(0)} \cong \mathrm{Frac}(A) \otimes_A M.$$

The right-hand side is independent of $\mathfrak{p}$, and thus $n_{\mathfrak{p}}$ must be constant. Thus we see that $\mathfrak{p} \mapsto n_{\mathfrak{p}} = \dim_{k(\mathfrak{p})}(k(\mathfrak{p}) \otimes_A M)$ is constant, as desired.

Conversely suppose the map is constant, let its value be n. In particular we have that, for $K = \mathrm{Frac}(A) = A_{(0)}$,

$$n = \dim_{k(\mathfrak{p})}(k(\mathfrak{p}) \otimes_A M) = \dim_K(K \otimes_A M).$$

We also have that

$$k(\mathfrak{p}) \otimes_A M \cong (k(\mathfrak{p}) \otimes_{A_{\mathfrak{p}}} A_{\mathfrak{p}}) \otimes_A M \cong k(\mathfrak{p}) \otimes_{A_{\mathfrak{p}}} (A_{\mathfrak{p}} \otimes_A M) \cong k(\mathfrak{p}) \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}}.$$

Thus we have

$$\dim_{k(\mathfrak{p})}(k(\mathfrak{p}) \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}}) = \dim_{k(\mathfrak{p})}(k(\mathfrak{p}) \otimes_A M) = \dim_K(K \otimes_A M).$$

Now also we know

$$K \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}} = (A_{\mathfrak{p}})_{(0)} \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}} \cong (A_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}})_{(0)} \cong (M_{\mathfrak{p}})_{(0)} \cong M_{(0)} \cong K \otimes_A M.$$

Thus $\dim_K(K \otimes_A M) = \dim_K(K \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}})$, and so

$$\dim_{k(\mathfrak{p})}(k(\mathfrak{p}) \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}}) = \dim_K(K \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}}).$$

By the previous lemma, $M_{\mathfrak{p}}$ is free. ■

II. Algebraic Geometry

1 Affine Schemes

1.1 Basic definitions

1.1.1 Definition

Let R be a ring, then its **(prime) spectrum** is its set of prime ideals,

$$\operatorname{Spec}(R) = \{\mathfrak{p} \leq R \mid \mathfrak{p} \text{ is prime}\}.$$

Note:

We can similarly define the **maximal spectrum** of a ring to be its set of maximal ideals. This is in general less useful than the prime spectrum, but its definition is analogous:

$$\operatorname{Specm}(R) = \{\mathfrak{m} \leq R \mid \mathfrak{m} \text{ is maximal}\}.$$

We can give $\operatorname{Spec}(R)$ a topological structure. Instead of defining the open sets of a spectrum, we will define its closed sets.

1.1.2 Definition

Let $S \subseteq R$ be a subset, then define its **vanishing set** to be

$$V(S) = \{\mathfrak{p} \in \operatorname{Spec}(R) \mid S \subseteq \mathfrak{p}\}.$$

Notice that clearly we have

$$V(S) = V((S)),$$

so that every vanishing set is a vanishing set of an ideal. Note that $V(I) = V(J)$ does not imply $I = J$: for instance $I = 2\mathbb{Z}$ and $J = 4\mathbb{Z}$ in $R = \mathbb{Z}$; their vanishing sets are both $\{2\mathbb{Z}\}$.

1.1.3 Lemma

A proper ideal $\mathfrak{p} \leq R$ is prime iff $IJ \leq \mathfrak{p}$ implies $I \leq \mathfrak{p}$ or $J \leq \mathfrak{p}$ for all ideals $I, J \leq R$.

Proof

The "if" part is clear: if $ab \in \mathfrak{p}$, then $(a) \cdot (b) = (ab) \in \mathfrak{p}$ so either (a) or (b) is a subset of $\mathfrak{p}$. Conversely, suppose that $\mathfrak{p}$ is a prime ideal and $IJ \leq \mathfrak{p}$ but I, J are both not subsets of $\mathfrak{p}$. Then let $x \in I, y \in J$ be elements not in $\mathfrak{p}$: we then have $xy \in IJ \leq \mathfrak{p}$, a contradiction to $\mathfrak{p}$'s primality. ■

1.1.4 Proposition

The collection of vanishing sets forms the collection of closed sets of a topology on $\operatorname{Spec}(R)$.

Proof

We need to show that $\operatorname{Spec}(R), \emptyset$ are vanishing sets, the arbitrary intersection of vanishing sets is a vanishing set, and the binary union of vanishing sets is a vanishing set. Clearly we have

$$\operatorname{Spec}(R) = V(0), \qquad \emptyset = V(R)$$

so the first point is true.
Now consider a collection $\{S_\alpha\}_\alpha$ of subsets. Then we have:

$$\bigcap_\alpha V(S_\alpha) = V\left(\bigcup_\alpha S_\alpha\right),$$

since $S_\alpha \leq \mathfrak{p}$ for all α iff $\bigcup_\alpha S_\alpha \leq \mathfrak{p}$. In terms of ideals, we have that

$$\bigcap_\alpha V(I_\alpha) = V\left(\left(\bigcup_\alpha I_\alpha\right)\right) = V\left(\sum_\alpha I_\alpha\right).$$

For the final point on binary unions, we can restrict our attention to vanishing sets of ideals. Due to the above lemma we have that

$$V(I) \cup V(J) = V(IJ).$$

Thus the collection of vanishing sets forms a topology of closed sets. ■

1.1.5 Definition

The topology defined above on $\operatorname{Spec}(R)$ is called the **Zariski topology** of $\operatorname{Spec}(R)$.

1.1.6 Example

Consider $R = \mathbb{C}[x]$, a PID. Its prime ideals are the principal ideals generated by irreducible elements and zero, in this case

$$\operatorname{Spec}(\mathbb{C}[x]) = \{(x-\lambda) \mid \lambda \in \mathbb{C}\} \cup \{(0)\}.$$

For a polynomial $f \in \mathbb{C}[x]$, we have that $(x-\lambda) \in V(f)$ iff λ is a root of f. Thus

$$V(f) = \{(x-\lambda) \mid f(\lambda) = 0\}.$$

As we showed above, for $I \leq \mathbb{C}[x]$, we have that

$$V(I) = V\left(\bigcup_{f\in I}\{f\}\right) = \bigcap_{f\in I} V(f) = \{(x-\lambda) \mid \lambda \text{ is a root of every polynomial in } I\}.$$

Thus $\operatorname{Spec}(\mathbb{C}[x])$'s proper closed sets are precisely the finite subsets of $\operatorname{Spec}(\mathbb{C}[x]) - (0)$, i.e. it induces the cofinite topology on this subspace. The ideal (0) is not contained in any proper closed set, so its closure is all of $\operatorname{Spec}(\mathbb{C}[x])$.

1.1.7 Example

Consider $R = \mathbb{Z}$, a PID. Its prime ideals are the principal ideals generated by primes and zero, so

$$\operatorname{Spec}(\mathbb{Z}) = \{(p) \mid p \text{ prime}\} \cup \{(0)\}.$$

For $n \in \mathbb{Z}$, $(p) \in V(n)$ iff p divides n so that

$$V(n) = \{(p) \mid p \text{ is a prime divisor of } n\}.$$

Thus $\operatorname{Spec}(\mathbb{Z})$'s proper closed sets are precisely the finite subsets of $\operatorname{Spec}(\mathbb{Z}) - 0$, so it again induces the cofinite topology on this subspace. As before, (0)'s closure is all of $\operatorname{Spec}(\mathbb{Z})$.

In the two examples above, we saw that the point (0)'s closure is all of $\operatorname{Spec}(R)$. If (0) is a prime ideal, then this is always true: $(0) \in V(I)$ iff $I = (0)$ in which case $V(0) = \operatorname{Spec}(R)$. Thus $\operatorname{Spec}(R)$ is the only closed set containing the zero ideal, and is thus its closure.

1.1.8 Example

Consider the case that $R = k$ is a field. The only prime ideal of k is the zero ideal, so $\operatorname{Spec}(k)$ is a single-point space. There is only one topology on such a space: the trivial and discrete topology coincide.

The map $R \mapsto \operatorname{Spec}(R)$ is functorial.

1.1.9 Definition

Let $\varphi\colon R \to S$ be a ring morphism, then define $\varphi^*\colon \operatorname{Spec}(S) \to \operatorname{Spec}(R)$ by

$$\varphi^*(\mathfrak{q}) = \varphi^{-1}(\mathfrak{q}), \qquad \mathfrak{q} \in \operatorname{Spec}(S).$$

This is well-defined, as we showed before that the preimage of a prime ideal is prime.

1.1.10 Proposition

φ^* is continuous, and functorial.

Proof

To show that φ^* is continuous, we need to show that its preimage on closed sets is closed. So let $I \leq R$ be an ideal, and consider $(\varphi^*)^{-1}V(I)$:

$$\mathfrak{q} \in (\varphi^*)^{-1}V(I) \iff \varphi^{-1}\mathfrak{q} \in V(I) \iff I \leq \varphi^{-1}\mathfrak{q} \iff \varphi(I) \leq \mathfrak{q},$$

so that we have

$$(\varphi^*)^{-1}V(I) = V(\varphi(I)),$$

which is closed as desired. Note that while $\varphi(I) \subseteq S$ may not be an ideal, all vanishing sets are closed. Functorality is clear. ∎

Note:

As $\mathrm{Specm}(R) \subseteq \mathrm{Spec}(R)$, we endow $\mathrm{Specm}(R)$ with the subspace topology. In particular its closed sets are

$$V_m(S) = \{\mathfrak{m} \in \mathrm{Specm}(R) \mid S \subseteq \mathfrak{m}\}.$$

But note that $R \mapsto \mathrm{Specm}(R)$ is not functorial: the above φ^* does not necessarily restrict to maximal ideals.

Let us consider more the Zariski topology. Clearly $V(\bullet)$ is contravariant (order reversing).

1.1.11 Proposition

Let $I, J \leq R$ be ideals, then $V(I) = V(J)$ iff $\sqrt{I} = \sqrt{J}$. In particular $V(I) = V(\sqrt{I})$ and $\mathrm{Spec}(R) = V(\mathrm{nil}(R))$.

Proof

We know that $\sqrt{I}$ is the intersection of all prime ideals containing I, meaning

$$\sqrt{I} = \bigcap_{\mathfrak{p} \in V(I)} \mathfrak{p}.$$

So the first implication is clear: if $V(I) = V(J)$ then $\sqrt{I} = \sqrt{J}$. Now if $\sqrt{I} = \sqrt{J}$, let $\mathfrak{p} \in V(I)$. If $\mathfrak{p} \notin V(J)$ then there is an $f \in J$ not in $\mathfrak{p}$, but then $f \in \sqrt{J}$ and $f \notin \mathfrak{p}$ so $f \notin \sqrt{I}$, a contradiction. ∎

Note:
Let the **Jacobson radical** of an ideal $I \leq R$ be the intersection of all maximal ideals containing I:

$$\mathrm{Jac}_R(I) = \bigcap_{I \leq \mathfrak{m}} \mathfrak{m} = \bigcap_{\mathfrak{m} \in V_m(I)} \mathfrak{m}.$$

We can alter the proof above to show that $V_m(I) = V_m(J)$ iff $\mathrm{Jac}_R(I) = \mathrm{Jac}_R(J)$. Thus $V_m(I) = V_m(\mathrm{Jac}_R(I))$ and $\mathrm{Specm}(R) = V_m(\mathrm{Jac}(R))$, where $\mathrm{Jac}(R) = \mathrm{Jac}_R(R)$ is the usual Jacobson radical of a ring.

1.1.12 Definition

Let $Z \subseteq \mathrm{Spec}(R)$ be a subset. Define its **ideal** to be

$$\mathcal{I}(Z) = \{f \in R \mid Z \subseteq V(f)\}.$$

We can characterize $\mathcal{I}(Z)$ more concretely:

$$\mathcal{I}(Z) = \bigcap_{\mathfrak{p} \in Z} \mathfrak{p}.$$

Indeed, if $f \in \mathcal{I}(Z)$ then $Z \subseteq V(f)$ so for every $\mathfrak{p} \in Z$ we have that $\mathfrak{p} \in V(f)$, meaning $f \in \mathfrak{p}$. Conversely if $f \in \mathfrak{p}$ for every $\mathfrak{p} \in Z$, then $\mathfrak{p} \in V(f)$ for every $\mathfrak{p}$ and so $Z \subseteq V(f)$.
From this characterization it is obvious that $\mathcal{I}(\bullet)$ is also order reversing.

1.1.13 Proposition

(1) Let $Z \subseteq \operatorname{Spec}(R)$ be a subset, then $V(\mathcal{I}(Z)) = \overline{Z}$, Z's closure.

(2) Let $J \leq R$ be an ideal, then $\mathcal{I}(V(J)) = \sqrt{J}$.

Proof

For the first point we need to show that $V(\mathcal{I}(Z))$ is the smallest vanishing set containing Z. It contains Z since if $\mathfrak{p} \in Z$ then $\mathcal{I}(Z) \leq \mathfrak{p}$ by definition (it is the intersection of all such $\mathfrak{p}$), so $\mathfrak{p} \in V(\mathcal{I}(Z))$. Now if $Z \subseteq V(J)$ then for $\mathfrak{p} \in Z$ we have that $J \leq \mathfrak{p}$, so that $J \subseteq \bigcap_{\mathfrak{p} \in Z} \mathfrak{p} = \mathcal{I}(Z)$. Since V is contravariant, we have that $V(\mathcal{I}(Z)) \subseteq V(J)$, as desired.

For the second point, we have

$$\mathcal{I}(V(J)) = \bigcap_{\mathfrak{p} \in V(J)} \mathfrak{p} = \bigcap_{J \leq \mathfrak{p}} \mathfrak{p} = \sqrt{J},$$

as desired. ■

1.1.14 Corollary

Let $\mathfrak{p} \in \operatorname{Spec}(R)$ be a point, then its closure is $\overline{\{\mathfrak{p}\}} = V(\mathfrak{p})$. In particular $\{\mathfrak{p}\}$ is closed iff $\mathfrak{p}$ is maximal.

Proof

This is because $\mathcal{I}\{\mathfrak{p}\} = \mathfrak{p}$, so apply the above proposition. And $V(\mathfrak{p}) = \{\mathfrak{p}\}$ iff $\mathfrak{p}$ is maximal. ■

1.1.15 Example

This shows that $\operatorname{Spec}(R)$ need not be T_1. Recall that a space X is T_1 if for every two distinct $x \neq y \in X$, there is an open set U such that $x \in U$ and $y \notin U$. Equivalently a space is T_1 iff its singletons are closed: if singletons are closed let $U = \{y\}^c$, and conversely if for every $y \neq x$ we choose a U_y such that $y \in U_y$ and $x \notin U_y$ we see that

$$X - \{x\} = \bigcup_{y \neq x} U_y$$

is open and thus $\{x\}$ is closed.

In any case, clearly if we have prime ideals which are not maximal, then $\operatorname{Spec}(R)$ is not T_1. For instance, take $X = \operatorname{Spec}(\mathbb{Z})$. Since $(0) \in X$ is not maximal, it is not closed and thus the space cannot be T_1.

$\operatorname{Spec}(R)$ *is* T_0 though: all distinct points are topologically distinct. Indeed, if $\mathfrak{p}, \mathfrak{q} \in \operatorname{Spec}(R)$ have the same neighborhoods they have the same closure and thus $V(\mathfrak{p}) = V(\mathfrak{q})$. Thus $\mathfrak{p} \in V(\mathfrak{q})$ and $\mathfrak{q} \in V(\mathfrak{p})$ so that $\mathfrak{p} \leq \mathfrak{q} \leq \mathfrak{p}$ so that they are equal.

1.1.16 Definition

Let P, Q be two posets. An **antitone Galois connection** are two order-reversing functions $F \colon P \to Q$ and $G \colon Q \to P$ such that for every $p \in P, q \in Q$,

$$q \leq Fp \iff p \leq Gq.$$

An alternative method of looking at antitone Galois connections is via category theory. Recall that we can view posets as categories, where there exists a unique morphism $a \to b$ iff $a \leq b$.

1.1.17 Proposition

An antitone Galois connection between two posets P, Q consists of two functors $F\colon P \to Q^{\mathsf{op}}\colon G$ such that $F \dashv G$ forms an adjunction.

Proof

Order reversing functions between posets are just contravariant functors. So if F, G form an antitone Galois connection, then they are functors $F\colon P^{\mathsf{op}} \to Q$ and $G\colon Q^{\mathsf{op}} \to P$. Then $F^{\mathsf{op}}\colon P \to Q^{\mathsf{op}}\colon G$ forms an adjunction: we have $q \leq Fp$ iff $F^{\mathsf{op}}p \leq^{\mathsf{op}} q$ iff $p \leq Gq$. The converse is shown similarly. ∎

1.1.18 Corollary

$F\colon P \to Q\colon G$ forms an antitone Galois connection iff they are order reversing and $p \leq GFp$ and $q \leq FGq$ for all $p \in P, q \in Q$.

Proof

This is just the unit-counit formulation of an adjunction (remember contravariance!). A natural transformation between functors X, Y between posets amounts to saying $X(\bullet) \leq Y(\bullet)$. ∎

1.1.19 Corollary

The two functions $V\colon \mathcal{P}(R) \to \mathcal{P}(\mathrm{Spec}(R))\colon \mathcal{I}$ form an antitone Galois connection.

Proof

The maps are order reversing and $S \subseteq \sqrt{(S)} = \mathcal{I}(V(S))$ and $Z \subseteq \overline{Z} = V(\mathcal{I}(Z))$, so they form an antitone Galois connection. Showing directly that $Z \subseteq V(S)$ iff $S \subseteq \mathcal{I}(Z)$ is easy to do as well. ∎

1.2 Topological properties of affine schemes

1.2.1 Definition

Let R be a ring, $X = \mathrm{Spec}(R)$ its spectrum. For $f \in R$, define

$$X_f = X - V(f) = \{\mathfrak{p} \in \mathrm{Spec}(R) \mid f \notin \mathfrak{p}\}.$$

1.2.2 Proposition

The collection $\{X_f\}_{f \in R}$ forms a basis of open sets for $X = \mathrm{Spec}(R)$.

Proof

This is equivalent to showing that $\{V(f)\}_{f \in R}$ forms a basis of closed sets; that they are closed and every closed set is an intersection of closed basis sets. Clearly $V(f)$ are closed, and for $S \subseteq R$ we have

$$V(S) = V\left(\bigcup_{f\in S}\{f\}\right) = \bigcap_{f\in S} V(f)$$

as desired. ∎

This shows that for an open set $X - V(S)$, we have

$$X - V(S) = \bigcup_{f\in S} X_f.$$

1.2.3 Proposition

Let R be a ring and $X = \operatorname{Spec}(R)$.

(1) Let $\{f_\alpha\}_\alpha$ be a collection of elements from R. Then $X = \bigcup_\alpha X_{f_\alpha}$ iff $(\{f_\alpha\}_\alpha) = R$.

(2) Let $f \in R$, then $X_f = X$ iff $f \in R^\times$.

(3) Let $f \in R$, then $X_f = \emptyset$ iff $f \in \operatorname{nil}(R)$.

Proof

We saw that

$$\bigcup_\alpha X_{f_\alpha} = X - V(\{f_\alpha\}_\alpha)$$

which is equal to X iff $V(\{f_\alpha\}_\alpha) = \emptyset$. This is iff

$$V(\{f_\alpha\}_\alpha) = V((\{f_\alpha\}_\alpha)) = V(R) \iff \sqrt{(\{f_\alpha\}_\alpha} = R.$$

Now $\sqrt{J} = R$ iff $J = R$, as $1 \in \sqrt{J}$ iff $1 \in J$. So this is iff $(\{f_\alpha\}_\alpha) = R$, as desired.
The second point follows from the first: $X_f = X$ iff $(f) = R$ iff $f \in R^\times$.
For the third point, $X_f = \emptyset$ iff $V(f) = X = V(0)$ iff $\sqrt{(f)} = \operatorname{nil}(R)$. This occurs iff $f \in \operatorname{nil}(R)$: clearly it implies this, and conversely $0 \leq (f) \leq \operatorname{nil}(R)$ and radicals are order-preserving so we get $\sqrt{(f)} = \operatorname{nil}(R)$. ∎

1.2.4 Corollary

$X = \operatorname{Spec}(R)$ is compact.

Proof

We need to show every open cover of X has a finite subcover. Let $\mathcal{U}$ be an open cover of X, then we can assume that $\mathcal{U}$ consists of basic open sets: $\mathcal{U} = \{X_{f_\alpha}\}_\alpha$. By the first point above, since $\mathcal{U}$ is a cover, $(\{f_\alpha\}_\alpha) = R$. Thus $1 \in (\{f_\alpha\}_\alpha)$ so there are finitely many $f_1, \dots, f_n$ such that $1 = x_1 f_1 + \cdots + x_n f_n$. Then $(f_1, \dots, f_n) = R$ and so again by the above proposition, $\{X_{f_1}, \dots, X_{f_n}\}$ is a finite subcover of $\mathcal{U}$, as desired. ∎

Note:
This result shows that the usual topological concept of compactness is in general not strong enough for algebraic geometry. Thus in algebraic geometry topological compactness is usually called **quasicompactness**, and compactness is reserved for a stronger concept.

1.2.5 Proposition

Let $f \in R$, and consider R_f, the localization of R at $\{f^n\}_{n<\omega}$. $\mathrm{Spec}(R_f)$ is homeomorphic to X_f.

Proof

Let $\iota\colon R \to R_f$ be the structure map. Then $\mathfrak{q} \mapsto \iota^{-1}\mathfrak{q}$ is a bijective correspondence between prime ideals of R_f and prime ideals of R which do not intersect $\{f^n\}$. A prime ideal $\mathfrak{p} \leq R$ does not intersect $\{f^n\}$ iff $f \notin \mathfrak{p}$. Thus $\iota^*\colon Y = \mathrm{Spec}(R_f) \to \mathrm{Spec}(R)$ is injective and its image are all prime ideals such that $f \notin \mathfrak{p}$, i.e. $\mathrm{im}\iota^* = X_f$.

We claim that ι^* is a homeomorphism onto its image. We already know it is continuous and bijective, so we need only show it is open; we will show it sends basic open sets to basic open sets. So let $g/f^n \in R_f$, with the basic open set

$$Y_{g/f^n} = \{\mathfrak{q} \in Y \mid g/f^n \notin \mathfrak{q}\}\,.$$

We claim that $\iota^*(Y_{g/f^n}) = X_g \cap X_f$. Indeed, if $\mathfrak{q} \in Y_{g/f^n}$ then $g/f^n \notin \mathfrak{q}$ and so $g \notin \iota^*\mathfrak{q}$ (since g/f^n is $\iota(g)$ times a unit, so if $g \in \iota^*\mathfrak{q}$ then $\iota(g) \in \mathfrak{q}$ and then $g/f^n \in \mathfrak{q}$); thus $\iota^*\mathfrak{q} \in X_g$ as desired. Conversely if $\mathfrak{p} \in X_f \cap X_g$ then $g/f^n \notin \mathfrak{p}R_f$: otherwise $g/f^n = h/f^m$ for $h \in \mathfrak{p}$ and thus $f^k(f^m g - f^n h) = 0$ so that $f^{m+k}g \in \mathfrak{p}$, but $f, g \notin \mathfrak{p}$ a contradiction. Now, $\mathfrak{p}R_f$ is prime and $\iota^*\mathfrak{p}R_f = \mathfrak{p}$, showing the equality. ∎

We now focus on the connectedness of topological spaces, in particular spectra of rings. Recall that a topological space is **disconnected** if it can be written as the disjoint union of two open subsets, equivalently if it has a non-trivial clopen subset.

1.2.6 Lemma

A surjective ring morphism $\varphi\colon R \to S$ induces a closed embedding $\varphi^*\colon \mathrm{Spec}(S) \to \mathrm{Spec}(R)$ (i.e. its image is closed, and it is a topological embedding). In particular if $I \leq R$ is an ideal, then the projection $\pi\colon R \to R/I$ induces a closed embedding $\pi^*\colon \mathrm{Spec}(R/I) \to \mathrm{Spec}(R)$ whose image is $V(I)$.

Proof

φ induces an isomorphism of rings $R/\ker\varphi \to S$, which carries prime ideals of $R/\ker\varphi$ bijectively to prime ideals of S. Thus φ^*'s image is the set of all prime ideals containing $\ker\varphi$, i.e. $\mathrm{im}\varphi^* = V(\ker\varphi)$ which is closed by definition. Since φ is surjective, φ^* is injective and thus a bijection onto its image. So to show it is a topological embedding, we need only show that it is a closed map.

Indeed, for $J \leq S$,

$$\varphi^*V(J) = \left\{\varphi^{-1}\mathfrak{q} \;\middle|\; J \leq \mathfrak{q}\right\} = \left\{\varphi^{-1}\mathfrak{q} \;\middle|\; \varphi^{-1}J \leq \varphi^{-1}\mathfrak{q}\right\} = V(\varphi^{-1}J) \cap \mathrm{im}\varphi^*,$$

which is closed. (The second equality is due to φ's surjectivity.) ∎

1.2.7 Proposition

If $R = S \times T$ is a product ring, then $\mathrm{Spec}(R)$ is homeomorphic to $\mathrm{Spec}(S) \sqcup \mathrm{Spec}(T)$, and thus disconnected.

Conversely if $\mathrm{Spec}(R) = Y \sqcup Z$ is disconnected, then there exist rings S, T such that Y, Z are homeomorphic to $\mathrm{Spec}(S), \mathrm{Spec}(T)$ respectively, and $R \cong S \times T$.

Proof

Let $X = \mathrm{Spec}(R)$. Consider $(1,0),(0,1) \in R$, so that R's unit is their sum. Then we showed above that $X = X_{(1,0)} \cup X_{(0,1)}$, and this is a disjoint union since $V(0,1) \cup V(1,0) = X$ (if $\mathfrak{p} \in X$ and $(0,1),(1,0) \notin \mathfrak{p}$ then their product 0 is not in $\mathfrak{p}$, a contradiction). We also showed that $X_{(1,0)} \cong \mathrm{Spec}(R_{(1,0)})$.
So let $\pi\colon R \to S$ be the projection, so $\pi(1,0) = 1 \in S^\times$ so that by the universal property of localization π induces a map $\pi'\colon R_{(1,0)} \to S$ which is surjective since π is. Now if $\pi'((a,b)/(1,0)) = \pi'((x,y)/(1,0))$ then $a = x$ so $(a,b)(1,0) = (x,y)(1,0)$ and thus $(a,b)/(1,0) = (x,y)/(1,0)$ showing that π' is also injective. Thus $S \cong R_{(1,0)}$, and similarly $T \cong R_{(0,1)}$ so that we have desired result.
Since Y, Z are closed, there are $I, J \leq R$ such that $Y = V(I)$ and $Z = V(J)$. Then let $S = R/I$ and $T = R/J$; we showed above that $\pi_1\colon R \to R/I$ induces a homeomorphism $\pi_1^*\colon \mathrm{Spec}(R/I) \to V(I) \subseteq \mathrm{Spec}(R)$ and similar for R/J. Now, $Y \cap Z = V(I) \cap V(J) = V(I+J)$ is empty so that $I + J = R$, and thus I, J are coprime. By the chinese remainder theorem,

$$R \cong R/I \times R/J = S \times T,$$

as desired. ∎

1.2.8 Definition

An element $f \in R$ is **idempotent** if $f^2 = f$. The **trivial idempotents** of R are 0 and 1.

If $f \in R$ is idempotent, then fR is a ring. Indeed, it is closed under addition and multiplication (it is the ideal generated by f), and f is its multiplicative identity since it is idempotent.

1.2.9 Proposition

$R \cong S \times T$ for non-trivial rings S, T iff R has nontrivial idempotents.

Proof

If $R \cong S \times T$ then $(1,0),(0,1)$ are nontrivial idempotents. Conversely let $f \in R$ be a nontrivial idempotent, then define $S = fR$ and $T = (1-f)R$. Then $R \to S \times T$ given by $g \mapsto (fg, (1-f)g)$ is clearly an injective since if $fg = (1-f)g = 0$ then $g = 0$, and it is surjective since for $(fg, (1-f)h) \in S \times T$ then we want a $x \in R$ such that $fg = fx$ and $(1-f)h = (1-f)x$, so $(1-f)h = x - fg$ and thus $x = (1-f)h + fg$ gives the desired element. They are also non-trivial since $f \neq 0$. ∎

1.2.10 Definition

A ring R is **connected** if it has no nontrivial idempotents.

1.2.11 Corollary

$\mathrm{Spec}(R)$ is connected iff R is.

1.2.12 Definition

A topological space X is **reducible** if there exist two proper closed subsets $Z, W \subsetneq X$ such that $X = Z \cup W$. Otherwise X is **irreducible**.

Note that irreducibility implies connectedness; if X is disconnected then its union $Z \sqcup W$ of open sets are also closed. The converse is not true in general (take the axes of $\mathbb{R}^2$).

1.2.13 Proposition

$X = \operatorname{Spec}(R)$ is irreducible iff $\operatorname{nil}(R)$ is prime.

Proof

If $\operatorname{nil}(R)$ is not prime, then there are $f, g \notin \operatorname{nil}(R)$ such that $fg \in \operatorname{nil}(R)$. Let $Z = V(f)$ and $W = V(g)$, so that

$$Z \cup W = V(f) \cup V(g) = V(fg) = X,$$

where the second equality is because $V(f) \cup V(g) = V((f)(g)) = V(fg)$, and the final equality is because $V(fg) \supseteq V(\operatorname{nil}(R)) = X$.

Conversely if X is reducible, then $X = V(I) \cup V(J)$ so that $V(IJ) = X$ for $V(I), V(J) \neq X$. But then $IJ \leq \operatorname{nil}(R)$ and I and J cannot be contained in $\operatorname{nil}(R)$, so $\operatorname{nil}(R)$ is not prime. ■

1.2.14 Corollary

The spectrum of a domain is irreducible.

1.2.15 Example

The affine space $\mathbb{A}^n_{\mathbb{C}} = \operatorname{Spec}(\mathbb{C}[x_1, \ldots, x_n])$ is irreducible since $\mathbb{C}[x_1, \ldots, x_n]$ is a domain.

Similarly the spectrum of $R = \mathbb{C}[x_1, \ldots, x_n]/(f)$ is irreducible when f is irreducible (and then (f) is prime and R is a domain). But the converse does not hold: let $R = \mathbb{C}[x]/(x^2)$, x^2 is reducible, and $\operatorname{nil}(R) = \sqrt{(x^2)} = (x)$ is prime.

1.2.16 Definition

A topological space X is **Noetherian** if its poset of closed subsets has the DCC.

1.2.17 Proposition

The spectrum of a Noetherian ring is a Noetherian space.

Proof

Let R be Noetherian, then a descending chain of closed subsets of $\operatorname{Spec}(R)$ corresponds (via $\mathcal{I}$) to an ascending chain of radical ideals of R. Since R is Noetherian, these must stabilize, and thus so too must the chain of closed subsets (since $V \circ \mathcal{I}$ is the identity on closed sets). ■

Note that $\operatorname{Spec}(R)$ being Noetherian does not imply that R is.

1.2.18 Lemma (Noetherian induction)

Let Σ be a poset with DCC, and P be a boolean function $\Sigma \to \{0, 1\}$ (think of it as a property of elements of Σ). Suppose further that

(1) if $p \in \Sigma$ is minimal then $P(p) = 1$,

(2) if $p \in \Sigma$ is not minimal and $P(q) = 1$ for all $q < p$, then $P(p) = 1$.

Then P is identically 1 on Σ (so the property holds for all elements of Σ).

Proof

If we assume that, Σ' the subset of Σ consisting of elements such that $P(p) = 0$, is nonempty, then since Σ has DCC it has a minimal element $p \in \Sigma'$. It cannot be minimal in Σ since then $P(p) = 1$ by condition (1). But for every $q < p$ since $q \notin \Sigma'$ (by minimality) we have $P(q) = 1$ and thus by (2) we have $P(p) = 1$ a contradiction. ■

1.2.19 Proposition

Let X be a Noetherian space, then X has finitely many irreducible components.

Proof

We use Noetherian induction on the closed subsets of X to show that they have finitely many irreducible components. Since X itself is closed, the result follows.

If $Y \subseteq X$ is a minimal closed subset then it has no proper closed subsets and is in particular irreducible (has a single irreducible component).

Now suppose $Y \subseteq X$ is non-minimal closed and every proper closed subset $Z \subsetneq Y$ has finitely many irreducible components. If Y itself is irreducible, we are done. Otherwise suppose $Y = Z \cup W$ for Z, W proper closed subsets. Let $C_1, \ldots, C_n$ be the irreducible components of Z and $D_1, \ldots, D_m$ those of W. Let $T \subseteq Y$ be a an irreducible component, so

$$T = Y \cap T = (Z \cup W) \cap T = \left(\bigcup_i C_i \cap T \right) \cup \left(\bigcup_i D_i \cap T \right).$$

Since each intersection is closed and T is irreducible, we must have that $T \subseteq C_i$ or $T \subseteq D_i$ for some i. Since T is a component and is thus maximal, we must have equality. So the irreducible components of Y are contained in $C_1, \ldots, C_n, D_1, \ldots, D_m$, and thus there are finitely many. ■

1.2.20 Lemma

The following are equivalent:

(1) X is Noetherian,

(2) Every subspace of Y is Noetherian,

(3) Every subspace of X is quasi-compact,

(4) Every open subspace of X is quasi-compact.

Proof

For a subspace $Y \subseteq X$, its closed sets are of the form $Y \cap F$ for $F \subseteq X$ closed. So let Σ be a set of closed subsets of Y, then let $\Sigma' = \{F \subseteq X \text{ closed} \mid F \cap Y \in \Sigma\}$. Then Σ' must have a minimal element F, then clearly $F \cap Y \in \Sigma$ is minimal, so Y is Noetherian. So (1) and (2) are equivalent.

To show that (1) implies (3), it is sufficient to show that X is quasi-compact, and the general case follows from (2). Being quasicompact is equivalent to a collection of closed subsets having the finite intersection property has nonempty intersection. So suppose $\{F_i\}_i \subseteq X$ has the finite intersection property. Let Σ be the set of finite intersections of F_i, so a poset of closed subsets. Since X is Noetherian, let $F \in \Sigma$ be minimal, and since F_i have the finite intersection property, $F \neq \emptyset$. Then clearly the intersection of all F_i is equal to $F \neq \emptyset$ as desired.

To show (4) implies (1), let $U_1 \subseteq U_2 \subseteq \cdots$ be increasing open subsets in X. Then $U = \bigcup_i U_i$ is open in X and thus quasi-compact, so there is a finite cover $U = \bigcup_{i=1}^n U_i$, but then $U = U_n$. So the chain terminates at n, as desired. ■

1.3 Categorical constructions

In this section we consider the category of affine schemes, i.e. spectrums of rings. A morphism of affine schemes $\operatorname{Spec}(R) \to \operatorname{Spec}(S)$ corresponds to a ring morphism $S \to R$.

Recall the following case of a categorical limit.

1.3.1 Definition

A **pullback** (or **fiber product**) is the limit of a diagram of shape $\bullet \to \bullet \leftarrow \bullet$ (if one exists). Explicitly, given two morphisms $A \xrightarrow{f} C \xleftarrow{g} B$, their fiber product is an object $A \times_C B$ along with morphisms $A \xleftarrow{i} A \times_C B \xrightarrow{j} B$ such that the following diagram commutes and is universal (in the sense that it is the terminal such diagram which commutes):

$$\begin{tikzcd} A \times_C B \arrow[r, "i"'] \arrow[d, "j"'] & A \arrow[d, "f"'] \\ B \arrow[r, "g"] & C \end{tikzcd}$$

Finite products and equalizers can be written as pullbacks with a terminal object. In particular if 1 is a terminal object, then the pullback of $A \xrightarrow{!} 1 \xleftarrow{!} B$, where ! denotes the unique morphism, is the product of A and B. And if $f, g \colon A \rightrightarrows B$ are parallel morphisms, the pullback of $A \xrightarrow{(f,g)} B \times B \xleftarrow{\Delta} B$, where Δ is the diagonal morphism, is the equalizer of f, g.

Thus any category with pullbacks and a terminal object has all finite limits. We use this to demonstrate now that the category of affine schemes has all finite limits.

1.3.2 Proposition

Let $X = \operatorname{Spec}(A), Y = \operatorname{Spec}(B), Z = \operatorname{Spec}(R)$ be affine schemes, along with morphisms $X \to Z \leftarrow Y$, then this diagram has a pullback. Explicitly, its pullback is $X \times_Z Y = \operatorname{Spec}(A \otimes_R B)$.

Proof

The morphisms $X \to Z \leftarrow Y$ are induced by morphisms $A \leftarrow R \to B$, so A, B are R-algebras. Thus $A \otimes_R B$ is an R-algebra as well, and the obvious maps $A, B \to A \otimes_R B$ given by $a \mapsto a \otimes 1, b \mapsto 1 \otimes b$ give us the cone $X \times_Z Y \to X, Y$. We need to show that this is indeed a limit cone, i.e. that it is terminal.

We need to show that if $W = \operatorname{Spec}(C)$ is any other affine scheme with morphisms $W \to X, Y$ commuting with $X \to Z, Y \to Z$ then there is a unique morphism fro W to $X \times_Z Y = \operatorname{Spec}(A \otimes_R B)$. Translating this to the category of rings: if there are morphisms $A \xrightarrow{\varphi} C \xleftarrow{\psi} B$ which commute with $R \to A, B$, then there is a unique morphism $A \otimes_R B \to C$ making the diagram commute

$$\begin{tikzcd}
C & & \\
& A \otimes_R B \arrow[ul, dashed, "\exists!"] & A \arrow[l] \arrow[ull, bend right, "\varphi"'] \\
& B \arrow[u] \arrow[uul, bend left, "\psi"] & R \arrow[l] \arrow[u]
\end{tikzcd}$$

Such a map must map $a \otimes b$ to $\varphi(a) \otimes \psi(b)$. This map is R-bilinear (because φ, ψ commute with the maps $R \to A, B$), and thus is well-defined, as desired. ■

1.3.3 Corollary

The category of affine schemes has all finite limits.

Proof

The ring $\mathbb{Z}$ is initial: there is a unique ring morphism $\mathbb{Z} \to A$ for all rings A. Thus its spectrum is terminal, so the category of affine schemes has a terminal object. The previous proposition showed the category has all pullbacks, so it has all finite limits. ■

2 Varieties

2.1 Affine varieties

In this section let k be an algebraically closed field, $\operatorname{Fun}(k^n, k)$ denote the k-algebra of functions $k^n \to k$, and $k[x_1, \ldots, x_n]$ be the k-algebra of polynomials over variables $x_1, \ldots, x_n$.
Note that there is a natural k-algebra morphism

$$\operatorname{ev}\colon k[x_1, \ldots, x_n] \to \operatorname{Fun}(k^n, k)$$

defined by $\operatorname{ev}(f)(P) = p(a_1, \ldots, a_n)$ for a polynomial f and $P = (a_1, \ldots, a_n) \in k^n$.
For a subset $S \subseteq k[x_1, \ldots, x_n]$ define its **zero set**

$$Z(S) = \{P \in k^n \mid f(P) = 0 \text{ for all } f \in S\} \subseteq k^n.$$

For $f \in k^n$ we write $Z(f)$ for $Z(\{f\})$ so that $Z(S) = \bigcap_{f \in S} Z(f)$.

2.1.1 Lemma

(1) Z is order reversing: if $S_1 \subseteq S_2$ then $Z(S_1) \supseteq Z(S_2)$.

(2) $Z(S)$ is equal to $Z((S))$ ((S) is the ideal generated by S).

(3) For a collection of subsets $\{S_\alpha\}_\alpha$:

$$\bigcap_\alpha Z(S_\alpha) = Z\Big(\bigcup S_\alpha\Big).$$

(4) For two subsets S_1, S_2, define $S_1S_2 = \{fg \mid f \in S_1, g \in S_2\}$, then

$$Z(S_1) \cup Z(S_2) = Z(S_1S_2).$$

(5) $Z(1) = \emptyset$ and $Z(0) = k^n$.

Proof

All of these points are clear. We prove only the fourth: $Z(S_1) \cup Z(S_2) = Z(S_1S_2)$. Note that the ideal generated by S_1S_2 is precisely the product of the ideals generated by S_1 and S_2. Thus $Z(S_1)$ and $Z(S_2)$ are contained in $Z(S_1S_2)$ by order-reversion. Conversely if $P \notin Z(S_1) \cup Z(S_2)$, then $f(P), g(P) \neq 0$ for all $f \in S_1, g \in S_2$ so $(fg)(P) = f(P)g(P) \neq 0$ and thus $P \notin Z(S_1S_2)$. ■

2.1.2 Corollary

The set of zero sets forms a topology of closed sets on k^n.

2.1.3 Definition

Define $\mathbb{A}_k^n$ to be the topological space k^n whose underlying topology is given by the zero sets. This is called the **Zariski topology**, and $\mathbb{A}_k^n$ is the n**-dimensional affine space**. When k is understood, we omit it and just write $\mathbb{A}^n$.
We also define $k[\mathbb{A}^n] = k[x_1, \ldots, x_n]$.

Note that $Z(f)$ forms a basis of closed sets: every closed set is an intersection of these basic closed sets. Thus

$$D(f) = \mathbb{A}^n - Z(f) = \{P \in k^n \mid f(P) \neq 0\}.$$

forms a basis of open sets.
Note that for every $S \subseteq k[x_1, \ldots, x_n]$, (S) is generated by finitely many $f_1, \ldots, f_n \in S$ since $k[x_1, \ldots, x_n]$ is Noetherian. Then

$$Z(S) = Z((S)) = Z(f_1, \ldots, f_n) = \bigcap_{i=1}^{n} Z(f_i).$$

Thus every open set is a finite union of basic open sets. And so $\mathbb{A}^n$ is quasicompact: if $D(f_i)$ cover $\mathbb{A}^n$ then

$$\mathbb{A}^n = \bigcup_i D(f_i) = D(\{f_i\}),$$

and as we showed (for zero sets, thus for open sets) $D(\{f_i\})$ is the finite union of $D(f_i)$s.

2.1.4 Proposition

$\mathbb{A}^n$ is quasicompact.

2.1.5 Example

The closed subsets of $\mathbb{A}^1$ are precisely the finite subsets of $\mathbb{A}^1$, or $\mathbb{A}^1$ itself. Indeed, $k[x]$ is a PID so all of its ideals are of the form $Z(f)$ for a polynomial f. If $f \neq 0$, then f has finitely many roots so $Z(f)$ is finite. And for a finite subset $\{c_1, \ldots, c_n\}$ of $\mathbb{A}^1 = k$, $f = (x - c_1) \cdots (x - c_n)$ has this set as its zero set.

Thus the open subsets of $\mathbb{A}^1$ are empty or cofinite. Since k is infinite (it is algebraically closed), $\mathbb{A}^1$ is not Hausdorff.

2.1.6 Definition

For a subset $Y \subseteq \mathbb{A}^n$, define

$$\mathcal{I}(Y) = \{f \in k[x_1, \ldots, x_n] \mid f(P) = 0 \text{ for all } P \in Y\}.$$

2.1.7 Proposition

(1) $\mathcal{I}$ is order-reversing.

(2) $\mathcal{I}(Y)$ is a radical ideal of $k[x_1, \ldots, x_n]$.

(3) $\mathcal{I}(Y_1 \cup Y_2) = \mathcal{I}(Y_1) \cap \mathcal{I}(Y_2)$.

(4) For every $Y \subseteq \mathbb{A}^n$, $Z(\mathcal{I}(Y)) = \overline{Y}$ is the closure of Y in $\mathbb{A}^n$.

(5) **Nullstellensatz:** for every ideal $J \leq k[x_1, \ldots, x_n]$, $\mathcal{I}(Z(J)) = \sqrt{J}$.

Proof

The first three points are clear. For the fourth, notice $Y \subseteq Z(\mathcal{I}(Y))$ and $Z(\mathcal{I}(Y))$ is closed, so $\overline{Y} \subseteq Z(\mathcal{I}(Y))$. Since $\overline{Y}$ is closed, $\overline{Y} = Z(J)$ for some ideal J. So

$$Y \subseteq \overline{Y} = Z(J) \Longrightarrow \mathcal{I}(\overline{Y}) = \mathcal{I}(Z(J)) \subseteq \mathcal{I}(Y).$$

And $J \subseteq \mathcal{I}(Z(J))$ clearly, so $J \subseteq \mathcal{I}(Y)$, and thus $\overline{Y} = Z(J) \supseteq Z(\mathcal{I}(Y))$ as desired.

For the fifth point, since $J \leq \mathcal{I}(Z(J))$ which is radical, we see $\sqrt{J} \leq \mathcal{I}(Z(J))$. The inverse inclusion is far deeper, and we give a proof next section. ∎

2.1.8 Corollary

Z and $\mathcal{I}$ form an antitone Galois connection.

2.1.9 Corollary

For every proper ideal $J < k[x_1, \ldots, x_n]$, $Z(J)$ is nonempty.

Proof

If $J \neq (1)$ then $\mathcal{I}(Z(J)) = \sqrt{J} \neq (1)$, thus $Z(J)$ cannot be empty (since $\mathcal{I}(\emptyset) = (1)$). ∎

Note:
Recall that a topological space is irreducible iff there are no two proper closed subsets whose union is X. In homework this was shown to be equivalent to the condition that all open subsets intersect.

2.1.10 Example

Every closed subset of $\mathbb{A}^1$ is either $\mathbb{A}^1$ or finite. Since k is infinite, $\mathbb{A}^1$'s irreducible closed subsets are its closed points.

2.1.11 Corollary

A closed subset $Y \subseteq \mathbb{A}^n$ is irreducible iff the ideal $\mathcal{I}(Y) \subseteq k[\mathbb{A}^n]$ is either prime or all $k[\mathbb{A}^n]$.

Proof

If $Y \subseteq \mathbb{A}^n$ is irreducible and $fg \in \mathcal{I}(Y)$, then we want to show either $f \in \mathcal{I}(Y)$ or $g \in \mathcal{I}(Y)$. Notice that since $fg \in \mathcal{I}(Y)$, we have $Y = \overline{Y} = Z(\mathcal{I}(Y)) \subseteq Z(fg) = Z(f) \cup Z(g)$. Since Y is irreducible either $Y \subseteq Z(f)$ or $Y \subseteq Z(g)$, meaning $f \in \mathcal{I}(Y)$ or $g \in \mathcal{I}(Y)$ as desired.

If $\mathcal{I}(Y)$ is prime or all $k[\mathbb{A}^n]$, suppose $Y = Y_1 \cup Y_2$ where $Y_1, Y_2 \subseteq Y$ are closed. Then

$$\mathcal{I}(Y_1)\mathcal{I}(Y_2) \subseteq \mathcal{I}(Y_1) \cap \mathcal{I}(Y_2) = \mathcal{I}(Y).$$

Thus either $\mathcal{I}(Y_1)$ or $\mathcal{I}(Y_2)$ is contained in $\mathcal{I}(Y)$. Suppose $\mathcal{I}(Y_1) \subseteq \mathcal{I}(Y)$, then applying Z gives $Y_1 \supseteq Y$ (since they are closed), so $Y = Y_1$ as desired. ∎

2.1.12 Example

$\mathbb{A}^n$ is irreducible since $\mathcal{I}(\mathbb{A}^n) = (0)$ is prime.

If $f \in k[\mathbb{A}^n]$ is irreducible, then $Z(f)$ is irreducible in $\mathbb{A}^n$. This is because $\mathcal{I}(Z(f)) = \sqrt{(f)} = (f)$ is prime.

Proof of the Nullstellensatz

2.1.13 Lemma

Let $f \in k[x_1, \ldots, x_n]$ be nonconstant and $A = k[x_1, \ldots, x_n]/(f)$. Then there exist integers $k_1, \ldots, k_{n-1} > 0$ such that the k-algebra homomorphism

$$\varphi \colon k[y_1, \ldots, y_{n-1}] \to A, \qquad y_i \mapsto x_i - x_n^{k_i}$$

is injective and finite.

Proof

For an n-tuple $I = (d_1, \ldots, d_n)$ let $x^I = x_1^{d_1} \cdots x_n^{d_n}$ and call I the **multidegree** of x^I. We define k_j for $1 \leq j \leq n$ by descending recursion and setting $k_n = 1$, and letting

$$k_j = 1 + \max_I \left\{ \sum_{i=j+1}^{n} k_i d_i \right\},$$

where $I = (d_1, \dots, d_n)$ ranges over all the multidegrees in f with nonzero coefficient. We claim that k_j satisfy the claim.

Let $y_i = x_i - x_n^{k_i}$ for $1 \le i \le n-1$, and order the multidegrees in f lexicographically. That is, $(d_1, \dots, d_n) > (d'_1, \dots, d'_n)$ iff $d_j > d'_j$ for some index j and $d_i = d'_i$ for all $i < j$. This is a total order, and so f has a maximum multidegree $I_f = (D_1, \dots, D_n)$.

Now define $N = \sum_{i=1}^n k_i D_i$. Then we claim that f is a polynomial in $B[x_n]$ of degree $N > 0$ in x_n, whose leading coefficient is in $k^\times$, where $B = k[y_1, \dots, y_{n-1}] \subseteq k[x_1, \dots, x_n]$.

Let $m = x^I$ be a monomial with $I = (d_1, \dots, d_n)$. Substitute $x_i = y_i + x_n^{k_i}$, which shows that m is a monic polynomial in $B[x_n]$ of x_n-degree $\sum_{i=1}^n k_i d_i$. We claim that under this substitution, the x_n-degree of x^{I_f} is greater than x^I's for any other multidegree I in f. That is,

$$\sum_{i=1}^n k_i D_i > \sum_{i=1}^n k_i d_i$$

for all other multidegrees I. This will then imply the claim regarding f, as it is of the form

$$f = \sum_I a_I x^I = \sum_I a_I (y_I - x_n^{k_I})^I,$$

and considering the combination of monomials.

So let $I = (d_1, \dots, d_n)$ be another multidegree of f. By I_f's maximality there is a j such that $D_j \ge d_j + 1$ and $D_i = d_i$ for $i < j$. Then

$$\sum_{i=1}^n k_i D_i \ge \sum_{i=1}^j k_i D_i \ge \sum_{i=1}^{j-1} k_i d_i + k_j(d_j + 1) = \sum_{i=1}^j k_i d_i + k_j.$$

By definition of k_j,

$$> \sum_{i=1}^j k_i d_i + \sum_{i=j+1}^n k_i d_i = \sum_{i=1}^n k_i d_i$$

as desired.

Now we need to show that these k_i actually prove the lemma. First we show that φ is integral; let $C = k[y_1, \dots, y_{n-1}]$ be φ's domain. Let $c \in k^\times$ be f's leading coefficient as a polynomial in $B[x_n]$, then $c^{-1} f$ vanishes in A. Since $c^{-1} f$ is a monic polynomial in $B[x_n]$ which is zero, so $\overline{x}_n \in A$ is integral over C, as desired. Since $\overline{x}_i = \overline{y}_i + \overline{x}_n^{k_i}$, $\overline{x}_i \in A$ is also integral over C, and thus A is integral over C via φ. Furthermore, A is fg over C and thus finite.

Note that $x_n \mapsto x_n$, $x_i \mapsto y_i = x_i - x_n^{k_i}$ is invertible, and thus $k[x_1, \dots, x_n] = k[y_1, \dots, y_{n-1}][x_n]$, and $y_1, \dots, y_{n-1}, x_n$ are algebraically independent. Now suppose $\varphi(g) = 0$ for some $g \in C = k[t_1, \dots, t_{n-1}]$. That is, $g(\overline{y}_1, \dots, \overline{y}_{n-1}) = 0 \in A$ so $g(y_1, \dots, y_{n-1}) = hf$ for some $h \in k[y_1, \dots, y_{n-1}][x_n]$. Since g has zero in x_n, but f has positive degree, we must have that $g = 0$. So φ is injective. ■

2.1.14 Theorem (Noether normalization lemma)

Let $A \ne 0$ be a fg algebra over a field k. Then there exists a k-subalgebra $B \subseteq A$ such that B is isomorphic as a k-algebra to a polynomial algebra $k[y_1, \dots, y_m]$ and A is a finite B-algebra.

Proof

Since A is fg, $A \cong k[x_1, \dots, x_n]/I$ for some ideal I. If $I = 0$, then take $B = A$. Otherwise there is a nonconstant $f \in I$ (since otherwise I is the whole polynomial ring and $A = 0$). By the previous lemma, there is a subalgebra $B' \subseteq A' = k[x_1, \dots, x_n]/(f)$ such that $B' \cong k[x'_1, \dots, x'_{n-1}]$ and A' is finite over B'.

Let $I' = I/(f) \subseteq A'$ and $J = I' \cap B'$ be ideals. Then $B'/J \cong k[x'_1, \dots, x'_{n-1}]/J$ is a finite subalgebra over $A'/I' = A$. By induction on the number of generators, the claim holds for B'/J, so there is a subalgebra

$B \subseteq B'/J \subseteq A$ such that B is a polynomial ring $k[y_1, \ldots, y_m]$ and B'/J is a finite B-algebra. Since being a finite algebra is transitive, A is a finite B-algebra, as desired. ■

The Noether normalization lemma then implies the following result, which can be seen to be a sort of generalization of Nullstellensatz.

2.1.15 Proposition

Let L/k be a field extension such that L is a fg k-algebra. Then L is a finite extension of k (i.e. it is a finite k-algebra).

Proof

By Noether, L contains a polynomial subalgebra over which it is finite, and thus integral. Let $B = k[x_1, \ldots, x_n] \subseteq L$ be this algebra. Since L is integral over B, B must be a field (if $A \subseteq B$ is integral, A is a field iff B is a field), and so n must be zero. Thus $k \subseteq L$ is finite, as desired. ■

2.1.16 Corollary (Weak Nullstellensatz)

Let $A \neq 0$ be a fg algebra over an algebraically closed field k. Then for any maximal ideal $\mathfrak{m} \leq A$, $A/\mathfrak{m} \cong k$ as k-algebras.

Proof

Let $L = A/\mathfrak{m}$, which is a fg k-algebra, so by the above proposition it must be a finite. Algebraically closed fields admit no proper extensions, so $A/\mathfrak{m} \cong k$. ■

2.1.17 Corollary

Let $A \neq 0$ be a fg algebra over an algebraically closed field k. Then there exists a k-algebra homomorphism $\psi\colon A \to k$.

Proof

Choose a maximal ideal $\mathfrak{m} \leq A$, and consider $\psi\colon A \to A/\mathfrak{m} \cong k$. ■

Recall that the preimage of a prime ideal is always prime, but the preimage of a maximal ideal need not be maximal. For example under the inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$, consider the preimage of the maximal ideal $(0) \leq \mathbb{Q}$, which is $(0) \leq \mathbb{Z}$; not maximal.

2.1.18 Proposition

Let A, B be fg k-algebras, and $f\colon A \to B$ a k-algebra morphism. Then for any maximal ideal $\mathfrak{m} \leq B$, $\mathfrak{m}_A = f^{-1}\mathfrak{m} \leq A$ is maximal.

Proof

The map f induces an injective k-algebra morphism $A/\mathfrak{m}_A \hookrightarrow B/\mathfrak{m}$. The latter is a field and a fg k-algebra, hence a finite k-algebra and in particular integral. Thus $A/\mathfrak{m}_A$ is an integral k-algebra as well, and since it is a domain ($\mathfrak{m}_A$ is prime), it must then be a field. So $\mathfrak{m}_A$ is maximal. ■

2.1.19 Definition

A ring A is **Jacobson** when $\mathrm{Jac}(A) = \mathrm{nil}(A)$.

In general we have $\mathrm{nil}(A) \subseteq \mathrm{Jac}(A)$, since $\mathrm{nil}(A)$ is the intersection of all prime and $\mathrm{Jac}(A)$ all maximal ideals.

2.1.20 Proposition

Every fg algebra over a field k is Jacobson.

Proof

Let A be a fg algebra over k, and $f \notin \mathrm{nil}(A)$. We then claim A has a maximal ideal not containing f. Since $f \notin \mathrm{nil}(A)$, $A_f \neq 0$ as the localizing set does not contain zero, and thus contains a maximal ideal $\mathfrak{m}$. Recall that $A_f \cong A[x]/(fx-1)$, so A_f is a fg k-algebra. Let $\iota\colon A \to A_f$ be the canonical map, then $\iota^{-1}\mathfrak{m}$ is maximal in A. Since $\iota f \in A_f^\times$, f is not in $\iota^{-1}\mathfrak{m}$ as desired. ■

2.1.21 Corollary

If A is a fg algebra over a field k, then for every ideal $I \leq A$,

$$\sqrt{I} = \mathrm{Jac}_A(I),$$

where $\mathrm{Jac}_A(I)$ is the Jacobson radical of I, the intersection of all maximal ideals containing I. In particular for a prime $\mathfrak{p} \leq A$,

$$\mathfrak{p} = \bigcap_{\mathfrak{p} \leq \mathfrak{m} < A} \mathfrak{m}.$$

Proof

This amounts to saying that A/I is a Jacobson ring, since pulling back $\mathrm{nil}(A/I)$ and $\mathrm{Jac}(A/I)$ give $\sqrt{I}, \mathrm{Jac}_A(I)$ respectively. Since A is a fg algebra over k, so is A/I and the result follows the above proposition. ■

Note:

Some authors define a Jacobson ring to be one where $\sqrt{I} = \mathrm{Jac}_A(I)$ for all ideals, or $\mathfrak{p} = \mathrm{Jac}_A(\mathfrak{p})$ for all prime ideals (these are equivalent). Note that this definition is more useful, but not how it was defined in this course. The above corollary shows that a fg algebra over a field is Jacobson in this regard as well.

Note that trhere is a natural bijection between $\mathbb{A}^n$ and k-algebra morphisms $\varphi\colon k[\mathbb{A}^n] \to k$.

(1) For every $P = (a_1, \ldots, a_n) \in \mathbb{A}^n$ define φ by $\varphi(x_i) = a_i$. That is, φ is the evaluation map ev_P given by $\mathrm{ev}_P(f) = f(P)$. In particular,

$$\ker \mathrm{ev}_P = \mathcal{I}(P).$$

(2) Conversely for $\varphi\colon k[\mathbb{A}^n] \to k$, let $P = (a_1, \ldots, a_n)$ by $a_i = \varphi(x_i)$. Then $\varphi = \mathrm{ev}_P$, so these are inverses.

Let $J \leq k[\mathbb{A}^n]$ be an ideal, then for every $P \in \mathbb{A}^n$, we have $P \in Z(J)$ iff $J \leq \mathcal{I}(P) = \ker \mathrm{ev}_P$ iff ev_P quotients over J: it induces $\overline{\mathrm{ev}}_P\colon k[\mathbb{A}^n]/J \to k$ where $\overline{\mathrm{ev}}_P(f+J) = f(P)$.

2.1.22 Proposition

For $P = (a_1, \dots, a_n) \in \mathbb{A}^n$ we have

$$\ker \mathrm{ev}_P = (x_1 - a_1, \dots, x_n - a_n),$$

which is a maximal ideal.

Proof

Since $\mathrm{ev}_P(x_i - a_i) = 0$ we have inclusion $\supseteq$. Since $k[\mathbb{A}^n]/\ker \mathrm{ev}_P \cong \mathrm{im}\,\mathrm{ev}_P \cong k$, $\ker \mathrm{ev}_P$ is maximal, it is sufficient to show $k[\mathbb{A}^n]/(x_1 - a_1, \dots, x_n - a_n) \cong k$. Equivalently, the natural map $\iota\colon k \to k[\mathbb{A}^n]/(x_1 - a_1, \dots, x_n - a_n)$ is surjective.

The projection $k[\mathbb{A}^n] \to k[\mathbb{A}^n]/(x_1 - a_1, \dots, x_n - a_n)$ maps x_i to $\overline{x}_i = \iota(a_i)$. Thus ι must be surjective as the projection is. ■

The following is another version of the Nullstellensatz:

2.1.23 Proposition

Let k be algebraically closed. Then the maximal ideals of $k[\mathbb{A}^n]$ are precisely the ideals $\ker \mathrm{ev}_P$ for $P \in \mathbb{A}^n$.

Proof

We saw that $\ker \mathrm{ev}_P$ are maximal, so let $\mathfrak{m}$ be maximal. By the weak Nullstellensatz, $k[\mathbb{A}^n]/k \cong k$, which gives a projection $f\colon k[\mathbb{A}^n] \to k[\mathbb{A}^n]/\mathfrak{m} \cong k$ with $\ker f = \mathfrak{m}$. But k-algebra morphisms $k[\mathbb{A}^n] \to k$ are all of the form ev_P for some P, so $f = \mathrm{ev}_P$ and thus $\mathfrak{m} = \ker \mathrm{ev}_P$. ■

Now we can finally prove Nullsellensatz itself.

Proof (Nullstellensatz)

We already showed $\sqrt{J} \leq \mathcal{I}(Z(J))$, we now show the other direction. Let $f \in \mathcal{I}(Z(J))$, and consider $A = k[\mathbb{A}^n]/J$. By the above proposition, the maximal ideals of A are the ideals of the form $\ker \mathrm{ev}_P$ containing J, so $\ker \mathrm{ev}_P$ with $P \in Z(J)$. Because $f \in \mathcal{I}(Z(J))$, f vanishes at every point $P \in Z(J)$, so $f \in \ker \mathrm{ev}_P$ for every $P \in Z(J)$. Thus f is in every maximal ideal of A, so $\overline{f} \in \mathrm{Jac}(A)$. A is a fg k-algebra it is Jacobson, so $\overline{f} \in \mathrm{Jac}(A) = \mathrm{nil}(A) = \sqrt{J}/J$, thus $f \in \sqrt{J}$ as desired. ■

Let us summarize the following formulations of Nullstellensatz:

2.1.24 Theorem (Equivalent formulations of Nullstellensatz)

The following are all equivalent formulations of Nullstellensatz, and all are true.

(1) Nullstellensatz: $\mathcal{I}(Z(J)) = \sqrt{J}$,

(2) If $J \neq (1)$ then $Z(J) \neq \emptyset$,

(3) Weak Nullstellensatz: if A is fg, $A/\mathfrak{m} \cong k$,

(4) If $A \neq 0$ is fg, then there is a $\varphi\colon A \to k$,

(5) Every maximal ideal of $k[\mathbb{A}^n]$ of the form $(x_1 - a_1, \dots, x_n - a_n)$.

2.1.25 Definition

For a topological space X, let $|X|$ denote the subspace of closed points.

Recall that for an affine scheme $\operatorname{Spec}(A)$, a closed point is just a maximal ideal, so $|X| \cong \operatorname{Specm}(A)$.

2.1.26 Corollary

Let k be algebraically closed, and $I \leq k[\mathbb{A}^n]$ an ideal. Then there is a bijection between the closed points of $\operatorname{Spec}(k[\mathbb{A}^n]/I)$ and $Z(I)$:

$$|\operatorname{Spec}(k[\mathbb{A}^n]/I)| \cong Z(I).$$

Proof

The closed points of the spectrum of $k[\mathbb{A}^n]/I$ are in bijection with the maximal ideals of $k[\mathbb{A}^n]$ containing I. These are of the form $(x_1 - a_1, \dots, x_n - a_n)$, and if it contains I then for $f \in I$ we must have that $f(a_1, \dots, a_n) = 0$. So mapping $(x_1 - a_1, \dots, x_n - a_n)$ to $(a_1, \dots, a_n) \in Z(I)$ is well-defined and clearly injective. Furthermore it is clearly surjective. ∎

2.1.27 Corollary

Let A, B be fg k-algebras for some field k, and let $\varphi\colon A \to B$ be a k-algebra morphism. Then $\varphi^*\colon \operatorname{Spec}(B) \to \operatorname{Spec}(A)$ restricts to a morphism of closed points $|\varphi^*|\colon |\operatorname{Spec}(B)| \to |\operatorname{Spec}(A)|$.

Proof

This is just a restatement of the preimage of a maximal ideal being maximal. ∎

2.2 Algebraic varieties

2.2.1 Definition

Let k be a field, then denote by $\operatorname{Fun}(U, k)$ the k-algebra of set-theoretic functions $U \to k$.

A **space with functions** (**SWF**) over k is a pair $(X, \mathcal{O}_X)$ where X is a topological space and for every $U \subseteq X$ open, $\mathcal{O}_X(U)$ is a subset of $\operatorname{Fun}(U, k)$. Furthermore it must satisfy:

(1) **presheaf condition**: if $f \in \mathcal{O}_X(U)$ and $V \subseteq U$ then the restriction of f to V, $f|_V$, is in $\mathcal{O}_X(V)$.

(2) **sheaf condition**: if U_i is an open covering of $U \subseteq X$, and $f\colon U \to k$ is a function such that $f|_{U_i} \in \mathcal{O}_X(U_i)$ for all i, then $f \in \mathcal{O}_X(U)$.

(3) **local condition**: for every $f \in \mathcal{O}_X(U)$, the subset $D_U(f) = \{x \in U \mid f(x) \neq 0\}$ is open, and the function $1/f\colon D_U(f) \to k$ is in $\mathcal{O}_X(D_U(f))$.

$\mathcal{O}_X$ is called the **structure sheaf** of the SWF.

Note:

In general a **presheaf** valued in a category $\mathcal{C}$ over a topological space X is a contravariant functor

$$\mathcal{O}\colon \mathrm{Op}(X)^{\mathsf{op}} \to \mathcal{C},$$

where $\mathrm{Op}(X)$ is the category of open sets of X (viewed as a preorder, ordered by inclusion). For $U \subseteq V$ open we denote the map $\mathcal{O}(V) \to \mathcal{O}(U)$ by res^V_U, if $\mathcal{C}$ is a concrete category (its objects are sets) then we write $\mathrm{res}^V_U(f) = f|_U$ for $f \in \mathcal{O}(V)$.

The objects $\mathcal{O}_X(U)$ are called **sections** of the presheaf.

If $\mathcal{C}$ is complete (has all limits), then a presheaf $\mathcal{O}$ is a **sheaf** if for every open cover U_i of U, the following is an equalizer diagram:

$$\mathcal{O}(U) \xrightarrow{\mathrm{res}_0} \prod_i \mathcal{O}(U_i) \underset{\mathrm{res}_2}{\overset{\mathrm{res}_1}{\rightrightarrows}} \prod_{i,j} \mathcal{O}(U_i \cap U_j),$$

where res_0 is the map $f \mapsto (f|_{U_i})$ (i.e. its component $\mathcal{O}(U) \to \mathcal{O}(U_i)$ is $\mathrm{res}^U_{U_i}$). The map res_1 is the map $(f_i) \mapsto (f_i|_{U_i \cap U_j})$ and res_2 is the map $(f_i) \mapsto (f_j|_{U_i \cap U_j})$. That is, their components $\prod_i \mathcal{O}(U_i) \to \mathcal{O}(U_i \cap U_j)$ are $\mathrm{res}^{U_i}_{U_i \cap U_j} \circ \mathrm{proj}_i$ and $\mathrm{res}^{U_j}_{U_i \cap U_j} \circ \mathrm{proj}_j$ respectively.

For a concrete category $\mathcal{C}$ being a sheaf means that if $f_i \in \mathcal{O}_X(U_i)$ are sections such that $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ for all i, j, then there exists a unique $f \in \mathcal{O}_X(U)$ such that $f|_{U_i} = f_i$ for all i.

Now a **subsheaf** of a sheaf $\mathcal{O}$ is another sheaf $\mathcal{O}'$ (on the same category) such that $\mathcal{O}'(U)$ is a subobject of $\mathcal{O}(U)$ for all open U. Under sufficiently weak conditions (e.g. $\mathcal{C}$ being finitely complete) this is equivalent to $\mathcal{O}'$ being a subfunctor of $\mathcal{O}$ (i.e. there being a monic natural transformation $\mathcal{O}' \Rightarrow \mathcal{O}$). Conditions (1) and (2) above are equivalent to saying that $\mathcal{O}$ is a subsheaf of $\mathrm{Fun}(\bullet, k)$ (defined in the obvious way).

A pair $(X, \mathcal{O}_X)$ where $\mathcal{O}_X$ is a sheaf on X valued in the category of rings (resp. k-algebras) is called a **ringed space** (resp. a k-ringed space).

Condition (3) is a replacement for the requirement that $(X, \mathcal{O}_X)$ be a **locally ringed space**. This condition says that the stalks of $\mathcal{O}_X$ (the ring of germs of a point $p \in X$, not defined here) is a local ring.

2.2.2 Definition

A morphism of SWFs $(X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a continuous map $\varphi\colon X \to Y$ such that for every open $V \subseteq Y$ and every $f \in \mathcal{O}_Y(V)$, the pullback $\varphi^*(f) = f \circ \varphi\colon \varphi^{-1}V \to V \to k$ belongs to $\mathcal{O}_X(\varphi^{-1}V)$.

The identity is a morphism of SWFs because $\mathrm{id}^* f = f$. And the composition of morphisms of SWFs is a morphism of SWFs clearly as well, thus SWFs form a category.

2.2.3 Example

Let k be a Hausdorff topological field, and X a topological space. Define $(X, \mathcal{O}_{cont})$ to be the SWF where $\mathcal{O}_{cont}(U)$ is the set of continuous functions $U \to k$. This is easily seen to be a SWF.

If X is a real manifold, then we have SWFs $(X, \mathcal{O}_{C^n})$, $(X, \mathcal{O}_{C^\infty})$, and $(X, \mathcal{O}_{anal})$ which map an open set to the algebra of real valued C^n-, C^∞-, or analytic maps respectively. These are SWFs and the identity maps

$$(X, \mathcal{O}_{C^n}) \xrightarrow{\mathrm{id}} (X, \mathcal{O}_{C^\infty}) \xrightarrow{\mathrm{id}} (X, \mathcal{O}_{anal})$$

are morphisms of SWFs.

For a complex manifold we can define the SWF of holomorphic functions on an open subset.

2.2.4 Definition

If $(X, \mathcal{O})$ is a space with functions and $Y \subseteq X$ is *open*, we can restrict $\mathcal{O}$ to open subsets of Y to obtain another SWF. That is, define $(Y, \mathcal{O}|_Y)$ to be the SWF defined by $\mathcal{O}|_Y(U) = \mathcal{O}(U)$ for every open $U \subseteq Y \subseteq X$.

In general if $Y \subseteq X$ is a subspace, we define the **induced space** $(Y, \mathcal{O}|_Y)$ as follows. For $V \subseteq Y$ open, let $\mathcal{O}|_Y(V)$ be the set of functions $f\colon V \to k$ such that there is an open cover $U_i \subseteq X$ of V (so $V = \bigcup U_i \cap V$) and $g_i \in \mathcal{O}(U_i)$ such that $f|_{V\cap U_i} = g|_{V\cap U_i}$ for all i.

Equivalently $f \in \mathcal{O}|_Y(V)$ iff for every $y \in V$ there is a neighborhood W of y in X and a function $g \in \mathcal{O}_X(W)$ such that $g|_{W\cap V} = f|_{W\cap V}$. If we think of $\mathcal{O}_X$ as locally smooth maps, then this says that a map is smooth on a subspace if it can be extended locally.

2.2.5 Proposition

The pair $(Y, \mathcal{O}|_Y)$ is an SWF and inclusion $\iota\colon X \hookrightarrow Y$ is a morphism $(Y, \mathcal{O}|_Y) \to (X, \mathcal{O}_X)$.

Furthermore if $(Z, \mathcal{O}_Z)$ is another SWF and $\varphi\colon Z \to Y$ a set-theoretic map, then φ is a morphism of SWFs $(Z, \mathcal{O}_Z) \to (Y, \mathcal{O}|_Y)$ iff $\iota \circ \varphi\colon (Z, \mathcal{O}_Z) \to (X, \mathcal{O}_X)$ is a morphism of SWFs.

Proof

Let $V \subseteq V' \subseteq Y$ be open subsets, and $f \in \mathcal{O}|_Y(V')$. Then there is an open cover of V' in X, $V' = \bigcup_i U_i \cap V'$ and $g_i \in \mathcal{O}_X(U_i)$ which restrict to f on $U_i \cap V'$. These U_i cover V as well then, and g_i restrict to $f|_V$ on $U_i \cap V$, so $f|_V \in \mathcal{O}|_Y(V)$, so it satisfies the presheaf condition.

Let $V \subseteq Y$ be open and $f \in \operatorname{Fun}(V, k)$ such that there is an open cover $V = \bigcup_i V_i$ with $f|_{V_i} \in \mathcal{O}_Y(V_i)$. Then we cover each V_i by U_i^j such that there are $g_i^j \in \mathcal{O}_X(U_i^j)$ with $g_i^j|_{U_i^j\cap V_i} = f|_{V_i \cap U_i^j}$. But then the collection of all U_i^j as i, j ranges and g_i^j show that $f \in \mathcal{O}|_Y(V)$. So $\mathcal{O}|_Y$ is a sheaf.

Now we need to show the locality axiom. Let $V \subseteq Y$ be open and $f \in \mathcal{O}|_Y(V)$ so there are $U_i \subseteq X$ open covering V such that $f|_{U_i\cap V} = g_i|_{U_i\cap V}$ for $g_i \in \mathcal{O}_X(U_i)$. Then

$$D_V(f) = \{x \in V \mid f(x) \neq 0\} = \bigcup_i \{x \in V \cap U_i \mid f(x) \neq 0\} = \bigcup_i D_{U_i}(g_i) \cap V,$$

which is relatively open in V. And we see the $D_{U_i}(g_i)$ cover $D_V(f)$ and $1/f$ restricts to $1/g_i \in \mathcal{O}_X(D_{U_i}(g_i))$ their common domain, so $1/f \in \mathcal{O}|_Y(D_U(f))$. Thus $\mathcal{O}|_Y$ is indeed a SWF.

The inclusion is continuous by virtue of the subspace topology. Let $U \subseteq X$ open so $\iota^{-1}U = U \cap Y$, and let $f \in \mathcal{O}_X(U)$ then $\iota^*(f) = f \circ \iota = f|_{U\cap V}$. This is in $\mathcal{O}|_Y(U \cap Y)$ by considering the open cover U and $f \in \mathcal{O}_X(U)$.

The composition of morphisms of SWFs is a morphism of SWFs, so if φ is a morphism so is $\iota \circ \varphi$. Conversely if $\iota \circ \varphi$ is continuous, then so is φ by virtue of the subspace topology. And if $f \in \mathcal{O}|_Y(V)$ for $V \subseteq Y$ open, then let $g_i \in \mathcal{O}_X(U_i)$ such that $g_i|_{U_i\cap V} = f|_{U_i\cap V}$ with U_i covering V. Then

$$(\iota \circ \varphi)^*(g_i) = g_i \circ \iota \circ \varphi = g_i|_{U_i\cap V} \circ \varphi = f|_{U_i\cap V} \circ \varphi = f \circ \varphi|_{W_i}$$

is in $\mathcal{O}_X(W_i)$ with $W_i = (\iota \circ \varphi)^{-1}(U_i)$. These W_i cover $\varphi^{-1}V$ and thus by definition $f \circ \varphi \in \mathcal{O}|_Y(\varphi^{-1}V)$, showing φ is a morphism of SWFs. ■

2.2.6 Example

Let $B^n \subseteq \mathbb{R}^n$ or $\mathbb{C}^n$ be the open unit ball. Then the above proposition gives us SWFs

$$(B^n, \mathcal{O}_{cont}), (B^n, \mathcal{O}_{C^k}), (B^n, \mathcal{O}_{C^\infty}), (B^n, \mathcal{O}_{anal}), (B^n, \mathcal{O}_{holo}).$$

We can then define a real/complex topological/C^k/C^∞/analytic manifold as a SWF $(X, \mathcal{O})$ such that X is Hausdorff and second countable and there is an open cover $X = \bigcup_i U_i$ such that there is a single SWF from one of the above $((B_n, \mathcal{O}_\bullet))$ such that each i $(U_i, \mathcal{O}|_{U_i})$ is isomorphic to it.

This is an equivalent definition to the standard definition of a manifold via charts. Then a smooth map between manifolds is just a morphism of SWFs.

We use this example to define general algebraic varieties. We will define affine algebraic varieties as a special family of SWFs, and then a general algebraic variety will be a SWF locally isomorphic to an affine algebraic variety.

2.2.7 Definition

Let $Y \subseteq \mathbb{A}^n$ be a subspace. A function $f\colon Y \to k$ is **regular** when it is locally rational (the product of polynomials). That is, for every $P \in Y$ there is a neighborhood $U_P \subseteq Y$ of Y and polynomials $g_P, h_P \in k[\mathbb{A}^n]$ such that $h_P(Q) \neq 0$ for $Q \in U_P$ and $f(Q) = g_P(Q)/h_P(Q)$ for all $Q \in U_P$.

Denote the set of regular functions $Y \to k$ by $\mathcal{O}_{reg}(Y)$.

2.2.8 Lemma

The pair $(Y, \mathcal{O}_{reg})$ is a SWF for every $Y \subseteq \mathbb{A}^n$.

Proof

It is easy to see that $\mathcal{O}_{reg}(U) \subseteq \mathrm{Fun}(U,k)$ is a subalgebra; i.e. that it is closed under addition and multiplication, and contains the unit (and thus all constants). Furthermore since being regular is a local condition, it is again easy to see that $\mathcal{O}_{reg}$ is a sheaf.

We just need to show locality. Let $f \in \mathcal{O}_{reg}(U)$ then for $P \in U$ there is an open $P \in U_P \subseteq U$ such that $f = g_P/h_P$ for polynomials g_P, h_P in U_P. Then

$$D_U(f) = \{x \in U \mid f(x) \neq 0\} = \bigcup_{P \in U} \{x \in U_P \mid g_P(x) \neq 0\} = \bigcup_{P \in U_P} D_{U_P}(g_P)$$

is open since each g_P is continuous. And on $D_{U_P}(g_P)$ we have $1/f = h_P/g_P$, so $1/f$ is regular on $D_U(f)$ as desired. ∎

Note:

It is easy to see that $\mathcal{O}_{reg}$ is the smallest SWF containing all the polynomial functions in $k[\mathbb{A}^n]$. Such a SWF must contain regular functions because of locality.

2.2.9 Lemma

Let $U \subseteq \mathbb{A}^n$ be open, and $f \in \mathcal{O}_{reg}(U)$ a regular function. Then f is rational: there are $g, h \in k[\mathbb{A}^n]$ such that $h \neq 0$ on U and $f = g/h$.

Proof

By definition there is an open covering $U = \bigcup_i U_i$ and polynomials g_i, h_i such that $h_i \neq 0$ on U_i and $f|_{U_i} = g_i/h_i$ on U_i. Further we can assume that g_i, h_i are relatively prime and U_i are nonempty.

We claim that for any i, h_i is nonzero on all of U, and $f = g_i/h_i$ on all of U. Let $P \in U_j$, so we have

$$\left.\frac{g_i}{h_i}\right|_{U_i \cap U_j} = f|_{U_i \cap U_j} = \left.\frac{g_j}{h_j}\right|_{U_i \cap U_j}.$$

Thus $g_ih_j - g_jh_i = 0$ on $U_i \cap U_j$. Since $\mathbb{A}^n$ is irreducible, any nonempty open subset is dense (this is equivalent) and so the intersection of nonempty open subsets is then dense (as each is dense and thus their intersection is nonempty, and thus a nonempty open subset, so dense). Thus g_ih_j and g_jh_i agree on a dense set, so they are equal (two maps into a Hausdorff space agreeing on a dense set are equal).

Since $k[\mathbb{A}^n]$ is a UFD and $\{g_i, h_i\}$ are all relatively prime, we must have $h_i = ch_j$ and $g_i = cg_j$ for some $c \in k[\mathbb{A}^n]^\times$ (which is $k^\times$). Thus we have $h_i(P) = c(P)h_j(P) \neq 0$ for all $P \in U_j$, so h_i is nonzero on all of U. And so for $P \in U_j$,

$$\frac{g_i(P)}{h_i(P)} = \frac{g_j(P)}{h_j(P)} = f(P),$$

so that $f = g_i/h_i$ on all of U as desired. ∎

So $(\mathbb{A}^n, \mathcal{O}_{reg})$ is a much simpler object than the general case.

2.2.10 Definition

An **affine algebraic variety** (or **affine variety**) is a SWF $(X, \mathcal{O}_X)$ which is isomorphic to a SWF $(Y, \mathcal{O}_{reg})$ where $Y \subseteq \mathbb{A}^n$ is closed.

An **algebraic variety** is a SWF $(X, \mathcal{O}_X)$ which has a finite local covering $X = \bigcup_{i=1}^n U_i$ such that each $(U_i, \mathcal{O}_X|_{U_i})$ is an affine variety. Elements of a section of an algebraic variety (elements of $\mathcal{O}_X(U)$) are called **regular functions**.

A morphism of algebraic varieties is just a morphism of SWFs.

2.2.11 Lemma

Every algebraic variety is Noetherian (as a topological space).

Proof

First we show that $\mathbb{A}^n$ is Noethrian. Indeed, if $\{Z_i\}_i$ is a collection of closed subsets of $\mathbb{A}^n$ then $\{\mathcal{I}(Z_i)\}_i \subseteq k[\mathbb{A}^n]$ has a maximal element since $k[\mathbb{A}^n]$ is Noetherian. Since $Z \circ \mathcal{I}$ is the identity on closed sets, this means there is a maximal Z_i, and thus $\mathbb{A}^n$ is Noetherian.

An affine variety is homeomorphic to a closed subset of $\mathbb{A}^n$ which is Noetherian.

Thus an algebraic variety is locally Noetherian, i.e. there is a finite open covering $\bigcup_{i=1}^n U_i$ where each U_i is Noetherian. Then X is Noetherian. ∎

2.2.12 Corollary

Every algebraic variety is quasicompact.

2.2.13 Definition

Let $Y \subseteq \mathbb{A}^n$ be closed, then define $k[Y] = k[\mathbb{A}^n]/\mathcal{I}(Y)$. Then there is a natural homomorphism

$$\mathrm{ev}_Y \colon k[\mathbb{A}^n] \xrightarrow{\mathrm{ev}} \mathrm{Fun}(\mathbb{A}^n, k) \xrightarrow{\mathrm{res}} \mathrm{Fun}(Y, k).$$

By definition the kernel of this is $\mathcal{I}(Y)$, and so ev_Y induces an injection $k[Y] \hookrightarrow \mathrm{Fun}(Y, k)$.

This notation is compatible with our previous definition of $k[\mathbb{A}^n]$. This is because $\mathcal{I}(\mathbb{A}^n) = 0$ so $k[\mathbb{A}^n]/\mathcal{I}(\mathbb{A}^n) = k[\mathbb{A}^n]$.

Note that $k[Y]$ is a fg k-algebra. Furthermore it is reduced, as it can be embedded into $\mathrm{Fun}(Y,k)$ which is reduced as k is (also because $\mathcal{I}(Y)$ is a radical ideal). Furthermore, $k[Y]$ is a domain iff $\mathcal{I}(Y)$ is prime iff Y is irreducible.

If B is a reduced fg k-algebra, then $B \cong k[Y]$ for some closed $Y \subseteq \mathbb{A}^n$. Indeed, $B \cong k[\mathbb{A}^n]/I$ for some ideal I, and since B is reduced I is a radical ideal and thus equal to $\mathcal{I}(Y)$ for $Y = Z(\mathcal{I})$, so $B \cong k[Y]$.

For $f \in k[\mathbb{A}^n]$ let $\overline{f} = f + \mathcal{I}(Y) \in k[Y]$ denote its projection. Then viewed as functions $\mathbb{A}^n \to k$ and $Y \to k$ respectively, we have the equality

$$D_Y(\overline{f}) = Y \cap D(f).$$

Since the $D(f)$ form a base of open subsets of $k[\mathbb{A}^n]$, we then see that $\left\{D_Y(\overline{f})\right\}_{\overline{f}\in k[Y]}$ is a basis of Y.

2.2.14 Lemma

For every collection $\left\{\overline{f}_i\right\}_i \subseteq k[Y]$, $\left\{D_Y(\overline{f}_i)\right\}_i$ form an open cover of Y iff the ideal $(\{f_i\}_i)$ is $k[Y]$.

Proof

We have that $D_Y(\overline{f}_i)$ cover Y iff $D(f_i)$ cover Y. Since $D(f_i) = Z(f_i)^c$, this is iff $\bigcap_i Z(f_i) \cap Y = \emptyset$, equivalently $Z((f_i)_i) \cap Y$ is empty. But

$$Z((f_i)_i) \cap Y = Z((f_i)_i) \cap Z(\mathcal{I}(Y)) = Z((f_i)_i + \mathcal{I}(Y)),$$

and this is empty iff $Z((f_i)_i) + \mathcal{I}(Y)$ is all of $k[\mathbb{A}^n]$ (by Nullstellensatz). Quotienting by $\mathcal{I}(Y)$ this is equivalent to $Z((\overline{f}_i)_i) = k[Y]$. ■

Recall that $f \in \mathcal{O}_{reg}(Y)$ iff there is an open cover $Y = \bigcup_i U_i$ such that $f|_{U_i}$ is rational for each i. Since Y is quasicompact, we can assume this covering is finite, and taking basic open sets, that $U_i = D_Y(t_i)$ for $t_i \in k[Y]$. So $f = g_i/h_i$ for $g_i, h_i \in k[\mathbb{A}^n]$ and h_i nonzero. Equivalently, $f = \overline{g}_i/\overline{h}_i$ (as $\overline{g}$ agrees with g on Y).

2.2.15 Theorem (Serre)

For every closed $Y \subseteq \mathbb{A}^n$ there is an equality $k[Y] = \mathcal{O}_{reg}(Y)$ (viewing $k[Y]$ as a subalgebra of $\mathrm{Fun}(Y,k)$).

Proof

Since $\mathcal{O}_{reg}(Y)$ consists of all polynomial functions, clearly $k[Y] \subseteq \mathcal{O}_{reg}(Y)$. By the above if $f \in \mathcal{O}_{reg}(Y)$ then there is a finite set of triples $\{t_i, g_i, h_i\}$ of polynomials in $k[Y]$ such that $D_Y(t_i)$ cover Y and $f_i = g_i/h_i$ on $D_Y(t_i)$ and $h_i \neq 0$ on $D_Y(t_i)$.

By assumption we have that $D_Y(t_i) \subseteq D_Y(h_i)$, and so $D_Y(t_i) = D_Y(t_i) \cap D_Y(h_i) = D_Y(t_i h_i)$. Since $g_i/h_i = t_i g_i/t_i h_i$, we can replace our collection with $\{t_i h_i, t_i g_i, t_i h_i\}_i$. In particular we can assume that $t_i = h_i$.

Now we want to show that $f \in k[Y]$. We first give a short, but tricky, proof and afterward we will give a more insightful one. Notice that for every i we have

$$f h_i^2 = g_i h_i \in \mathrm{Fun}(Y,k),$$

so $(f h_i - g_i) h_i = 0$. This is true in $D_Y(h_i)$ clearly, and for $P \in Y - D_Y(h_i)$ we have that $h_i(P) = 0$ so the equality holds trivially.

Now since $Y = \bigcup_i D_Y(h_i) = \bigcup_i D_Y(h_i^2)$ we conclude that $(h_i^2) = k[Y]$ as per the previous lemma. So there are $r_i \in k[Y]$ such that

$$\sum_i r_i h_i^2 = 1.$$

(The algebraic analogue of a partition of unity.) Thus

$$f = f \sum_i r_i h_i^2 = \sum_i r_i f h_i^2 = \sum_i r_i g_i h_i \in k[Y]$$

as desired. ■

We now give a more insightful proof of Serre's theorem.

2.2.16 Lemma

Let A be a ring and suppose $f_1, \dots, f_n$ generate the unit ideal. Then

$$0 \to A \xrightarrow{\varphi} \prod_i A_{f_i} \xrightarrow{\psi} \prod_{i,j} A_{f_i f_j},$$

is exact, where $\varphi(a) = (a/1)_i$ and $\psi((a_i)_i) = (a_i - a_j)_{ij}$.

Proof

First we show that the sequence is exact at A, meaning φ is injective. If $a/1 = 0$ for all i, then there are n_i such that $f_i^{n_i} a = 0$ for each i. But

$$f_i \in \sqrt{(f_i^{n_i})},$$

and so $\sqrt{(f_i^{n_i})}$ is equal to the unit ideal. But this means that $(f_i^{n_i})$ is equal to the unit ideal, and thus $a = 0$. Now we show that $\mathrm{im}\varphi = \ker \psi$. Firstly,

$$\psi \circ \varphi(a) = \psi((a/1)_i) = (a/1 - a/1)_{i,j} = 0,$$

so the sequence is a chain complex. We now need only show that $\ker \psi \subseteq \mathrm{im}\varphi$. So let $(a_i/f_i^{n_i})_i$ lie in $\ker \psi$. Since $a_i f^{r_i}/f_i^{n_i+r_i} = a_i/f_i^{n_i}$, we may enlarge the n_i so that they are equal.

So

$$0 = \psi(a_i/f_i^n) = (a_i/f_i^n - a_j/f_j^n)_{i,j}.$$

This means that $a_i/f_i^n = a_j/f_j^n$ in $A_{f_i f_j}$ for all i, j and thus there is an M_{ij} such that $(f_i f_j)^{M_{ij}}(f_j^n a_i - f_i^n a_j) = 0$. Again by potentially enlarging the M_{ij} we can assume they are all equal to M. So we have that for all i, j,

$$(f_i f_j)^M (f_j^n a_i - f_i^n a_j) = 0.$$

As before f_j^{M+n} generate the unit ideal so there are $b_j \in A$ such that

$$\sum_j b_j f_j^{M+n} = 1,$$

and so if we define $a = \sum_j b_j f_j^M a_j$, then

$$f_i^{M+r} a = \sum_j b_j f_j^M f_i^{M+r} a_j = \sum_j b_j f_i^M f_j^{M+r} a_i = f_i^M a_i \sum_j b_j f_j^{M+r} = f_i^M a_i.$$

Thus $a/1 = a_i/f_i^n$, and so $\varphi(a) = (a_i/f_i^n)_i$ as desired. ■

Now in the proof of Serre's theorem, since $(h_i)_i = k[Y]$ we have the exact sequence

$$0 \to k[Y] \xrightarrow{\varphi} \prod_i k[Y]_{h_i} \xrightarrow{\psi} \prod_{i,j} k[Y]_{h_i h_j}.$$

Now consider $\overline{f} = (g_i/h_i) \in \prod_i k[Y]_{h_i}$, then $g_i/h_i - g_j/h_j = 0$ in $k[Y]_{h_i h_j}$ because as we will show,

$$(g_i h_j - g_j h_i) h_i h_j = 0 \text{ in } k[Y].$$

Since $k[Y] \subseteq \mathrm{Fun}(Y, k)$ we can show that the above expression vanishes for every $P \in Y$, and this is sufficient. If $P \in D_Y(h_i) \cap D_Y(h_j)$ then

$$(g_i/h_i)(P) = (g_j/h_j)(P) = f(P),$$

and the equality follows. For all other P either $h_i(P)$ or $h_j(P)$ is zero, and the equality follows trivially.

Thus by exactness there is a $\tilde{f} \in k[Y]$ such that $\varphi(\tilde{f}) = \overline{f}$, meaning $\tilde{f}/1 = g_i/h_i \in k[Y]_{h_i}$ for all i. So for some n, $h_i^n(h_i\tilde{f} - g_i) = 0$ In particular for $P \in D_Y(h_i)$ since $h_i(P) \neq 0$ we have that $h_i(P)\tilde{f}(P) = g_i(P)$ so

$$\tilde{f}(P) = g_i(P)/h_i(P) = f(P).$$

Thus $f = \tilde{f} \in k[Y]$ as desired, showing that $\mathcal{O}_{reg}(Y) = k[Y]$ as desired.

Note:
Abusing terminology, for *any* SWF $(X, \mathcal{O}_X)$ we call elements of $\mathcal{O}_X(U)$ **regular functions** on U.

2.2.17 Theorem

Let $(X, \mathcal{O}_X)$ be a SWF and $Y \subseteq \mathbb{A}^n$ a subset. Let $\psi \colon X \to Y$ be a map of sets, and set $\psi_i = \mathrm{proj}_i \circ \psi \in \mathrm{Fun}(X, k)$ to be the projections of ψ. Then ψ is a morphism of SWFs $(X, \mathcal{O}_X) \to (Y, \mathcal{O}_{reg})$ iff $\psi_i \in \mathcal{O}_X(X)$ for all i. Equivalently, $\psi^* \colon \mathrm{Fun}(Y, k) \to \mathrm{Fun}(X, k)$ maps $\mathcal{O}_Y(Y)$ into $\mathcal{O}_X(X)$.

Note:
Equivalently, $\psi = (\psi_1, \ldots, \psi_n) \colon X \to Y$ is a morphism iff each ψ_i is regular.

Proof

Notice that $\mathrm{proj}_i \colon \mathbb{A}^n \to k$ is given by a regular function x_i on $\mathbb{A}^n$. So if ψ is a morphism, then every $\psi_i = \psi^*(x_i)$ is a regular function on X.

Conversely assume $\psi_i \in \mathcal{O}_X(X)$ for all i. So we want to show that ψ is continuous, and for every $f \in \mathcal{O}_Y(U)$ we have $\psi^*(f) \in \mathcal{O}_X(\psi^{-1}U)$ for all $U \subseteq Y$ open.

First we show that for every $f \in k[\mathbb{A}^n]$ we have $\psi^*(f) \in \mathcal{O}_X(X)$. Notice that $\psi^* \colon \mathrm{Fun}(\mathbb{A}^n, k) \to \mathrm{Fun}(X, k)$ is a k-algebra morphism, while f belongds to the k-subalgebra generated by the x_i. Thus $\psi^* f$ belongds to the k-algebra generated by the $\psi^*(x_i) = \psi_i$. Since $\psi_i \in \mathcal{O}_X(X)$ by assumption, this means $\psi^* f \in \mathcal{O}_X(X)$ as desired (since it is a subalgebra).

Now we show that ψ is continuous. The basic open sets of Y are $D_Y(f)$ with $f \in k[\mathbb{A}^n]$, and so it is sufficient to show $\psi^{-1} D_Y(f)$ is open in X. Notice that

$$\psi^{-1} D_Y(f) = \{x \in X \mid f(\psi x) = 0\} = D_X(\psi^* f).$$

And since $\psi^* f \in \mathcal{O}_X(X)$ by locality of SWFs we have that $D_X(\psi^* f)$ is open as desired.

Now we need to show that for every open $U \subseteq Y$ and $f \in \mathcal{O}_Y(U)$ we have $\psi^* f \in \mathcal{O}_X(\psi^{-1}U)$. Replacing Y by U and X by $\psi^{-1}U$, we can assume $U = Y$.

First assume that f is rational, so $f = g/h$ on Y with $g, h \in k[\mathbb{A}^n]$ and h nonzero. Then $\psi^* f = \psi^* g / \psi^* h$ and $\psi^* h(x) \neq 0$ for every $x \in X$. By locality $1/\psi^* h$ is in $\mathcal{O}_X(X)$ and thus $\psi^* f$ is in $\mathcal{O}_X(X)$ as it is a k-algebra, as desired.

Now in general, locally $f \in \mathcal{O}_Y(Y)$ has the form $g/h|_Y$, the general case follows from the rational case and the sheaf axioms.

And the final equivalence is because if ψ^* maps $\mathcal{O}_Y(Y)$ into $\mathcal{O}_X(X)$ then it maps $x_i \in \mathcal{O}_Y(Y)$ to $\psi^*(x_i) = \psi_i$ into $\mathcal{O}_X(X)$. ∎

2.2.18 Corollary

For every SWF $X = (X, \mathcal{O}_X)$ there is a natural bijection

$$\hom_{\mathsf{SWF}}(X, \mathbb{A}^n) \xrightarrow{\sim} \mathcal{O}_X(X)^n, \qquad \psi \mapsto (\mathrm{proj}_1 \circ \psi, \ldots, \mathrm{proj}_n \circ \psi).$$

In particular there is a natural identification $\hom_{\mathsf{SWF}}(X, \mathbb{A}^1) \cong \mathcal{O}_X(X)$.
Furthermore, for $X = (X, \mathcal{O}_X)$ and a subset $Y \subseteq \mathbb{A}^n$, a set-theoretic map $\psi\colon X \to Y$ is a morphism of SWFs $(X, \mathcal{O}_X) \to (Y, \mathcal{O}_{reg})$ iff the composition $X \xrightarrow{\psi} Y \hookrightarrow \mathbb{A}^n$ is a morphism of SWFs.

Proof

The first point folows from the above theorem with $Y = \mathbb{A}^n$. The second follows from the theorem as well, since $\psi = (\psi_1, \ldots, \psi_n)$ being a morphism is equivalent to all ψ_i being regular which is equivalent to the composition of ψ with the inclusion being a morphism. ■

Note:
The second point follows directly from the fact that $(Y, \mathcal{O}_{reg})$ is the restriction of $(\mathbb{A}^n, \mathcal{O}_{reg})$ to Y.

Because for an affine scheme $(X, \mathcal{O}_X)$ by Serre we have an identifcation $\mathcal{O}_X(X) \cong k[X]$, the following is immediate.

2.2.19 Corollary

If $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ are affine varities, then $\psi\colon X \to Y$ is a morphism of SWFs iff it maps $k[Y]$ to $k[X]$.

There is an obvious functor $\mathsf{SWF}_k^{\mathrm{op}} \to \mathsf{Alg}_k$ given by mapping $(X, \mathcal{O}_X)$ to $\mathcal{O}_X(X)$ and $\varphi\colon X \to Y$ to $\varphi^*\colon \mathcal{O}_Y(Y) \to \mathcal{O}_X(X)$. This is called the **global sections** functor, as it takes global sections. It is denoted by Γ, so $\Gamma(X, \mathcal{O}_X)$ for objects (we may omit one of the two inputs when the other is obvious), and $\Gamma(\varphi) = \varphi^*$ on morphisms.

2.2.20 Lemma

A morphism of SWFs $\psi\colon X \to \mathbb{A}^n$ satisfies $\varphi(X) \subseteq Y$ for closed $Y \subseteq \mathbb{A}^n$ iff $\psi^*\colon k[\mathbb{A}^n] \to \mathcal{O}_X(X)$ satisfies $\mathcal{I}(Y) \subseteq \ker \psi^*$.

Proof

We have $\psi(X) \subseteq Y$ iff $\psi(x) \in Z(\mathcal{I}(Y))$ for all $x \in X$ and $f \in \mathcal{I}(Y)$, which is iff $f(\psi(x)) = \psi^*(f)(x) = 0$. So $\psi(X) \subseteq Y$ iff $\psi^*(f) = 0$ for all $f \in \mathcal{I}(Y)$, i.e. iff $\mathcal{I}(Y) \subseteq \ker \psi^*$. ■

2.2.21 Proposition

Let $(X, \mathcal{O}_X)$ be an SWF and $(Y, \mathcal{O}_Y)$ be an affine variety. Then Γ is bijective

$$\Gamma\colon \hom_{\mathsf{SWF}}(X, Y) \to \hom_{\mathsf{Alg}}(\mathcal{O}_Y(Y), \mathcal{O}_X(X)).$$

Proof

First take $Y = \mathbb{A}^n$, so $\mathcal{O}_Y(Y) = k[\mathbb{A}^n]$, the free k-algebra on n generators. Thus there is a natural bijection

$$\text{nat}\colon \hom_{\mathsf{Alg}}(k[\mathbb{A}^n], \mathcal{O}_X(X)) \to \mathcal{O}_X(X)^n, \qquad \varphi \mapsto (\varphi(x_1), \dots, \varphi(x_n)).$$

Now, the composition

$$\hom_{\mathsf{SWF}}(X, \mathbb{A}^n) \xrightarrow{\Gamma} \hom_{\mathsf{Alg}}(k[\mathbb{A}^n], \mathcal{O}_X(X)) \xrightarrow{\text{nat}} \mathcal{O}_X(X)^n$$

maps φ to $(\varphi^*(x_1), \dots, \varphi^*(x_n))$, and as we showed above this is a bijection. Thus Γ must be a bijection, as desired.

Now let $Y \subseteq \mathbb{A}^n$ be an arbitrary affine variety, so $\mathcal{O}_Y(Y) = k[Y] = k[\mathbb{A}^n]/\mathcal{I}(Y)$. Recall that $(Y, \mathcal{O}_Y)$ is induced from $(\mathbb{A}^n, \mathcal{O}_{reg})$ so SWF morphisms into Y are the same as SWF morphisms into $\mathbb{A}^n$ landing in Y. Thus by the previous lemma

$$\hom_{\mathsf{SWF}}(X, Y) \cong \{\varphi \in \hom_{\mathsf{SWF}}(X, \mathbb{A}^n) \mid \varphi(X) \subseteq Y\} = \{\varphi \in \hom_{\mathsf{SWF}}(X, \mathbb{A}^n) \mid \mathcal{I}(Y) \subseteq \ker \psi^*\},$$

and as we just showed, this is bijective to

$$\cong \{\varphi^* \in \hom_{\mathsf{Alg}}(k[\mathbb{A}^n], \mathcal{O}_X(X)) \mid \mathcal{I}(Y) \subseteq \ker \psi^*\},$$

and these are precisely the maps which quotient to $k[Y] = k[\mathbb{A}^n]/\mathcal{I}(Y) \to \mathcal{O}_X(X)$, i.e. this forms a bijection

$$\cong \hom_{\mathsf{Alg}}(k[Y], \mathcal{O}_X(X)).$$

Now it is easy to see that the composition of these isomorphisms is Γ. ■

2.2.22 Corollary

A SWF Y is an affine variety iff the k-algebra $\mathcal{O}_Y(Y)$ is fg and for every SWF X, Γ is bijective from $\hom_{\mathsf{SWF}}(X, Y)$.

Proof

We just showed for an affine variety, the property holds. Conversely, if $\mathcal{O}_Y(Y)$ is fg then it is equal to $\mathcal{O}_{Y'}(Y')$ for some affine variety Y' (we showed that reduced fg k-algebras B are all of the form $k[Y']$ for an affine Y', and $\mathcal{O}_Y(Y) \subseteq \text{Fun}(Y, k)$ is reduced always). Thus there is an isomorphism

$$\hom_{\mathsf{SWF}}(X, Y') \cong \hom_{\mathsf{Alg}}(\mathcal{O}_{Y'}(Y'), \mathcal{O}_X(X)) = \hom_{\mathsf{Alg}}(\mathcal{O}_Y(Y), \mathcal{O}_X(X)) \cong \hom_{\mathsf{SWF}}(X, Y),$$

where the final isomorphism is given by Γ. Thus $\hom(\bullet, Y') \cong \hom(\bullet, Y)$ naturally, so by Yoneda $Y' \cong Y$. ■

2.2.23 Corollary

Γ forms an anti-equivalence of categories from affine varieties to fg reduced k-algebras: $\mathsf{AffVar}_k \to \mathsf{Alg}_k^{\text{fg,red}}$. In particular Γ reflects isomorphisms, so $X \cong Y$ iff $\mathcal{O}_X(X) \cong \mathcal{O}_Y(Y)$ for affine X, Y.

Proof

We showed that Γ is fully faithful on affine varieties, and so it is an equivalence to its essential image. Its essential image are the fg reduced k-algebras.

We can explicitly construct its inverse: for a fg reduced k-algebra A, then we map it to $\operatorname{Specm}(A)$, and maps $\varphi\colon A \to B$ are mapped to $|\varphi^*|$. ■

2.2.24 Corollary

Let $X \subseteq \mathbb{A}^n$ and $Y \subseteq \mathbb{A}^m$ be affine varieties. A map $\varphi\colon X \to Y$ is a morphism of varieties iff $\varphi = (f_1, \ldots, f_m)$ for some $f_i \in k[\mathbb{A}^n]$.

Proof

φ is a morphism iff it is a morphism into $\mathbb{A}^m$, which is iff φ's projections are in $\mathcal{O}_X(X) = k[X]$, which are just the restriction of polynomial functions to X. ■

2.2.25 Definition

Let X be a topological space. A subset $Y \subseteq X$ is **locally closed** iff $Y = U \cap F$ for open $U \subseteq X$ and closed $F \subseteq X$.

The following lemma better explains the reasoning for the terminology of "locally closed".

2.2.26 Lemma

For $Y \subseteq X$ a subspace, the following are equivalent:

(1) Y is locally closed,

(2) Y is an open subset of a closed subset,

(3) Y is a closed subset of an open subset,

(4) for every $y \in Y$, there is a neighborhood U of y in X such that $Y \cap U$ is closed in U.

Proof

Recall that open (resp. closed) subsets of a subspace S are of the form $S \cap U$ for U open (resp. $S \cap F$ for F closed). So an open subset of a closed subset F are the sets of the form $U \cap F$ for U open, and similarly for closed subsets of open subsets. Thus we see the equivalence of (1), (2), and (3).

Now if Y is locally closed, then it is the closed subset of an open subset U. Thus every $y \in Y$ has neighborhood U in which Y is closed, showing (4). Conversely, for each $y \in Y$ let $y \in U_y$ be a neighborhood where $Y \cap U_y$ is closed in U_y. Let $U = \bigcup_y U_y$, so

$$U - Y = \bigcup_y (U_y - Y),$$

and since $Y \cap U_y$ is closed in U_y, we have $U_y - Y = U_y - Y \cap U_y$ is open, so $U - Y$ is open and thus Y is closed in U. ■

2.2.27 Corollary

A locally closed subset of a locally closed subset is locally closed.

Proof

So suppose $Z \subseteq Y \subseteq X$ is a chain of inclusions, where each subset is locally closed in the next. It is sufficient to show that an open/closed subset of Y is locally closed in X, since if $Z = U \cap F$ in Y, then U, F are locally closed in X and then clearly their intersection is too. But clearly by (2) an open subset of Y is locally closed, and by (3) the same for a closed subset. ■

2.2.28 Proposition

Let $X \subseteq \mathbb{A}^n$ be an affine variety, and let $f \in k[\mathbb{A}^n]$. Then $(D_X(f), \mathcal{O}_{reg})$ is an affine variety isomorphic to $(Y, \mathcal{O}_{reg})$ where $Y \subseteq \mathbb{A}^{n+1}$ is the closed subset $Z(\mathcal{I}(X) \cup \{fx_{n+1} - 1\})$.

Proof

We construct explicit inverse morphisms $\varphi\colon D_X(f) \to Y \colon \psi$. Define φ by mapping $(x_1, \ldots, x_n) \in X \subseteq \mathbb{A}^n$ to

$$\varphi(x_1, \ldots, x_n) = \left(x_1, \ldots, x_n, \frac{1}{f(x_1, \ldots, x_n)}\right).$$

Its image lies in Y clearly, and its components are rational functions on $D_X(f)$ so it forms a morphism into $\mathbb{A}^n$, thus into Y. Similarly define ψ by mapping $(x_1, \ldots, x_{n+1}) \in Y$ to

$$\psi(x_1, \ldots, x_{n+1}) = (x_1, \ldots, x_n).$$

Its image lies in X since $f(x_1, \ldots, x_n)x_{n+1} = 1$ so $f(x_1, \ldots, x_n) \neq 0$, and it is a morphism into $\mathbb{A}^n$ and thus into X. Clearly $\psi \circ \varphi = \mathrm{id}$ and for $(x_1, \ldots, x_{n+1}) \in Y$ we must have $x_{n+1} = 1/f(x_1, \ldots, x_n)$ by definition so $\varphi \circ \psi = \mathrm{id}$. ■

2.2.29 Example

The SWF $(D_{\mathbb{A}^1}(x), \mathcal{O}_{reg})$ is isomorphic to the hyperbola $Z(xy - 1) \subseteq \mathbb{A}^2$.

We can also prove the above proposition via Yoneda. Define $B = k[X]_f$. Now we know that Γ defines a bijection $\hom_{\mathsf{SWF}}(Z, X) \to \hom_{\mathsf{Alg}}(k[X], \mathcal{O}_Z(Z))$ for any SWF Z. Furthermore, the universal property of localization says that a morphism $\zeta\colon k[x] \to \mathcal{O}_Z(Z)$ factors through B iff $\zeta(f) \in \mathcal{O}_Z(Z)^\times$, and such a factorization is unique. And by locality, $\mathcal{O}_Z(Z)^\times$ are just the nowhere vanishing functions in $\mathcal{O}_Z(Z)$. Thus Γ restricts to a bijection

$$\hom_{\mathsf{Alg}}(k[X]_f, \mathcal{O}_Z(Z)) \cong \{\varphi \in \hom_{\mathsf{SWF}}(Z, X) \mid \varphi^*(f)(z) \neq 0 \text{ for all } z \in Z\}.$$

And we know

$$\varphi^*(f)(z) = f(\varphi(z)),$$

so $\varphi^*(f)(z) \neq 0$ iff $\varphi(z) \in D_X(f)$, thus

$$\hom_{\mathsf{Alg}}(k[X]_f, \mathcal{O}_Z(Z)) \cong \{\varphi \in \hom_{\mathsf{SWF}}(Z, X) \mid \varphi(Z) \subseteq D_X(f)\}.$$

And since $D_X(f) \subseteq X$ has the induced SWF structure (as they are both induced from $\mathbb{A}^n$), this means Γ defines a natural isomorphism

$$\hom_{\mathsf{Alg}}(k[X]_f, \mathcal{O}_Z(Z)) \cong \hom_{\mathsf{SWF}}(Z, D_X(f)).$$

Now $k[X]_f$ is a reduced fg k-algebra, and thus it is equivalent to some affine variety Y under Γ (so $\mathcal{O}_Y(Y) = k[X]_f$), so we have natural isomorphisms

$$\hom_{\mathsf{SWF}}(Z, Y) \cong \hom_{\mathsf{Alg}}(k[X]_f, \mathcal{O}_Z(Z)) \cong \hom_{\mathsf{SWF}}(Z, D_X(f)),$$

and by Yoneda $Y \cong D_X(f)$ as desired.

Notice that this means $k[D_X(f)] \cong k[X]_f$ (the left-hand side is $\mathcal{O}_{reg}(D_X(f))$ which are rational functions on $D_X(f)$). Explicitly this isomorphism is given by the universal property of localization applied to the restriction $k[X] \to k[D_X(f)]$.

This immediately shows the following two corollaries.

2.2.30 Corollary

For $X \subseteq \mathbb{A}^n$ affine and $f \in k[\mathbb{A}^n]$, we have

$$\mathcal{O}_{reg}(D_X(f)) = k[\mathbb{A}^n]|_{D_X(f)}[1/f] \subseteq \mathrm{Fun}(D_X(f), k),$$

i.e. regular functions on $D_X(f)$ are of the form p/f for $p \in k[\mathbb{A}^n]$.

2.2.31 Corollary

If $(X, \mathcal{O}_X)$ is an affine variety, then for every $f \in \mathcal{O}_X(X)$, the SWF $(D_X(f), \mathcal{O}_X|_{D_X(f)})$ is affine. Furthermore, the restriction map $\mathcal{O}_X(X) \to \mathcal{O}_X(D_X(f))$ induces an isomorphism of k-algebras

$$\mathcal{O}_X(X)_f \xrightarrow{\sim} \mathcal{O}_X(D_X(f)).$$

Note:
Since $D_X(f) \subseteq X$ is open, the restriction of $\mathcal{O}_X$ to $D_X(f)$ is equal to $\mathcal{O}_X$ but on the subposet of open subsets of $D_X(f)$.

2.2.32 Corollary

An affine variety has a basis consisting of affine varieties.

Proof

The $D_X(f)$ form an open basis for X (since they do for $\mathbb{A}^n$, and X is a subspace and $D_X(f) = D_{\mathbb{A}^n}(f) \cap X$), and they are affine. ■

2.2.33 Definition

Let $X = (X, \mathcal{O}_X)$ be an algebraic variety. A **closed/open/locally closed subvariety** of X is an SWF $(Y, \mathcal{O}_X|_Y)$ where $Y \subseteq X$ is closed/open/locally closed. A **quasi-affine variety** is an open subvariety of an affine variety.

2.2.34 Corollary

A closed/open/locally closed subvariety of an algebraic variety is an algebraic variety.

Proof

Let X_i be a finite subcovering of X of affine varieties. Then $Y = \bigcup_i (Y \cap X_i)$, and to show that Y is an algebraic variety it is sufficient to show that $Y \cap X_i$ is. So replacing X by X_i and Y by $Y \cap X_i$, we assume X is affine.

We consider the cases when Y is open and closed, and this gives us the case when Y is locally closed, since if Y is open in a closed subset then it is an open subset of an algebraic variety which is then an algebraic variety.

If Y is closed, then $(Y, \mathcal{O}_X|_Y)$ is affine since following the isomorphism to a closed subset of $\mathbb{A}^n$ we see that Y is a closed subset of $\mathbb{A}^n$ and thus an affine variety.

Now if $Y \subseteq X$ is open, then $\{D_X(f)\}_{f \in \mathcal{O}_X(X)}$ is a basis of X, and since X is Noetherian, Y is quasi-compact, so it is the finite union of $D_X(f)$s. Each of these are affine varieties, so Y is an algebraic variety by definition. ∎

2.2.35 Example

We have shown some quasi-affine varieties are affine, like $D_X(f)$. But not all!

For example let $X = \mathbb{A}^2 - 0$, which is quasi affine as it is equal to $D_{\mathbb{A}^2}(x_1) \cup D_{\mathbb{A}^2}(x_2)$. The inclusion $X \hookrightarrow \mathbb{A}^2$ is not an isomorphism, as it is not a homeomorphism, but the induced map $\mathrm{res}\colon k[\mathbb{A}^2] \to \mathcal{O}_X(X)$ is an isomorphism. This cannot happen for affine varieties (Γ reflects isomorphisms).

We see that X is the pushout of $D(x_1), D(x_2)$ along their intersection. Thus $\mathcal{O}_X(X)$ is the pullback of $k[x_1,x_2]_{x_1}, k[x_1,x_2]_{x_2}$ along $k[x_1,x_2]_{x_1x_2}$. That is, $\mathcal{O}_X(X)$ consists of pairs $f_i \in k[x_1,x_2]_{x_i}$ such that $f_1 = f_2$ in $k[x_1,x_2]_{x_1x_2}$. But a rational function whose denominator is both a power of x_1 and a power of x_2 must be a polynomial, so $\mathcal{O}_X(X) = k[\mathbb{A}^2]$.

Some notes on the category SWF_k

There is a clear forgetful functor from SWF_k to the category of topological spaces, $U\colon \mathsf{SWF}_k \to \mathsf{Top}$, which maps $(X, \mathcal{O}_X)$ to its underlying topological space X and a morphism $\varphi\colon (X,\mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ to its underlying continuous map φ. This functor is faithful, but not full (it is easy to construct a continuous map which is not a morphism of SWFs). Though it is a left-adjoint, as we will show.

2.2.36 Definition

For a topological space X and an arbitrary set S, a function $f\colon X \to S$ is **locally constant** if for every $p \in X$ there is a neighborhood $p \in U \subseteq X$ such that $f|_U$ is constant.

Note that $f\colon X \to S$ being locally constant is the same as it being continuous when S is endowed with the discrete topology. Indeed, if f is locally constant then for every $p \in f^{-1}\{x\}$ there is an open neighborhood of p contained in $f^{-1}\{x\}$ by definition, so $f^{-1}\{x\}$ is open. Thus f is continuous to discrete S. Conversely, f being continuous to discrete S means that $f^{-1}\{x\}$ is open and so for every $p \in X$ we take the neighborhood $U = f^{-1}(fp)$ to see it is locally constant.

2.2.37 Lemma

For an SWF $(X, \mathcal{O}_X)$, $\mathcal{O}_X(U)$ contains all locally constant functions $U \to k$.

Proof

Let $f\colon U \to k$ be locally constant, so there is an open cover U_i of U on which $f|_{U_i}$ is constant. Since $\mathcal{O}_X(U_i)$ is a k-algebra, it must contain all constant functions (since it contains the unit and all of its multiples), so $f|_{U_i} \in \mathcal{O}_X(U_i)$. By gluing, this means $f \in \mathcal{O}_X(U)$ as desired. ∎

2.2.38 Proposition

The forgetful functor $U\colon \mathsf{SWF}_k \to \mathsf{Top}$ is a left-adjoint.

Proof

Forgetful functors are in general "the most general" object, so in our case in lieu of the previous lemma it seems that the following construction is a good contender. For open $U \subseteq X$ define $\mathcal{O}_X(U)$ to be the collection of locally constant $U \to k$. Thus $\mathcal{O}_X$ is just the sheaf of continuous maps into k, which is always a sheaf.

For locality, let $f \in \mathcal{O}_X(U)$ then its preimage on all subsets of k is open, in particular $D_U(f)$. And if f is locally constant, so too is $1/f$ (f and $1/f$ are constant on the same open subsets). Thus $(X, \mathcal{O}_X)$ is an SWF which is denoted $\underline{X}$.

We claim that $U \dashv \underline{\bullet}$. This amounts to constructing a natural isomorphism

$$\hom_{\mathsf{Top}}(U(X, \mathcal{O}_X), Y) \cong \hom_{\mathsf{SWF}}((X, \mathcal{O}_X), \underline{Y}).$$

U defines an injective map from the right to the left, so we just need to show that it is surjective. That is, for $\varphi\colon X \to Y$ continuous, φ lifts to a morphism of SWFs $(X, \mathcal{O}_X) \to \underline{Y}$. So for $U \subseteq Y$ open we need to show φ^* maps $\mathcal{O}_Y(U) \to \mathcal{O}_X(\varphi^{-1}U)$.

For $f\colon U \to k$ locally constant, then $f \circ \varphi\colon \varphi^{-1}U \to k$ is locally constant as well, as the composition of continuous maps. Thus $\varphi^*(f) \in \mathcal{O}_X(\varphi^{-1}U)$ by the previous lemma. ■

In particular U preserves colimits. The following is a technical definition and lemma which we will not prove in full here (see homework).

2.2.39 Definition

Let $\mathcal{B}$ be a basis for a topology, then a **B-SWF** over k is a presheaf $\mathcal{O}\colon \mathcal{B}^{\mathsf{op}} \to \mathsf{Alg}_k$ satisfying the following axioms:

(1) $\mathcal{O}$ is a subpresheaf of the function sheaf, $\mathrm{Fun}(\bullet, k)$.

(2) **B-sheaf axiom**: for any basic set $B \in \mathcal{B}$ and cover of basic sets $B = \bigcup_i B_i$, if $f\colon B \to k$ is a function such that $f|_{B_i} \in \mathcal{O}(B_i)$ for all i, then $f \in \mathcal{O}(B)$.

(3) **locality**: for any $B \in \mathcal{B}$ and $f \in \mathcal{O}(B)$, $D_B(f)$ is open (i.e. the union of basic sets) and for every basic cover B_i of $D_B(f)$, we have that $1/f|_{B_i} \in \mathcal{O}(B_i)$.

2.2.40 Proposition

For every B-SWF $(\mathcal{B}, \mathcal{O})$ where $\mathcal{B}$ is a basis of the topology on X, there is a unique SWF $(X, \mathcal{O}_X)$ extending it, meaning $\mathcal{O}_X(B) = \mathcal{O}(B)$ for all $B \in \mathcal{B}$.

Proof

The construction is as follows: for every $U \subseteq X$ open and $\{B_i\} \subseteq \mathcal{B}$ covering U, define

$$\mathcal{O}_X(U) = \{f\colon U \to k \mid f|_{B_i} \in \mathcal{O}(B_i) \text{ for all } i\}\,.$$

Then one can show without much difficulty that $\mathcal{O}_X(U)$ is independent on the choice of basic covering. In particular this means that $\mathcal{O}_X(B) = \mathcal{O}(B)$ for all $B \in \mathcal{B}$ by choosing the covering $\{B\}$.

It is then easy to show that this is a sheaf, and locality follows easily from locality of $(\mathcal{B}, \mathcal{O})$. ■

Now we will show that SWF_k is cocomplete. Since U preserves colimits, we already know what the underlying topological spaces of the colimits are.

2.2.41 Lemma

Let $\pi\colon X \to Y$ be a quotient map, and suppose there is an SWF structure $(X, \mathcal{O}_X)$ on X. Then there is a unique SWF structure $(Y, \mathcal{O}_Y)$ given by the following universal property: $\varphi\colon (Y, \mathcal{O}_Y) \to (Z, \mathcal{O}_Z)$ is a morphism of SWFs iff $\varphi \circ \pi\colon (X, \mathcal{O}_X) \to (Z, \mathcal{O}_Z)$ is a morphism of SWFs.

Proof

This is a universal property and thus a solution is unique up to unique isomorphism. Define

$$\mathcal{O}_Y(U) = \left\{ f\colon U \to k \;\middle|\; \pi^*(f) \in \mathcal{O}_X(\pi^{-1}(U)) \right\}.$$

It is not hard to show that this satisfies the universal property. ■

2.2.42 Theorem

The category SWF_k is cocomplete.

Proof

We just need to show that SWF_k has coproducts and coequalizers. For coproducts, let $\{X_i\}_i$ be a collection of SWFs and let $X = \coprod_i X_i$. It has a basis of open subsets given by the union of all open subsets of X_i. Then define $\mathcal{O}_X$ on X as a B-SWF where for $U \subseteq X_i$ open we define

$$\mathcal{O}_X(U) = \mathcal{O}_{X_i}(U).$$

It is easy to see that this is a B-SWF, and thus defines an SWF on X. It is also not hard to see that this has the universal property of a coproduct.

For coequalizers, let $\varphi, \psi\colon X \rightrightarrows Y$ be parallel morphisms of SWFs. The underlying space of their coequalizer will be $Y/\!\sim$ where $\sim$ is the equivalence relation generated by $\varphi(x) \sim \psi(x)$ for all $x \in X$. This is a quotient space, so we just give it the quotient space SWF structure from the previous lemma. This clearly satisfies the universal property of coequalizers as a direct result from the universal property of quotients. ■

Note:

While the colimit of SWFs is an SWF, the colimit of algebraic varieties need not be a variety!

2.2.43 Corollary

The global sections functor Γ contravariantly preserves colimits.

Proof

Since Γ is contravariant, preserving colimits means mapping colimiting cones to limit cones. So it is sufficient to show that

$$\Gamma\left(\coprod_i (X_i, \mathcal{O}_{X_i})\right) \cong \prod_i \Gamma(X_i, \mathcal{O}_{X_i}), \qquad \Gamma(\mathrm{coeq}(\varphi, \psi)) \cong \mathrm{eq}(\Gamma(\varphi), \Gamma(\psi)),$$

in a way which maps the colimit cone to the limit cone, which we leave as an exercise.
For coproducts, by the above construction we see that for $X = \coprod_i X_i$,

$$\Gamma\left(\coprod_i (X_i, \mathcal{O}_{X_i})\right) = \{f\colon X \to k \mid f|_{X_i} \in \mathcal{O}_{X_i}(X_i)\},$$

so the map $\Gamma(X) \to \prod_i \Gamma(X_i)$ given by $f \mapsto (f|_{X_i})$ is a natural isomorphism.
For a quotient map $\pi\colon X \to Y$, we have

$$\Gamma(Y) = \{f\colon Y \to k \mid f \circ \pi \in \mathcal{O}_X(X)\},$$

meaning $\pi^*\colon \Gamma(Y) \to \Gamma(X)$ is an injection (by π's surjectivity), meaning $\Gamma(Y) \cong \mathrm{im}\pi^*$. So letting $Z = \mathrm{coeq}(\varphi, \psi)$ we have

$$\Gamma(\mathrm{coeq}(\varphi, \psi)) = \{f\colon Z \to k \mid f \circ \pi \in \mathcal{O}_Y(Y)\} \cong \mathrm{im}\pi^*.$$

We know that

$$\mathrm{eq}(\Gamma\varphi, \Gamma\psi) = \ker(\varphi^* - \psi^*).$$

So we want to show that

$$\mathrm{im}\pi^* = \ker(\varphi^* - \psi^*).$$

Firstly, we have $\pi \circ \varphi = \pi \circ \psi$ by definition so $\varphi^* \circ \pi^* = \psi^* \circ \pi^*$ and thus $(\varphi^* - \psi^*) \circ \pi^* = 0$. So we have left-to-right inclusion. Now suppose that $f \circ \varphi = f \circ \psi$ for $f \in \Gamma(Y)$, then by the equivalence on Z this means f lifts to a map $g\colon Z \to k$, i.e. $f = g \circ \pi$. So by definition $g \in \Gamma(Z)$ since $f \in \Gamma(Y)$, and this shows that $f \in \mathrm{im}\pi^*$ as desired. ■

2.3 Quasi-projective varieties

2.3.1 Definition

A **graded ring** is a ring S together with a decomposition $S = \bigoplus_{i \geq 0} S_i$, where each $S_i \subseteq S$ is an Abelian group and $S_i \cdot S_j \subseteq S_{i+j}$. In particular $S_0 \subseteq S$ is a subring, and $S_i \subseteq S$ is an S_0-submodule, and the direct sum $S_+ = \bigoplus_{i>0} S_i$ is an ideal of S.

Elements of S_i are called **homogeneous of degree** i, and an element is just **homogeneous** if it is homogeneous of some degree. Denote the set of homogeneous elements of S by S_{hom}.

An ideal $I \leq S$ is **homogeneous** if it respects the decomposition: $I = \bigoplus_{i \geq 0}(I \cap S_i)$.

2.3.2 Example

Let $S = k[x_1, \ldots, x_n]$, then S is graded by $S_d = \mathrm{span}_k \left\{x_1^{i_1} \cdots x_n^{i_n} \mid i_1 + \cdots + i_n = d\right\}$, called the space of homogeneous polynomials of degree d.

For S graded and $a \in S$, we can uniquely write a as $a = \sum_d a_d$ for $a_d \in S_d$, all but finitely many being zero. The element a_d is called the **homogeneous part of** a **of degree** d. Clearly $(a+b)_d = a_d + b_d$ so the map $S \to S_d$ mapping an element to its homogeneous part of degree d is an Abelian group morphism. Furthermore,

$$\left(\sum_d a_d\right)\left(\sum_d b_d\right) = \sum_d \sum_k a_k b_{d-k}, \qquad \sum_k a_k b_{d-k} \in S_d,$$

so that $(ab)_d = \sum_k a_k b_{d-k}$.

We let $\deg(a)$ denote the maximal d such that $a_d \neq 0$. So we see that if $\deg(a) \leq n$ and $\deg(b) \leq m$, then $\deg(ab) \leq n + m$ and $(ab)_{n+m} = a_n b_m$.

And clearly an ideal I is homogeneous iff for all $a \in I$ we have $a_d \in I$.

2.3.3 Lemma

(1) An ideal $I \leq S$ is homogeneous iff it is generated by homogeneous elements, so $I = (I \cap S_{\text{hom}})$.

(2) Homogeneous ideals are closed under sums, products, intersections, and radicals.

(3) A non-unit homogeneous ideal $I \leq S$ is prime iff for all $a, b \in S_{\text{hom}}$ such that $a, b \notin I$ we have $ab \notin I$.

(4) A finitely generated homogeneous ideal is generated by finitely many homogeneous elements.

Proof

(1) If $I \leq S$ is homogeneous, then

$$I = \sum_i (I \cap S_i) = \left(\bigcup I \cap S_i\right) = (I \cap S_{\text{hom}}).$$

Conversely suppose (b_i) generate I, where $b_i \in S_{d_i}$. Then elements of I are of the form $a = \sum_i a_i b_i$ with $a_i \in S$. Then

$$a_d = \sum_i (a_i b_i)_d = \sum_i (a_i)_{d-d_i} b_i \in I,$$

so I is homogeneous.

(2) This is clear from (1) for sums and products. And if $a \in I = \bigcap_i I_i$ then $a_d \in I_i$ for all i so $a_d \in I$ and thus I is homogeneous.

Now if I is homogeneous, we induct on d to show that for every $f \in \sqrt{I}$ such that $f = \sum_{i=0}^d f_i$, $f_i \in \sqrt{I}$. For $d = 0$ we have $f = f_0$ so this is clear. Now inductively, it is sufficient to show that $f_d \in \sqrt{I}$ since then $f - f_d = \sum_{i=0}^{d-1} f_i \in \sqrt{I}$ and we proceed by induction.

By definition there is some n such that $f^n \in I$. Now, we know $\deg(f) \leq d$ and thus inductively $\deg(f^n) \leq dn$ so $(f^n)_{dn} = f_d^n$. Since $f^n \in I$ we have $f_d^n = (f^n)_{dn} \in I$ and thus $f_d \in \sqrt{I}$ as desired.

(3) One direction is clear. Now suppose we have that $ab \notin I$ when $a, b \in S_{\text{hom}} - I$. For $a, b \in S - I$ we have $a = \sum_i a_i$ and $b = \sum_j b_j$. Let us write $a_I = \sum_{i, a_i \in I} a_i$ and $a^I = \sum_{i, a_i \notin I} a_i$ so $a = a_I + a^I$ and similarly $b = b_I + b^I$. So $a_I, b_I \in I$ and

$$ab = a_I b_I + a_I b^I + a^I b_I + a^I b^I,$$

so it is sufficient to show that $a^I b^I \notin I$.

That is, it is sufficient to consider the case when $a_i \notin I$ and $b_i \notin I$ to show $ab \notin I$. Let $n = \deg(a)$ and $m = \deg(b)$ so $(ab)_{n+m} = a_n b_m$, and since $a_n, b_m \in S_{\text{hom}} - I$ we have $a_n b_m \notin I$, and by homogeneity $ab \notin I$.

(4) Let $T \subseteq I$ be finite and generate I. Then let

$$T_{\text{hom}} = \{a_i \mid a \in T, i \geq 0\}.$$

Since I is homogeneous, $T_{\text{hom}} \subseteq I$. And since elements of T are finite sums of elements from T_{hom}, we have $I = (T) \subseteq (T_{\text{hom}}) \subseteq I$ so $(T_{\text{hom}}) = I$. Clearly T_{hom} is a finite subset of S_{hom}. ■

For an algebraically closed field k, the group $k^\times$ acts on $\mathbb{A}^n = k^n$ via multiplication: for $P = (a_1, \dots, a_n) \in k^n$ and $\lambda \in k^\times$ we have the action $\lambda P = (\lambda a_1, \dots, \lambda a_n)$.

For the graded ring $S = k[\mathbb{A}^n]$ with grading $S_d = \operatorname{span}_k \left\{x^I \;\middle|\; \sum I = d\right\}$, we see that for $f \in S_d$ we have $f(\lambda P) = \lambda^d f(P)$. This is clear for monomials x^I, and follows in general by linearity.

2.3.4 Lemma

A closed subset $Z \subseteq \mathbb{A}^n$ is invariant under $k^\times$'s action iff $\mathcal{I}(Z) \subseteq S$ is homogeneous. For a general subset $X \subseteq \mathbb{A}^n$, if X is $k^\times$-invariant then $\mathcal{I}(X) \leq S$ is homogeneous (only one direction of the previous equivalence holds).

Proof

Suppose $\mathcal{I}(Z) \subseteq S$ is homogeneous, and let $P \in Z$ and $\lambda \in k^\times$. Since Z is closed, $Z = Z(\mathcal{I}(Z))$ it is sufficient to show that for $f \in \mathcal{I}(Z)$ we have $f(\lambda P) = 0$. If f is homogeneous, this is clear as $f(\lambda P) = \lambda^d f(P) = 0$. Then in general we have $f = \sum_d f_d$ and $f_d \in \mathcal{I}(Z)$ so $f_d(\lambda P) = 0$ and thus $f(\lambda P) = 0$.

Now if X is $k^\times$-invariant, we want to show that if $f \in \mathcal{I}(X)$ then $f_d \in \mathcal{I}(X)$. Fix $P \in X$, so we want to show $f_d(P) = 0$ for all d. For every $\lambda \in k^\times$, by invariance we have $\lambda P \in X$ so $f(\lambda P) = 0$. Thus

$$0 = f(\lambda P) = \sum_d f_d(\lambda P) = \sum_d \lambda^d f_d(P).$$

That is, the polynomial $Q(t) = \sum_d f_d(P) x^d$ satisfies $Q(\lambda) = 0$ for all $\lambda \in k^\times$. So Q has infinitely many roots, and thus must be the zero polynomial, so $f_d(P) = 0$ for all d, as desired. ■

Notice that since $\overline{X} = Z(\mathcal{I}(X))$ and thus $\mathcal{I}(\overline{X}) = \mathcal{I}(X)$, we see that if X is $k^\times$-invariant so too is $\overline{X}$.

We view $S = k[\mathbb{A}^{n+1}]$ as the graded k-algebra $k[x_0, \ldots, x_n]$, so we begin indexing at 0 here. Since $\mathbb{A}^{n+1} - 0$ is $k^\times$-invariant, we can quotient by $k^\times$ to get $\mathbb{P}^n_k = \mathbb{P}^n$. That is, we define $\mathbb{P}^n$ to be the set of all $k^\times$-orbits of $\mathbb{A}^{n+1} - 0$. We denote the orbit of $(a_0, \ldots, a_n) \in \mathbb{A}^{n+1} - 0$ by $[a_0 : \ldots : a_n] \in \mathbb{P}^n$. Note that equivalently, $\mathbb{P}^n$ can be viewed as the set of all lines in $\mathbb{A}^{n+1}$ passing through the origin, and then $[a_0 : \ldots : a_n]$ corresponds to the line $t(a_0, \ldots, a_n)$.

Let $\pi \colon \mathbb{A}^{n+1} - 0 \to \mathbb{P}^n$ be the canonical projection map: $\pi(a_0, \ldots, a_n) = [a_0 : \ldots : a_n]$. This is surjective, and thus we can endow $\mathbb{P}^n$ with the quotient topology. In particular this makes $\mathbb{P}^n$ into an SWF $(\mathbb{P}^n, \mathcal{O}_{reg})$ where

$$\mathcal{O}_{reg}(U) = \left\{ f \colon U \to k \;\middle|\; f \circ \pi \in \mathcal{O}_{reg}(\pi^{-1}U) \right\}.$$

That is, $f \colon U \to k$ is regular iff the pullback $\pi^*(f) \colon \pi^{-1}U \to k$ is regular.

2.3.5 Definition

For a homogeneous polynomial $f \in S_{\text{hom}}$ and $P \in \mathbb{P}^n$ we write $f(P) = 0$ to mean that $f(\tilde{P}) = 0$ for some representative (equivalently every representative) $\tilde{P} \in \pi^{-1}(P)$. For $f \in S_{\text{hom}}$ let

$$Z_{\mathbb{P}^n}(f) = \{ P \in \mathbb{P}^n \mid f(P) = 0 \},$$

and in general for $T \subseteq S_{\text{hom}}$ let

$$Z_{\mathbb{P}^n}(T) = \bigcap_{f \in T} Z_{\mathbb{P}^n}(f).$$

For an homogeneous ideal $I \leq S$ we define $Z_{\mathbb{P}^n}(I) = Z_{\mathbb{P}^n}(I \cap S_{\text{hom}})$. If T is any of the above (an homogeneous polynomial/set/ideal) then we define $D_{\mathbb{P}^n}(T) = \mathbb{P}^n - Z_{\mathbb{P}^n}(T)$.

Finally for any $Y \subseteq \mathbb{P}^n$ we define the homogeneous ideal $\mathcal{I}_{\text{hom}}(Y) \leq S$ by

$$\mathcal{I}_{\text{hom}}(Y) = \langle f \in S_{\text{hom}} \mid f(P) = 0 \text{ for all } P \in Y \rangle.$$

As it is generated by homogeneous elements, it is an homogeneous ideal.

2.3.6 Lemma

(1) $Z_{\mathbb{P}^n}$ is order reversing,

(2) $Z_{\mathbb{P}^n}(0) = \mathbb{P}^n$ and $Z_{\mathbb{P}^n}(1) = \emptyset$,

(3) $Z_{\mathbb{P}^n}(T_1 \cdot T_2) = Z_{\mathbb{P}^n}(T_1) \cup Z_{\mathbb{P}^n}(T_2)$ and $Z_{\mathbb{P}^n}(\bigcup_i T_i) = \bigcap_i Z_{\mathbb{P}^n}(T_i)$,

(4) For every $T \subseteq S_{\text{hom}}$, there is an equality $Z_{\mathbb{P}^n}(T) = Z_{\mathbb{P}^n}((T))$.

Proof

Points (1) and (2) are clear. Point (3) is shown in essentially the same way as in the affine case. For point (4) we have

$$Z_{\mathbb{P}^n}((T)) = Z_{\mathbb{P}^n}((T) \cap S_{\text{hom}}),$$

and $T \subseteq (T) \cap S_{\text{hom}}$ so we have an inclusion $Z_{\mathbb{P}^n}((T)) \subseteq Z_{\mathbb{P}^n}(T)$. Conversely let $P \in Z_{\mathbb{P}^n}(T)$ and $f \in (T) \cap S_{\text{hom}}$ then $f = \sum_i a_i t_i$ for $a_i \in S$ and $t_i \in T$. Suppose $\deg(f) = d$ then $\deg(a_i t_i) = d$ for all i which means that each a_i must be homogeneous as well and $\deg(a_i) = d - \deg(t_i)$. Since everything is homogeneous, since $t_i(P) = 0$ for all i we have $f(P) = 0$ so we have the other inclusion as well. ■

2.3.7 Proposition

(1) For $T \subseteq S_{\text{hom}}$ we have the equality

$$\pi^{-1} Z_{\mathbb{P}^n}(T) = Z_{\mathbb{A}^{n+1}-0}(T).$$

(2) $Z \subseteq \mathbb{P}^n$ is closed iff $Z = Z_{\mathbb{P}^n}(T)$ for some $T \subseteq S_{\text{hom}}$.

(3) The collection $\{D_{\mathbb{P}^n}(f)\}_{f \in S_{\text{hom}}}$ forms a basis of open sets of $\mathbb{P}^n$.

Proof

We know that for $f \in S_{\text{hom}}$ and $\tilde{P} \in \mathbb{A}^{n+1} - 0$, $f(\tilde{P}) = 0$ iff $f(\pi(\tilde{P})) = 0$. Thus $\pi(\tilde{P}) \in Z_{\mathbb{P}^n}(T)$ iff $f(\pi(\tilde{P})) = 0$ for all $f \in T$ iff $f(\tilde{P}) = 0$ for all $f \in T$ iff $\tilde{P} \in Z_{\mathbb{A}^{n+1}-0}(T)$ as desired.

The preimage $\pi^{-1} Z_{\mathbb{P}^n}(T) = Z_{\mathbb{A}^{n_1}-0}(T)$ is closed in $\mathbb{A}^{n+1} - 0$. Since $\mathbb{P}^n$ has the quotient topology, this means $Z_{\mathbb{P}^n}(T)$ is closed in $\mathbb{P}^n$. Conversely if $Z \subseteq \mathbb{P}^n$ is closed, then $\pi^{-1} Z \subseteq \mathbb{A}^{n+1} - 0$ is closed and $k^\times$-invariant. Since $\overline{\pi^{-1} Z} = Z_{\mathbb{A}^{n+1}}(\mathcal{I}(\pi^{-1} Z))$, we see that $I = \mathcal{I}(\pi^{-1} Z)$ is homogeneous. So $I = (I \cap S_{\text{hom}})$ and thus $\overline{\pi^{-1} Z} = Z_{\mathbb{A}^n}(I) = Z_{\mathbb{A}^n}(I \cap S_{\text{hom}})$ so we have equality

$$\pi^{-1} Z = \overline{\pi^{-1} Z} \cap (\mathbb{A}^{n+1} - 0) = Z_{\mathbb{A}^{n+1}-0}(I \cap S_{\text{hom}}) = \pi^{-1} Z_{\mathbb{P}^n}(I \cap S_{\text{hom}}).$$

Thus we see $Z = Z_{\mathbb{P}^n}(I \cap S_{\text{hom}})$ as desired. The reverse direction is clear since $\pi^{-1} Z_{\mathbb{P}^n}(T) = Z_{\mathbb{A}^{n+1}-0}(T)$ is closed.

Every closed set in $\mathbb{P}^n$ is of the form $Z_{\mathbb{P}^n}(T) = \bigcap_{f \in T} Z_{\mathbb{P}^n}(f)$ so all open sets are unions of $D_{\mathbb{P}^n}(f)$. ■

For homogeneous polynomials of the same degree $g, h \in S_d$ and $\tilde{P} \in D_{\mathbb{A}^{n+1}}(h)$ and $\lambda \in k^\times$ we see

$$\frac{g}{h}(\lambda \tilde{P}) = \frac{g(\lambda \tilde{P})}{h(\lambda \tilde{P})} = \frac{\lambda^d g(\tilde{P})}{\lambda^d h(\tilde{P})} = \frac{g(\tilde{P})}{h(\tilde{P})}.$$

So homogeneous rational functions g/h define maps $D_{\mathbb{P}^n}(h) \to k$.

This allows us to better describe the SWF $(\mathbb{P}^n, \mathcal{O}_{reg})$.

2.3.8 Lemma

For $U \subseteq \mathbb{P}^n$ open, a function $f\colon U \to k$ is regular iff there are homogeneous polynomials $g, h \in S_d$ such that $U \subseteq D_{\mathbb{P}^n}(h)$ and $f = g/h|_U$.

Proof

We know that f is regular iff $\pi^*(f)\colon \pi^{-1}U \to k$ is regular. This is iff it is rational, so there are polynomials $g, h \in S$ such that $\pi^{-1}U \subseteq D(h)$ and for everu $\tilde{P} \in \pi^{-1}U$ we have

$$\pi^*(f)(P) = f(\pi(P)) = \frac{g(\tilde{P})}{h(\tilde{P})}.$$

In particular, for $g, h \in S_d$ the function $f = g/h$ is regular.

Conversely we want to show that two polynomials satisfying the above equality must be homogeneous of the same degree. First we claim that $\pi^{-1}U \subseteq D(h)$ implies h homogeneity. Suppose not, so there are $i_1 \neq i_2$ such that h_{i_1}, h_{i_2} are nonzero. We have open non-empty subsets $D(h_{i_1}), D(h_{i_2}), \pi^{-1}U \subseteq \mathbb{A}^{n+1}$, and since $\mathbb{A}^{n+1}$ is irreducible this means that their intersection is nonempty. So there is $\tilde{P} \in \pi^{-1}U$ such that $h_{i_1}(\tilde{P}), h_{i_2}(\tilde{P}) \neq 0$.

Then for every $\lambda \in k^\times$ we have $\lambda\tilde{P} \in \pi^{-1}U \subseteq D(h)$ so $h(\lambda\tilde{P}) \neq 0$. Hence

$$0 \neq h(\lambda\tilde{P}) = \sum_i h_i(\lambda\tilde{P}) = \sum_i \lambda^i h_i(\tilde{P}),$$

then as before this means $Q(t) = \sum_i h_i(\tilde{P})t^i$ has no zeroes in $k^\times$. Since k is algebraically closed, this means its only zero is 0 and thus $Q(t) = at^d$ for some $a \in k$. This means that $h_i(\tilde{P}) = 0$ for all $i \neq d$, so there is at most a single index where $h_i(\tilde{P}) \neq 0$, a contradiction.

Now we claim that g is homogeneous. For every $\tilde{P} \in \pi^{-1}U$ and $\lambda \in k^\times$ we have

$$\frac{g(\lambda\tilde{P})}{\lambda^d h(\tilde{P})} = \frac{g(\lambda\tilde{P})}{h(\lambda\tilde{P})} = f(\pi(\lambda\tilde{P})) = f(\pi(\tilde{P})) = \frac{g(\tilde{P})}{h(\tilde{P})}.$$

Thus we see $g(\lambda\tilde{P}) = \lambda^d g(\tilde{P})$. Thus for all $\lambda \in k^\times$,

$$\sum_i \lambda^i g_i(\tilde{P}) = \lambda^d \sum_i g_i(\tilde{P}).$$

Defining $Q(t) = t^d \sum_i g_i(\tilde{P}) - \sum_i t^i g_i(\tilde{P})$ we see that this has infinitely many zeroes and is thus identically zero, so $g_i(\tilde{P}) = 0$ for all $i \neq d$. So $g(\tilde{P}) = g_d(\tilde{P})$ for all $\tilde{P} \in \pi^{-1}U$. Since $\mathbb{A}^{n+1}$ is Noetherian, $\pi^{-1}U$ is dense and so $g = g_d$ on all $\mathbb{A}^{n+1}$ as desired. ■

Note:

The above results, namely that $Z \subseteq \mathbb{P}^n$ is closed iff it is of the form $Z_{\mathbb{P}^n}(T)$ and $f\colon U \to k$ is regular iff it is homogeneous rational, are many times taken to be definitions of projective space. We had to work a little harder to show that the more obvious definition, induced from affine space, agrees with this.

2.3.9 Corollary

Let $Y \subseteq \mathbb{P}^n$ be an arbitrary subspace, so $(\mathbb{P}^n, \mathcal{O}_{reg})$ induces an SWF $(Y, \mathcal{O}_{reg})$.

(1) The closed sets of Y are those of the form $Z_Y(T) = \{P \in Y \mid f(P) = 0 \text{ for all } f \in T\}$.

(2) The collection $D_Y(f) = \{P \in Y \mid f(P) \neq 0\}$ forms a basis of open sets of Y.

(3) A function $f\colon Y \to k$ is regular iff it is locally given by homogeneous rational functions. That is, for every $P \in Y$ there is an open subset $P \in U \subseteq \mathbb{P}^n$ such that $f|_U$ is homogeneous rational.

Proof

Points (1) and (2) are clear from the definition of the subspace topology, the obvious equalities

$$Z_Y(T) = Z_{\mathbb{P}^n}(T) \cap Y, \qquad D_Y(T) = D_{\mathbb{P}^n}(T) \cap Y,$$

and the results we proved above. Point (3) is immediate from the definition of induced SWFs and the previous result. ∎

2.3.10 Lemma

The SWF $(\mathbb{P}^n, \mathcal{O}_{reg})$ is an algebraic variety.

Proof

Cover $\mathbb{A}^{n+1} - 0$ by $\{D(x_i)\}_{i=0}^n$, so through π we cover $\mathbb{P}^n$ by $\{D_{\mathbb{P}^n}(x_i)\}_{i=0}^n$. And clearly we have

$$U_i = D_{\mathbb{P}^n}(x_i) = \{[a_0 : \ldots : a_n] \in P \mid a_i \neq 0\}.$$

We claim that these U_i are affine.
Define $\varphi\colon U_i \to \mathbb{A}^n$ by

$$\varphi[a_0 : \ldots : a_n] = \left(\frac{a_0}{a_i}, \ldots, \frac{\hat{a}_i}{a_i}, \ldots, \frac{a_n}{a_i}\right).$$

Each map $[a_0 : \ldots : a_n] \mapsto a_j/a_i$ is regular, as it is homogeneous rational (given by x_j/x_i), so this is a morphism into $\mathbb{A}^n$. φ is bijective with inverse

$$\psi(a_1, \ldots, a_n) = [a_1 : \ldots : a_i : 1 : a_{i+1} : \ldots : a_n].$$

To show that ψ is a morphism of SWFs, we note that it is equal to the composition

$$\mathbb{A}^n \xrightarrow{\tilde{\psi}} \mathbb{A}^{n+1} \xrightarrow{\pi} \mathbb{P}^n,$$

where $\tilde{\psi}$ maps $(a_1, \ldots, a_n)$ to $(a_1, \ldots, a_i, 1, a_{i+1}, \ldots, a_n)$, and this is clearly a morphism of SWFs as its components are regular (polynomial even). Thus ψ is the composition of morphisms of SWFs and thus a morphism itself. ∎

2.3.11 Definition

A **projective variety** is an SWF X which is isomorphic to an SWF $(Y, \mathcal{O}_{reg})$ where $Y \subseteq \mathbb{P}^n$ is closed.

A **quasi-projective variety** is an SWF X which is isomorphic to an open subspace of an open subvariety of a projective variety.

Clearly projective varieties are quasi-projective. Notice that quasi-projective varieties are just SWFs which are isomorphic to locally closed subvarieties of $\mathbb{P}^n$.

2.3.12 Corollary

(1) Quasi-projective varieties are algebraic varieties.

(2) Every locally closed subvariety of a quasi-projective variety is quasi-projective.

(3) Every quasi-affine variety is quasi-projective.

Proof

The first point follows from the fact that locally closed subvarieties of an algebraic variety are algebraic. A locally closed subvariety of a quasi-projective variety is isomorphic to a locally closed subvariety of a locally closed subvariety of $\mathbb{P}^n$. Local closure is a transitive property, so this is just a locally closed subvariety of $\mathbb{P}^n$ and thus quasi-projective, showing (2).

Since $\mathbb{A}^n \cong U_0 \subseteq \mathbb{P}^n$, we see that affine space $\mathbb{A}^n$ is projective. Quasi-affine varieties are locally closed subvarieties of $\mathbb{A}^n$, and thus locally closed subvarieties of a projective variety, and thus quasi-affine by (2).

■